

**Centre for International Macroeconomic Studies
School of Economics, University of Surrey**



The Science and Art of DSGE Modelling: A Foundations Course

Cristiano Cantore

Vasco Gabriel

Paul Levine

Maryam Mirfatah

Ricardo Nunes

Luciano Rispoli

Kirill Shakhnov

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INTRODUCTION

This foundations course is aimed at early researchers with some knowledge of Real Business Cycle (RBC) or Dynamic Stochastic General Equilibrium (DSGE) macroeconomic models, but little or no experience of Dynare. The material covers the underlying theory proceeding in steps from the Real Business Cycle Model through to a medium-sized New Keynesian Model. It also covers the theoretical basis for Bayesian Estimation. This is a hands-on course based on available facilities in the software platform **Dynare**. Model software is provided consisting of models set-up in Dynare in a manner suitable for simulation, calibration, estimation and policy analysis.

Taken over the four days the participants will be guided through a seamless methodology for the construction, solution, estimation and policy analysis of micro-founded macroeconomic models summarized by the following steps:

1. The construction of DSGE models describing the first-order conditions for economic agents in the form of a set of non-linear difference equations
2. Solution of the deterministic (non-stochastic) steady state to be used for both solution and calibration
3. The stochastic solution by a perturbation method about this deterministic of the steady state
4. The Perfect Foresight Solution
5. Bayesian estimation methods
6. Model comparisons between different models or variants of same model
7. Model validation by comparison with the second moments of data
8. Stability using the Global Sensitivity Analysis toolbox
9. Optimal policy

1.1 READING

There are a number of excellent books on modern dynamic macroeconomics that provide background reading for the course. Herbst and Schorfheide (2015) and Dejong and Dave (2007) cover in depth the empirical aspects of DSGE modelling. To understand the models themselves, good recent text-books are Miao (2014) (which all covers empirical aspects and use of dynare), and Galí (2015), covering closed and open-economy models. Other useful books are Wickens (2012), Koop (2003), McCandless (2008), Lim and McNelis (2008) and Geweke (2010). On optimal policy, rules and discretion, Currie and Levine (1993) may also prove useful.

At some stage of their researchers in this area will need to consult four seminal books: one on New Keynesian models, Woodford (2003), two cover the empirical side, Geweke (2005) and Canova (2007), and Ljungqvist and Sargent (2012) covers dynamic optimization; but they are all challenging reads!

1.2 INSTRUCTORS

Profiles for the instructors are as follows.

Cristiano Cantore is a Senior Research Economist at the Research Hub of the Bank of England and a Reader in the School of Economics at the University of Surrey. He graduated from the Bocconi University (Milan, Italy) in 2004. He then completed his MSc degree in Economics at Pompeu Fabra University (Barcelona, Spain). In 2006, he started the PhD in Economics at University of Kent where he was awarded an ESRC Scholarship. Cristiano's PhD focussed on financial frictions and capital labour substitution in dynamic stochastic general equilibrium (DSGE) models. Cristiano has also worked at the OECD and at the ECB as a trainee and visited the Bank of Spain in 2012 as a research fellow. In September 2009 he was appointed as full time Lecturer and in 2013 he was promoted to Senior Lecturer at the University of Surrey. His research fields mainly include macroeconomic theory, computational economics, monetary economics and production theory.

Vasco Gabriel is a Reader and until recently was the Head of the School of Economics at the University of Surrey. He graduated in Economics from the Technical University of Lisbon in 1995, where he was awarded the ICEP prize. He received a masters degree in Econometrics in 1998 from the same institution. In 2002, he completed his PhD in Economics at Birkbeck College,

University of London. He taught at the University of Minho, Portugal, before being appointed as a Lecturer at the University of Surrey in 2004 and Senior Lecturer in 2010. Vasco's main field of specialization is Macroeconometrics, focusing on the application of non-linear methods, as well as general inference issues in macro models. He has published extensively in these areas including publications in *Economic Letters*, the *Journal of Macroeconomics*, the *Journal of Money, Credit and Banking* and the *Oxford Handbook of the Indian Economy*.

Paul Levine is a Professor in the School of Economics at the University of Surrey. He received a first-class BSc and a PhD, both in Mathematics, from the University of Manchester and an MSc in economics (distinction) at Queen Mary, London. In 1984 he became a senior research officer at the Centre for Economic Forecasting, London Business School and was appointed Professor of Economics at the University of Leicester in 1989. In 1994 he moved to the University of Surrey where he now leads the Centre for International Macroeconomic Studies (CIMS). He has acted as a consultant and/or visiting researcher at the IMF, the ECB, the central banks of Peru and Nigeria, and the World Bank. His main research activity is in constructing empirically-based DSGE models for the purpose of macroeconomic policy analysis. He has published over 100 refereed articles or chapters and 2 books.

Maryam Mirfatah is a Postdoctoral Research Fellow in Economics at the School of Economics, University of Surrey. She joined the Centre for International Macroeconomic Studies (CIMS) in 2018 on the project of "Macroeconomic Modelling and Policy Analysis for Emerging Economies". Currently, she is working on optimized simple monetary policy rules for open economies. Her research interests are in Macroeconomics, in particular the construction and estimation of DSGE models for the purpose of macroeconomic policy analysis. Maryam obtained her BSc in Statistics from Isfahan University of Technology, MSc in Economics from Azad University and a PhD in Economics (2019) from Yazd University, focusing on Money Growth Rules in Emerging Economies.

Ricardo Nunes is a Professor in the School of Economics at the University of Surrey. He graduated from Universitat Pompeu Fabra (Barcelona, Spain) obtaining a MSc in Economics in 2003 and a PhD in Economics in 2007. After graduating he spent 10 years in the Federal Reserve System under various roles. In 2007 he joined the Board of Governors of the Federal Reserve System, where he worked as an economist and senior economist. In 2014 he moved to the Federal Reserve Bank of Boston working as a senior economist and policy advisor. He was also a visiting researcher at the Bank of Portugal and the IMF and has

given talks at various central banks. His main research is on monetary and fiscal policy. He has published extensively in these areas including the Quarterly Journal of Economics, Journal of Monetary Economics, Journal of Economic Theory, Journal of European Economic Association, among others. For further details of publications see <https://sites.google.com/view/ricardonunes/home>

Luciano Rispoli received his undergraduate degree in Economics in 2009 from Bocconi University. During his undergraduate degree he spent 6 months as an exchange student at the Australian National University. He then obtained a MSc in International Economics and Finance with distinction from the University of Sheffield in 2011. He then pursued his doctoral studies at Birkbeck College - University of London where he also worked as a Teaching Assistant and sessional Associate Lecturer. In 2016 he joined the University of Warwick as a Teaching Fellow where he taught Monetary Economics (MSc) and undergraduate micro and macroeconomics. In 2019 he joined the School of Economics at Surrey University where he currently teaches Quantitative methods (BSc), Corporate finance (BSc) and Financial Econometrics (MSc). His research fields are macro-econometrics and monetary policy.

Kirill Shakhnov is a Lecturer in the School of Economics at the University of Surrey. In 2015, he completed his Ph.D. in Economics at European University Institute, Italy. He then held the Foscolo Europe Fellowship position at the Einaudi Institute for Economics and Finance and taught at LUISS University, before being appointed as a Lecturer at the University of Surrey in 2019. Kirill's primary research interests lie in the fields of international macroeconomics with a focus on sovereign bond markets and cryptocurrencies. The research is organized around three related topics: asset pricing of internationally traded assets, public finance, and the sources of macroeconomic fluctuations. For further details, please see <https://sites.google.com/site/kshakhnov>.

1.3 COURSE OUTLINE

The contents of lectures given by the instructors over the four days are as follows:

- **Day 1: Dynare Basics and the RBC Model**
 - Instructors: Levine, Shakhnov
 - Introduction to the 4-Day Course
 - Dynare Basics

- RBC Model
- Calibration and Use of an External Steady State
- Linearisation
- Exercises
- **Day 2: The New Keynesian Model, Stability Analysis and the GSA Toolbox**
 - Instructors: Cristiano, Levine, Nunes
 - The New Keynesian (NK) Model
 - * The Workhorse NK Model
 - * The Smets-Wouters NK Model
 - * Monetary Policy
 - Solution-Stability-Indeterminacy
 - The Global Sensitivity Analysis toolbox in Dynare
 - Exercises
- **Day 3: Bayesian Estimation of the NK Model**
 - Instructors: Levine, Mirfatah, Rispoli,
 - Preparing the Data including use of various filters
 - An Introduction to Bayesian Methodology
 - Identification and Pre-estimation Checks
 - Direct Linear Estimation of the Linear NK Model
 - Exercises
- **Day 4: More on Estimation and Optimal Monetary Policy**
 - Instructors: Gabriel, Levine
 - Model Comparisons by Likelihood Races
 - Comparison of Second Moments of Model with Data
 - Optimal Monetary and Fiscal Policy for Linear-Quadratic Problems
 - * The Ramsey Problem
 - * Optimal Time-Consistent Policy
 - * Optimized Simple Rules

- Concluding Remarks
- Exercises

2 | THE BASICS OF DYNARE

2.1 WHAT IS DYNARE?

- A suite of programmes for the solution, simulation and estimation of rational expectations-DSGE models
- Also can carry out optimal policy exercises
- OS: Windows, Mac OS and Linux.
- Platforms: **MATLAB**, version 7.3 (R2006b) and later, and GNU Octave version 3.2
- Collection of over 300+ MATLAB routines (part of Dynare is programmed in C++)
- More information and to download Dynare (v4.4.3 is the latest (stable) version)
- User-friendly
- This course will be based on Dynare v4.4.3

2.2 LEARNING RESOURCES

- See Dynare *Reference Manual (and User Guide)*
- *Official Online Examples*: available also from the Dynare website and are usually well documented
- *Dynare Discussion Forum* (access from the website): for asking questions, posting comments and uploading examples and has more examples and latest developments
- *DSGE.net*: website run by the Dynare team and useful not only for the dynare users but also for all scholars working on DSGE modelling

- **Dynare Wiki:** various bits of information such as new or undocumented features and development plans

2.3 WHAT IS DYNARE ABLE TO DO?

- Specifies a rational expectations (RE) on entirely backward-looking non-RE model in linear or non-linear form, no need for state space matrices
- Computes the steady state (numerically) of the model
- Computes the solution of deterministic models
- Computes first, second and third order perturbation (Taylor series) approximation to solutions of stochastic models
- Estimates (either by maximum-likelihood(ML) or Bayesian approach) parameters of DSGE models and their distribution

2.4 HOW DOES DYNARE WORK?

- The user writes a .mod file (in an editor of your choice)
- That file is then called from MATLAB allowing the pre-processor to translate the .mod file into a suitable input for the MATLAB routines (i.e. Dynare at this stage produces .m files)
- MATLAB routines are used to either solve or estimate the model and present the output $\Rightarrow$.mat are matrices produced by MATLAB

– Installing (running) Dynare on Windows (in MATLAB):

- * First, at least MATLAB version 7.3 for most of Dynare features (sometimes the optimization toolbox needed)
- * Download and run the installation file (This will place Dynare in the root of C: e.g. *c : \dynare\4.4.y*)
- * Start MATLAB and click on menu File/Set Path to add the path to the Dynare MATLAB subdirectory (to store all the subroutines), e.g. the path would be set to *c : \dynare\4.4.y\matlab*

- * Save the changes
- * Note that you can have several versions of Dynare coexisting
- * To run in MATLAB, simply type in the command window: `dynare filename.mod`
- Octave can be used as an equivalent to MATLAB.
- **Installing (running) Dynare in Octave**
 - * First, the latest version of Octave should be installed. Information is on the Dynare website.
 - * Download and run the Dynare installation file.
 - * Run Octave by clicking on the icon and add the path to the Dynare subdirectory by typing `add path c : \dynare\4.x.y\matlab` into the command prompt.
 - * Ensure your working directory is correct by typing e.g. `cd c:\mydocuments\dynare` into the command prompt to find the folder containing your .mod file.
 - * Solve the model using Dynare in Octave by typing `dynare filename.mod` into the command prompt
 - * Note that if additional .m files are used such as external steady state solvers, these might need amending to work within the Octave environment.

2.5 STRUCTURE OF A BASIC .MOD FILE

- Variables and parameters declarations
- Parameter initialization
- Model declaration
- Shocks (and their properties)
- Initial and terminal conditions for simulations (where the terminal conditions are only applicable to specify permanent shocks that hit deterministic models)
- Initial guess values for non-linear solvers (optional)

- Computations to be performed, e.g.: `steady`, `check`, `forecast`, `simul`, `stoch_simul`, `estimation`, `ramsey_policy`, etc ...

2.5.1 Declarations of the variables and parameters

- Endogenous and exogenous variables are declared separately
- For endogenous variables use the command: `var` (followed by the list of endogenous in the model)
- Exogenous variables are declared with: `varexo` (followed by the list of exogenous variables that will be shocked)
- The parameters of the model are defined with the command: `parameters` and afterwards they are calibrated (values are assigned to each)
- **Note** that each instruction of the `.mod` file must be terminated by a semi-colon (;)
- Also Dynare uses 2 forward slashes (//) to comment out any line (whereas MATLAB uses %). (Note: for Dynare the two are equivalent!)

2.5.2 Specifying the model

- The equations of the model (non-linear) are defined within the commands `model;` and `end;`
- Different time indices in a dynamic model are abbreviated as:
 - $x_t = x \rightarrow$ variable x decided in the current period (t) is written plainly as x
 - $x_{t-1} = x(-1) \rightarrow$ the variable is decided in $t-1$ (predetermined), e.g. the capital stock, write it as $x(-1)$ instead of x
 - $x_{t+1} = x(+1) \rightarrow$ the variable is decided n periods ahead, write $x(+n)$. Dynare reads $x(+1)$ as a jumper (or forward-looking or non-predetermined) variable
- It is important to make sure when specifying the model:
 - there need to be as many equations as your endogenous variables declared (except for optimal policy)

- names are case sensitive
- The stability “Blanchard-Kahn” conditions are met only if the number of jumpers equals the number of eigenvalues greater than one. (See Topic 2).
- In case the model consists of linear equations use *model (linear)*; as opening command.

2.5.3 Properties of shocks

- The variances (and covariances) of the shocks are defined within the commands *shocks*; and *end*;
- e.g. one most common way is to use: the command ‘*var name of exogenous variable; stderr 0.02;*’ which sets the std. error of this exogenous variable = 0.02
- Notice that in Dynare the shocks statement is evaluated only once (for the model solution)

2.5.4 Task 1: finding the steady state

- Dynare solves for the steady state of the model (nonlinear Newton-type solver). It just needs good numerical initial values - simple to use?
- These are initial guesses specified within the commands *initval*; and *end*;
- Then, the command *steady* triggers the computation of exact values for the steady state using the initial guesses for the values of the endogenous variables
- Variables that are not initialized in *initval;...end*; are set to zero
- The routines (solvers) are very sensitive to your guess - the best guess is the analytically calculated steady state
- A useful command after *steady, check*, that checks the Blanchard and Kahn (1980) conditions that ensure the existence of a RE solution by computing the eigenvalues of the model
- Can have problems in finding the steady state depending on the model complexity and the initial values

- An alternative is using a MATLAB/Octave function doing the computation externally with a Matlab program `FILENAME_steadystate.m`
- The correct steady state is then computed and called by `FILENAME.mod`
- Useful (more efficient) when you know how to compute the steady state for your model
- Sometimes the steady state can be written recursively and put within the initial values commands. This is the case on this Course.

2.5.5 Task 2: solving (simulating) the model

- For the most part we construct *stochastic* models, hence DSGE. The command `stoch_simul` performs the solution using a perturbation (Taylor small-shock approximation) routine.
- However, for a perfect foresight (*deterministic*) simulations, `simul` gives an exact solution. See Day 2 applied to fiscal policy in the RBC model.
- `stoch_simul` computes a Taylor approximation around the steady state of order one, two or three
- Output includes a Taylor approximation of the decision and transition functions for the model as well as simulated IRFs, moments, autocorrelations
- It also simulates artificial data from the model (`datatomfile` - saves the simulated data in a m file. Example: `datatomfile('simuldata',[1])`)
- Some useful options set in brackets after `stoch_simul`:
 - `periods` - specifies the number of simulation periods. Example: `periods=1000;`
 - `irf` sets the number of periods for which to compute impulse responses.
 - `order = 1` sets the order of the Taylor approximation (default is two)
 - a complete description can be found from the manual

2.5.6 Task 3: estimating the model (MLE or Bayesian)

- Dynare estimates the structural parameters of a model based on a linear approximation
- Estimation steps:
 - computes the steady state
 - linearizes the model (only for non-linear models)
 - solves the linearized model
 - computes the log-likelihood via the Kalman filter
 - finds the maximum of the likelihood or posterior mode
 - simulates posterior distribution with Metropolis algorithm
 - computes various statistics on the basis of the posterior distribution (post. moments)
 - estimates the posterior marginal density to compare models.
 - computes smoothed values of unobserved variables
 - computes forecasts and confidence intervals

2.6 SIMPLE DYNARE MODELLING EXAMPLES

To illustrate the workings of dynare the folder **Dynare Basics** sets codes to solve the following

1. Example 1: AR1 processes:

$$x_t = \rho_x x_{t-1} + e_{x,t}$$

$$y_t = \rho_y y_{t-1} + e_{y,t}$$

$$e_{x,t} \sim i.i.d(0, \sigma_x^2)$$

$$e_{y,t} \sim i.i.d(0, \sigma_y^2)$$

2. Example 2: A first-order 2-variable VAR:

$$x_t = \rho_{xx} x_{t-1} + \rho_{xy} y_{t-1} + e_{x,t}$$

$$y_t = \rho_{yy} y_{t-1} + \rho_{yx} x_{t-1} + e_{y,t}$$

$$e_{x,t} \sim i.i.d(0, \sigma_x^2)$$

$$e_{y,t} \sim i.i.d(0, \sigma_y^2)$$

Dynare model files **Dynare_Example_1. mod** and **Dynare_Example_2** illustrate the stochastic solution where shocks are present.

2.7 DYNARE EXERCISE

Generalize example 2 above to a first-order, three-variable VAR:

$$\begin{aligned}
 x_t &= \rho_{xx}x_{t-1} + \rho_{xy}y_{t-1} + \rho_{xz}z_{t-1} + e_{x,t} \\
 y_t &= \rho_{yy}y_{t-1} + \rho_{yx}x_{t-1} + \rho_{yz}z_{t-1} + e_{y,t} \\
 z_t &= \rho_{zx}x_{t-1} + \rho_{zy}y_{t-1} + \rho_{zz}z_{t-1} + e_{z,t} \\
 e_{x,t} &\sim i.i.d(0, \sigma_x^2) \\
 e_{y,t} &\sim i.i.d(0, \sigma_y^2) \\
 e_{z,t} &\sim i.i.d(0, \sigma_z^2)
 \end{aligned}$$

Experiment with parameters to find unstable dynamics.

3 | THE REAL BUSINESS CYCLE MODEL

The construction of a DSGE model requires the specification of agents' preferences, the economy's technological constraints and the set of exogenous shocks to which the economy is subject. There is also an implicit or explicit assumption regarding the formation of expectations and the information available regarding future macroeconomic variable relevant to the agents. Throughout the course we assume rational, model consistent expectations with information assumptions made clear as we proceed.

The agents decision rules are derived from the first order conditions of the dynamic optimization problem for each agent. Aggregating over agents and assuming that markets clear allow us to derive a system of non-linear stochastic difference equations, involving the endogenous variables, the parameters and a set of shocks. The purpose is then to find a stable and unique solution to the model, which requires an additional set of procedures. The model should be written in stationary form, so that the variables are written as deviations from a balanced-growth steady-state. Often DSGE models are log-linearized and written in state-space form. Standard methods then result in a linear rational expectations solution of the model. We demonstrate this procedure for the models that follow, but using the software package Dynare. The modeller can simply set up the non-linear model and either compute the steady state or in the real business cycle (RBC) model write down an analytical solution. Dynare then performs the linearization to produce a first-order solution. These different stages are detailed next.

We consider a standard real business cycle (RBC) model with adjustment costs of investment (see Lucas (1987), Wickens (2008), Lim and McNelis (2008), chapter 6). This lies at the core of New Keynesian (NK), growth, banking, search-match and small open economy models, some of which will follow.

3.1 THE STANDARD RBC MODEL WITH INVESTMENT ADJUSTMENT COSTS

Households choose between work and leisure and therefore how much labour they supply. They also own the capital stock which is rented to firms at a rental rate r_t^K and choose an optimal investment path. Let L_t be the proportion of the total time available for work (say 16 hours per day) that consists of leisure time and H_t the proportion of this time spent at work. Then clearly $L_t + H_t = 1$. The single-period utility is

$$U = U(C_t, L_t) \quad (3.1)$$

and we assume that¹

$$U_C > 0, U_L > 0, U_{CC} \leq 0, U_{LL} \leq 0 \quad (3.2)$$

In a stochastic environment, the value function of the representative household at time t is given by

$$V_t = V_t(B_{t-1}) = \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta^s U(C_{t+s}, L_{t+s}) \right] \quad (3.3)$$

For the household's problem at time t is to choose paths for consumption $\{C_t\}$, leisure, $\{L_t\}$, labour supply $\{H_t = 1 - L_t\}$, capital stock $\{K_t\}$, investment $\{I_t\}$ and bond holdings to maximize V_t given by (3.3) given its budget constraint in period t

$$B_t = R_{t-1}B_{t-1} + r_t^K K_{t-1} + W_t H_t - C_t - I_t - T_t \quad (3.4)$$

where B_t is the value (price $\times$ number of bonds) of the net stock of financial assets at the end of period t , r_t^K is the rental rate, W_t is the wage rate and R_t is the gross interest rate paid on assets held at the beginning of period t , I_t is investment and T_t are lump-sum taxes; and given that capital stock accumulates according to

$$K_t = (1 - \delta)K_{t-1} + (1 - S(X_t))I_t; \quad (3.5)$$

$$X_t \equiv \frac{I_t}{I_{t-1}}; \quad S', S'' \geq 0; \quad S(1) = S'(1) = 0 \quad (3.6)$$

In (3.6), $S(X_t)$ are investment adjustment costs, I_t units of output converts to $(1 - S(X_t))I_t$ of new capital sold at a real price Q_t (Tobin's Q). All variables are expressed in real terms relative to the price of output.

¹ Our notation is $U_C \equiv \frac{\partial U}{\partial C}$, $U_{CC} \equiv \frac{\partial^2 U}{\partial C^2}$ etc.

From Appendix 1, the first-order conditions for this optimization problem are

$$\text{Euler Consumption : } U_{C,t} = \beta R_t \mathbb{E}_t [U_{C,t+1}] \quad (3.7)$$

$$\text{Labour Supply : } \frac{U_{H,t}}{U_{C,t}} = -\frac{U_{L,t}}{U_{C,t}} = -W_t \quad (3.8)$$

$$\text{Leisure and Hours : } L_t \equiv 1 - H_t \quad (3.9)$$

$$\begin{aligned} \text{Investment FOC : } Q_t(1 - S(X_t) - X_t S'(X_t)) \\ + E_t \left[\Lambda_{t,t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2 \right] = 1 \end{aligned} \quad (3.10)$$

$$\text{Capital Supply : } \mathbb{E}_t \left[\Lambda_{t,t+1} R_{t+1}^K \right] = 1 \quad (3.11)$$

where $\Lambda_{t,t+1} \equiv \beta \frac{U_{C,t+1}}{U_{C,t}}$ is the *real stochastic discount factor* over the interval $[t, t+1]$, $X_t = I_t/I_{t-1}$ is the rate of change of investment and R_t^K is the gross return on capital given by

$$R_t^K = \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}} \quad (3.12)$$

The Euler consumption equation, (3.13), where $U_{C,t} \equiv \frac{\partial U_t}{\partial C_t}$ is the marginal utility of consumption and $E_t[\cdot]$ denotes rational expectations based on agents observing all current macroeconomic variables (i.e., ‘perfect information’), describes the optimal consumption-savings decisions of the household. It equates the marginal utility from consuming one unit of income in period t with the discounted marginal utility from consuming the gross income acquired, R_t , by saving the income. For later use it is convenient to write the Euler consumption equation as

$$1 = R_t \mathbb{E}_t [\Lambda_{t,t+1}] \quad (3.13)$$

(3.8) equates the real wage with the marginal rate of substitution between consumption and leisure.

(3.10) is the first-order condition for investment where I_t units of output converts to $(1 - S(X_t))I_t$ of new capital sold at a real price Q_t (See Appendix 1). (3.11), (3.12) and (3.13) equate the expected discounted return on a riskless bond with that of capital over the period $[t, t+1]$.

Output and the firm’s behaviour is summarized by:

$$\text{Output : } Y_t = F(A_t, H_t, K_{t-1}) \quad (3.14)$$

$$\text{Labour Demand : } F_{H,t} = W_t \quad (3.15)$$

$$\text{Capital Demand : } F_{K,t} = r_t^K \quad (3.16)$$

(3.14) is a production function. Note here K_t is *end-of-period* t capital stock. Equation (3.15), where $F_{H,t} \equiv \frac{\partial F_t}{\partial H_t}$, equates the marginal product of labour with the real wage. (3.16), where $F_{K,t} \equiv \frac{\partial F_t}{\partial K_t}$, equates the marginal product of capital with the rental rate r_t^K .

The model is completed with an output equilibrium and a balanced budget constraint with lump-sum taxes.

$$\begin{aligned} Y_t &= C_t + G_t + I_t \\ K_t &= (1 - \delta)K_{t-1} + (1 - S(X_t))I_t; \quad S', S'' \geq 0; \quad S(1) = S'(1) = 0 \\ X_t &\equiv \frac{I_t}{I_{t-1}} \\ G_t &= T_t \end{aligned}$$

In the zero-growth steady state, $Q = X = 1$ and $S = 0$ and investment costs disappear.

We now specify *functional forms* for production and utility and AR(1) processes for exogenous variables A_t and G_t . For production we assume a Cobb-Douglas function. The utility function is non-separable and consistent with a balanced growth path when the inter-temporal elasticity of substitution, $1/\sigma_c$ is not unitary (see Barro and Sala-i-Martin (2004), chapter 9, section 9.4). These functional forms, the associated marginal utilities and marginal products, and exogenous processes are given by

$$F(A_t, H_t, K_{t-1}) = (A_t H_t)^\alpha K_{t-1}^{1-\alpha} \quad (3.17)$$

$$F_H(A_t, H_t, K_{t-1}) = \frac{\alpha Y_t}{H_t} \quad (3.18)$$

$$F_K(A_t, H_t, K_{t-1}) = \frac{(1 - \alpha)Y_t}{K_{t-1}} \quad (3.19)$$

$$\log A_t - \log \bar{A}_t = \rho_A (\log A_{t-1} - \log \bar{A}_{t-1}) + \epsilon_{A,t} \quad (3.20)$$

$$\log G_t - \log \bar{G}_t = \rho_G (\log G_{t-1} - \log \bar{G}_{t-1}) + \epsilon_{G,t} \quad (3.21)$$

$$\begin{aligned} U_t &= \frac{(C_t^{(1-\varrho)} (1 - H_t)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\ &\rightarrow (1 - \varrho) \log C_t + \varrho \log(1 - H_t) \text{ as } \sigma_c \rightarrow 1 \end{aligned} \quad (3.22)$$

$$U_{C,t} = (1 - \varrho) C_t^{(1-\varrho)(1-\sigma_c)-1} (1 - H_t)^{\varrho(1-\sigma_c)} \quad (3.23)$$

$$U_{H,t} = -\varrho C_t^{(1-\varrho)(1-\sigma_c)} (1 - H_t)^{\varrho(1-\sigma_c)-1} \quad (3.24)$$

$$S(X_t) = \phi_X (X_t - 1)^2 \quad (3.25)$$

3.2 SUMMARY OF EQUILIBRIUM

We can now summarize the equilibrium in terms of the following 18 independent endogenous equations in 18 macroeconomic variables U_t , $U_{C,t}$, $U_{H,t}$, $\Lambda_{t-1,t}$, R_t , R_t^K , I_t , X_t , W_t , Y_t , H_t , r_t^K , Q_t , C_t , K_t , $S(X_t)$, X_t and T_t , given parameter values and exogenous processes, A_t and G_t . For the latter we assume AR1 processes about possibly trending steady states $\bar{A}_t, \bar{G}_t$ driven by zero mean iid shocks $\epsilon_{A,t}$ and $\epsilon_{G,t}$.

The equilibrium is a system, but you can think of these variables being “assigned” to these equations in that order; for instance (3.38) “determines” Q_t and (3.39) “determines” C_t . In fact all variables are simultaneously determined which is what we mean by an equilibrium.

$$U_t = \frac{C_t^{(1-\varrho)}(1-H_t)^{\varrho}1^{1-\sigma_c} - 1}{1-\sigma_c} \quad (3.26)$$

$$U_{C,t} = (1-\varrho)C_t^{(1-\varrho)(1-\sigma_c)-1}(1-H_t)^{\varrho(1-\sigma_c)} \quad (3.27)$$

$$U_{H,t} = -\varrho C_t^{(1-\varrho)(1-\sigma_c)}(1-H_t)^{\varrho(1-\sigma_c)-1} \quad (3.28)$$

$$\Lambda_{t-1,t} \equiv \beta \frac{U_{C,t}}{U_{C,t-1}} \quad (3.29)$$

$$\mathbb{E}_t[\Lambda_{t,t+1}R_t] = 1 \quad (3.30)$$

$$\mathbb{E}_t[\Lambda_{t,t+1}R_{t+1}^K] = 1 \quad (3.31)$$

$$-\frac{U_{H,t}}{U_{C,t}} = W_t \quad (3.32)$$

$$Q_t(1 - S(X_t) - X_t S'(X_t)) + E_t[\Lambda_{t,t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2] = 1 \quad (3.33)$$

$$X_t \equiv \frac{I_t}{I_{t-1}} \quad (3.34)$$

$$Y_t = (A_t H_t)^\alpha K_{t-1}^{1-\alpha} \quad (3.35)$$

$$\frac{\alpha Y_t}{H_t} = W_t \quad (3.36)$$

$$\frac{(1-\alpha)Y_t}{K_{t-1}} = r_t^K \quad (3.37)$$

$$R_t^K = \frac{r_t^K + (1-\delta)Q_{t-1}}{Q_t} \quad (3.38)$$

$$Y_t = C_t + G_t + I_t \quad (3.39)$$

$$K_t = (1 - \delta)K_{t-1} + (1 - S(X_t))I_t \quad (3.40)$$

$$S(X_t) = \phi_X(X_t - 1)^2 \quad (3.41)$$

$$X_t \equiv \frac{I_t}{I_{t-1}} \quad (3.42)$$

$$T_t = G_t \quad (3.43)$$

$$\log A_t - \log \bar{A}_t = \rho_A(\log A_{t-1} - \log \bar{A}_{t-1}) + \epsilon_{A,t} \quad (3.44)$$

$$\log G_t - \log \bar{G}_t = \rho_G(\log G_{t-1} - \log \bar{G}_{t-1}) + \epsilon_{G,t} \quad (3.45)$$

3.3 THE ZERO-GROWTH STEADY STATE

We assume a zero-growth steady state with $\bar{A}_t = \bar{A}_{t-1} = A$ say and $\bar{G}_t = \bar{G}_{t-1} = G$. $K_t = K_{t-1} = K$, etc. Then the full steady state of the standard RBC model is given by:

$$\begin{aligned} Q &= 1 \\ X &= 1 \\ S &= 0 \\ R &= \frac{1}{\beta} \\ R^K &= R = \rho_t + 1 - \delta \\ \rho_t &= \frac{(1 - \alpha)Y}{K} \\ Y &= (AH)^\alpha K^{1-\alpha} \\ \frac{\varrho C}{(1 - \varrho)(1 - H)} &= W \\ \frac{\alpha Y}{H} &= W \\ \frac{K}{Y} &= \frac{1 - \alpha}{R - 1 + \delta} \\ I &= \delta K \\ Y &= C + I + G \\ G &= T \\ U &= \frac{(C^{(1-\varrho)}(1 - H)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \end{aligned}$$

$$\begin{aligned}
&\rightarrow (1 - \varrho) \log C_t + \varrho \log(1 - H_t) \text{ as } \sigma_c \rightarrow 1 \\
U_C &= (1 - \varrho) C^{(1-\varrho)(1-\sigma_c)-1} ((1 - H)^{\varrho(1-\sigma_c)}) \\
U_H &= -\varrho C^{(1-\varrho)(1-\sigma_c)} (1 - H)^{\varrho(1-\sigma_c)-1}
\end{aligned}$$

Given A and G , the steady state above gives 8 equations in 8 stationary variables R, C, Y, W, H, I, K, T . This describes the zero-growth steady-state equilibrium.

In *recursive form* this steady state can be written

$$\begin{aligned}
R &= \frac{1}{\beta} \\
R^K &= R \\
\rho^K &= R^K - 1 + \delta \\
\frac{K}{Y} &= \frac{1 - \alpha}{\rho_t} = \frac{(1 - \alpha)}{R - 1 + \delta} \\
\frac{I}{Y} &= \frac{\delta K}{Y} = \frac{(1 - \alpha)\delta}{R - 1 + \delta} \\
\frac{C}{Y} &= 1 - \frac{I}{Y} - \frac{G}{Y} = 1 - \frac{I}{Y} - g_y \\
\frac{H\varrho}{(1 - H)(1 - \varrho)} &= \frac{WH}{C} = \frac{WH/Y}{C/Y} = \frac{\alpha}{C/Y} \\
&\Rightarrow H = \frac{\alpha(1 - \varrho)}{\varrho C/Y + \alpha(1 - \varrho)} \\
Y &= (AH)^\alpha K^{1-\alpha} = (AH)^\alpha \left(\frac{K}{Y}\right)^{1-\alpha} (Y)^{1-\alpha} \Rightarrow Y = AH(K/Y)^{\frac{1-\alpha}{\alpha}} \\
G &= g_y Y \\
W &= \alpha \frac{Y}{H} \\
I &= \frac{I}{Y} Y \\
C &= \frac{C}{Y} Y \\
K &= \frac{K}{Y} Y
\end{aligned}$$

3.4 DYNARE EQUILIBRIUM FOR RBC MODEL WITHOUT INVESTMENT ADJUSTMENT COSTS

In the absence of investment adjustment costs we have $S = 0$ and $Q_t = 1$. Then the RBC model becomes the following equilibrium in terms of 14 independent endogenous equations in 14 macroeconomic variables $U_t, U_{C,t}, U_{H,t}, \Lambda_{t-1,t}, R_t, R_t^K, I_t, W_t, Y_t, H_t, r_t^K, C_t, K_t$ and T_t , given parameter values and exogenous processes, A_t and G_t :

$$U_t = \frac{C_t^{(1-\varrho)}(1-H_t)^\varrho^{1-\sigma_c} - 1}{1-\sigma_c} \quad (3.46)$$

$$U_{C,t} = (1-\varrho)C_t^{(1-\varrho)(1-\sigma_c)-1}(1-H_t)^\varrho(1-\sigma_c) \quad (3.47)$$

$$U_{H,t} = -\varrho C_t^{(1-\varrho)(1-\sigma_c)}(1-H_t)^{\varrho(1-\sigma_c)-1} \quad (3.48)$$

$$\Lambda_{t-1,t} \equiv \beta \frac{U_{C,t}}{U_{C,t-1}} \quad (3.49)$$

$$\mathbb{E}_t [\Lambda_{t,t+1} R_t] = 1 \quad (3.50)$$

$$\mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}^K] = 1 \quad (3.51)$$

$$-\frac{U_{H,t}}{U_{C,t}} = W_t \quad (3.52)$$

$$Y_t = (A_t H_t)^\alpha K_{t-1}^{1-\alpha} \quad (3.53)$$

$$\frac{\alpha Y_t}{H_t} = W_t \quad (3.54)$$

$$\frac{(1-\alpha)Y_t}{K_{t-1}} = r_t^K \quad (3.55)$$

$$R_t^K = \frac{(1-\alpha)Y_t}{K_{t-1}} + 1 - \delta \quad (3.56)$$

$$Y_t = C_t + G_t + I_t \quad (3.57)$$

$$K_t = (1-\delta)K_{t-1} + I_t \quad (3.58)$$

$$T_t = G_t \quad (3.59)$$

$$\log A_t - \log A = \rho_A(\log A_{t-1} - \log A) + \epsilon_{A,t}$$

$$\log G_t - \log G = \rho_G(\log G_{t-1} - \log G) + \epsilon_{G,t}$$

3.5 CALIBRATION

The deterministic steady state of the RBC (and NK) models can be used to calibrate a number of parameters. The idea is to assume an observed baseline steady state equilibrium. We then use this observed equilibrium to solve for model parameters consistent with this observation. In general terms, our baseline steady state can be described in terms of a vector $\underline{X} = f(\underline{\theta})$ of outcomes where $\underline{\theta}$ is a vector of parameters. The calibration strategy is to choose a subset $\underline{X}_1$ of n observed outcomes to calibrate a subset $\underline{\theta}_1$ of n parameters. Partition $\underline{X} = [\underline{X}_1, \underline{X}_2]$ and $\underline{\theta} = [\underline{\theta}_1, \underline{\theta}_2]$. Then $\underline{\theta}_1$ is found by solving

$$[\underline{X}_1, \underline{X}_2] = f([\underline{\theta}_1, \underline{\theta}_2]) \quad (3.60)$$

for $\underline{X}_2$ and $\underline{\theta}_1$, given $\underline{X}_1$ and $\underline{\theta}_2$. If such a solution exists for economically meaningful parameter values for $\underline{\theta}_1$ then a successful calibration has been achieved.

To illustrate this, suppose we have data for factor shares in the production sector, hours as a proportion of the available leisure time (H), the real interest rate (R) and expenditure shares $c_y \equiv \frac{C}{Y}$, $i_y \equiv \frac{I}{Y}$ and $g_y \equiv \frac{G}{Y}$. First we calibrate α to be the observed wage share in the wholesale sector.

We can choose units of output and capital stock so that $A = 1$. Then using $K/Y = \frac{1-\alpha}{R-1+\delta}$ from the RBC steady state we can now write

$$i_y \equiv \frac{I}{Y} = \frac{\delta K}{Y} = \frac{\delta K}{Y} = \frac{\delta(1-\alpha)}{R-1+\delta}$$

from which δ can be calibrated.

From the steady state equation $\frac{qC}{(1-q)(1-H)} = W$ we have seen that

$$H = \frac{\alpha(1-q)}{qC/Y + \alpha(1-q)}$$

from which the calibrated q is obtained as:

$$q = \frac{(1-H)\alpha}{(1-H)\alpha + c_y H} \quad (3.61)$$

Finally from an observation of R we can calibrate β from

$$R = \frac{1}{\beta}$$

Observed Equilibrium	Value
H	0.35
wage share = α	0.7
c_y	0.6
i_y	0.2
g_y	0.2
R	1.01
Calibrated Parameters	Value
ϱ	0.6842
δ	0.0202
β	0.990

Table 3.1: Calibration of RBC Model

Some remaining parameters need to be set ϕ_X needed with investment adjustment costs, the persistence parameters, ρ_A and ρ_G , and the standard deviations of the shocks. These are later estimated by Bayesian methods, but for the purposes of the simulations on the Course we choose values $\sigma_{es} = \phi_X = 2.0$ roughly reflecting the empirical literature. For all exogenous processes in our models, persistence parameters are set at 0.75, and shocks have a standard deviation of 1%. This completes the calibration – typical US observations and calibrated parameters are illustrated in Table 3.1.

3.6 MATCHING SECOND MOMENTS

The most basic RBC model has only private consumption ($G_t = 0$). This is the model studied in Dejong and Dave (2007), Chapter 6, Section 6.4. Estimating labour's share as $\alpha = 0.77$ and $\delta = 0.02$ they estimate an AR1 process for total factor productivity (not labour productivity as in our previous setup) giving $\rho_A = 0.78$ and $\text{sd}(\epsilon_A) = 0.0067$. The moments for HP-filtered data we computed on day one for the full sample (computed using the code *moments_matching.m* and the corresponding ones from our model with these features (*RBC_nogov.mod*) are shown in Table 3.2. Not surprising perhaps this very simple model performs rather well in reproducing the moments for output but it does not in the case of consumption, investment and hours.

HP Filtered Data				
j	σ_j	$\frac{\sigma_j}{\sigma_Y}$	$\varphi(1)$	$\varphi(j, Y)$
Y	1.7374	1.00	0.8478	1.0
C	1.1940	0.6872	0.7717	0.7885
I	4.9960	2.8755	0.8717	0.7791
H	2.8992	1.6687	0.9676	0.5135

RBC Model				
j	σ_j	$\frac{\sigma_j}{\sigma_Y}$	$\varphi(1)$	$\varphi(j, Y)$
Y	1.4432	1.00	0.7927	1.00
C	0.6409	0.4441	0.9454	0.8068
I	6.9042	4.7840	0.7441	0.9527
H	0.7047	0.4883	0.7385	0.9256

Table 3.2: Moment Comparisons

Notes: σ_Y denotes the standard deviation of $\frac{Y_t}{Y}$ etc; $\varphi(1)$ denotes first-order autocorrelation; $\varphi(j, Y)$ denotes cross-correlation with output;

3.7 THE SOCIAL PLANNER'S PROBLEM

The RBC model up to now is for a decentralized market economy. By contrast, the social (central planner's) problem² at time t to choose paths for consumption $\{C_t\}$, leisure $\{L_t\} = \{1 - H_t\}$ and capital stock $\{K_t\}$ to maximize the value function V_t given by

$$V_t = V_t(K_{t-1}) = E_t \left[\sum_{s=0}^{\infty} \beta^s U(C_{t+s}, L_{t+s}) \right] \quad (3.62)$$

given initial capital stock at the beginning of period t , K_{t-1} , and given the resource constraint

$$F(K_t, H_t) = C_t + G_t + \underbrace{\Delta K_t + \delta K_{t-1}}_{I_t} = C_t + \underbrace{K_t - (1 - \delta)K_{t-1}}_{I_t} \quad (3.63)$$

Exercise

Use the Lagrangian method of Appendix A to show that the allocation of $\{C_t\}$,

² In fact Kydland and Prescott (1982) considered this problem and not the decentralized market case.

$\{K_t\}, \{L_t\}, \{H_t\}$ over time is identical in the centralized and decentralized economies.

Another way of stating this equivalence is to say that the decentralized market equilibrium is *efficient*. This is known as the First Theorem of Welfare Economics (see, for example, Varian chapter 28) and formalizes the “invisible hand of the market” intuition advanced by Adam Smith in 1776 in “An Inquiry into the Nature and Causes of the Wealth of Nations”. It applies to a model economy such as that in this topic where all markets are functioning perfectly. If we introduce any form of *market failure* such as oligopolistic firms setting prices in the output market (as in the New Keynesian model), trade unions setting wages in the labour market, pollution externalities and information asymmetries in financial markets, then the First Theorem no longer holds. The simple RBC model is a starting point for studying real world economies, but no more.

3.8 JR PREFERENCES

The RBC model up to now with a Cobb-Douglas utility function displays a strong *wealth effect* in response to a positive technology shock. As a result household reduce their hours relative to the steady state and “consume” more leisure. Hours and output then do not co-move, a feature we do see in the data.

An alternative functional form for utility found in the literature from Jaimovich and Rebello (2008) controls the wealth effect. It takes the form:

$$U_t = \frac{(C_t - \kappa H_t^\theta \Xi_t)^{1-\sigma_{es}} - 1}{1 - \sigma_{es}} \quad (3.64)$$

$$\Xi_t = C_t^\gamma \Xi_{t-1}^{1-\gamma}; \quad \gamma \in [0, 1] \quad (3.65)$$

The parameter κ can be set to target $\bar{H}$ (as we did using ϱ with the Cobb-Douglas function previously). The parameter θ can be set to target the elasticity of labour supply with respect to the real wage.³ This leaves γ to control for wealth effects. With $\gamma = 1$ we have a utility function of the form

$$U_t = \frac{(C_t(1 - \kappa H_t^\theta))^{1-\sigma_{es}} - 1}{1 - \sigma_{es}} \quad (3.66)$$

³ See Bilbiie (2009) and Bilbiie (2011) for details.

How does (3.66) compare with the Cobb-Douglas utility function (3.22)? Writing the latter as

$$\begin{aligned} U_t &= \frac{(C_t^{(1-\varrho)}(1-H_t)^\varrho)^{1-\sigma_c} - 1}{1-\sigma_c} \\ &= \frac{(C_t(1-H_t)^{\varrho/(1-\varrho)})^{(1-\varrho)(1-\sigma_c)} - 1}{1-\sigma_c} \end{aligned} \quad (3.67)$$

this is of the form

$$U_t = \frac{(C_t(1-H_t)^\theta)^{1-\sigma_{es}} - 1}{1-\sigma_{es}} \quad (3.68)$$

where

$$\theta = \varrho/(1-\varrho) \quad (3.69)$$

$$\sigma_{es} = 1 - (1-\varrho)(1-\sigma_c) \quad (3.70)$$

The CD utility function is less flexible in that it can only target one steady state outcome $H = \bar{H}$ whereas the JR utility function can target labour supply elasticity and (as we shall see) wealth effects. Furthermore empirical evidence is obtained from utility function of the form (3.68) for the elasticity σ_{es} . From (3.69), we should therefore calibrate σ_c from

$$\sigma_c = \frac{\sigma_{es} - \varrho}{1 - \varrho} \quad (3.71)$$

From Appendix 1.1, the first-order conditions for the household now become:

$$\text{Euler Consumption : } 1 = R_t \mathbb{E}_t [\Lambda_{t,t+1}] \quad (3.72)$$

$$\text{Stochastic Discount Factor : } \Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t} \quad (3.73)$$

$$\text{where : } \lambda_t = U_{C,t} - \gamma \mu_t \frac{\Xi_t}{C_t} \quad (3.74)$$

$$\text{and : } \mu_t = -U_{\Xi,t} + \beta(1-\gamma) \mathbb{E}_t \frac{\mu_{t+1} \Xi_{t+1}}{\Xi_t} \quad (3.75)$$

$$\text{Labour Supply : } \frac{U_{H,t}}{\lambda_t} = -W_t \quad (3.76)$$

$$\begin{aligned} \text{Investment FOC : } & Q_t(1 - S(X_t) - X_t S'(X_t)) \\ & + \mathbb{E}_t \left[\Lambda_{t,t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2 \right] = 1 \end{aligned} \quad (3.77)$$

$$\text{Capital Supply : } \mathbb{E}_t \left[\Lambda_{t,t+1} R_{t+1}^K \right] = 1 \quad (3.78)$$

where R_t^K is the gross return on capital given by

$$R_t^K = \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}} \quad (3.79)$$

3.9 DYNARE AND MATLAB FILES

The codes can be found in the folder **RBC** for the CD utility function and in **RBC_JR** for JR preferences. A general graph plotter for comparing irfs from any number of different simulations of the RBC model is the matlab file **graphs_irfs_compare_RBC**.

3.10 DYNARE EXERCISE

1. Introduce *external habit* into the utility function by changing the single-period utility and marginal utilities of consumption and labour supply to:

$$\begin{aligned} U_t &= U(C_t, L_t) = \frac{((C_t - \chi C_{t-1})^{(1-\varrho)}(1 - H_t)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\ U_{C,t} &= (1 - \varrho)(C_t - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)-1}(1 - H_t)^{\varrho(1-\sigma_c)} \\ U_{H,t} &= -\varrho(C_t - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)}(1 - H_t)^{\varrho(1-\sigma_c)-1} \end{aligned}$$

where χ is a new habit parameter. Note that in evaluating the marginal utility of consumption, $U_{C,t}$, the household takes external habit as given.⁴

- a) Generalize the steady state relationship corresponding to the household foc (3.8) to the case of habit in consumption.
- b) Use **graphs_irfs_compare_RBC**. with appropriate changes to the file names of results to compare your RBC model with high, low and no habit.

⁴ By contrast, *internal habit* is taken into account in a dynamic optimization problem. External habit in fact is a negative externality which in equilibrium leads to over-consumption and too little leisure from a welfare point of view.

- c) Discuss your results.
2. Use the set-up with JR preferences and **graphs_irfs_compare_RBC** to reproduce the graphs 3.1 and 3.2 and discuss these irfs.

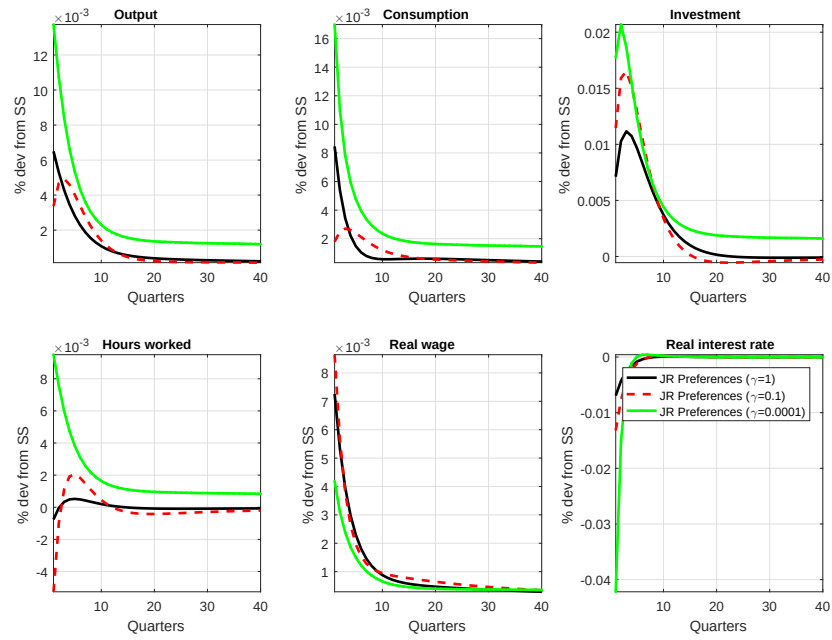


Figure 3.1: RBC Model with JR Preferences: IRFs to Technology Shock

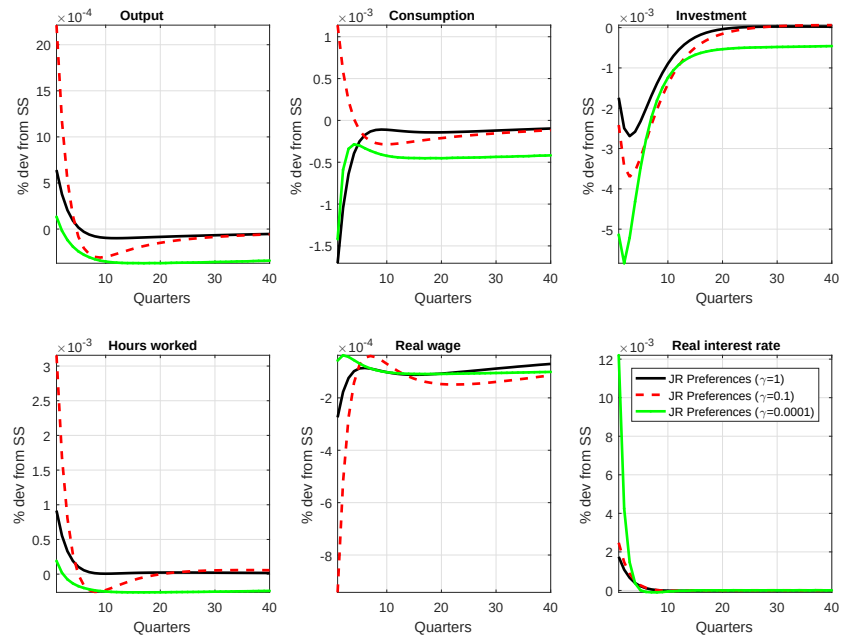


Figure 3.2: RBC Model with JR Preferences: IRFs to Government Spending Shock

4

CALIBRATION AND USE OF THE EXTERNAL STEADY STATE

We have carefully set out the steady state of the RBC (and NK) models because it is needed to calibrate a number of parameters. We have covered calibration in the previous section. We now explain how to code and use an external steady state file to accompany the .mod file.

4.1 THE EXTERNAL STEADY STATE

The steady state for the RBC and NK models is solved using matlab programs *MODFILENAME_steadystate.m* which can call *fun_RBC.m*. In order for Dynare to understand that it must look for the external file solving for the steady state the m-file should have exactly the same name as the mod file followed by *_steadystate.m*.

The file *MODFILENAME_steadystate.m* uses the matrix M_* in which Dynare stores the value of the parameters used in the mod file and uses the Matlab function **fsolve** to find the root of the equation in *fun_RBC.m*. The file *MODFILENAME_steadystate.m* then calculates the steady state values of all the other endogenous variables in the model and stores them in the array *ys* that is the name which Dynare uses for the vector of the steady state values of the endogenous variables in the mod file.

The codes that use this approach can be found in the folder **RBC_external_ss**. **RBC_1.mod** solves the steady state with an external steady-state matlab file simply repeating the analytical steady state used before in the set-up without an external steady state.

$$\begin{aligned} R &= \frac{1}{\beta} \\ \frac{K}{Y} &= \frac{(1 - \alpha)}{R - 1 + \delta} \\ Y &= (A)(K/Y)^{\frac{1-\alpha}{\alpha}} \\ K &= \frac{K}{Y}Y \end{aligned}$$

$$\begin{aligned}
I &= \delta K \\
G &= g_y Y \\
W &= \alpha \frac{Y}{H} \\
C &= W \frac{(1 - \varrho)(1 - H)}{\varrho}
\end{aligned}$$

leaving

$$Y = C + I + G$$

as the equation to solve for H in `fsolve` calling a matlab function **fun_RBC.m**.

In general one needs to use this approach for medium-sized or big models as we demonstrate with various models.¹ A further advantage of having the `_steadystate.m` file is that it can be used in a slightly modified form to carry out comparative statics. This is demonstrated in the matlab file **RBC_1_Compar_Statics.m**.

4.2 DYNARE CODES

In the subfolder **RBC_external_ss** in the RBC folder you can find other modifications of the main RBC code used for various calibrations:²

- **RBC_2.mod** : calibrates hours worked (*hobs*) and treats ϱ as a constant endogenous variable. The value of ϱ consistent with *hobs* is then found using `fsolve` which calls **fun_RBC_2.m**.
- **RBC_3.mod** : calibrates hours worked and investment as a share of GDP (*hobs*, *iobs*) and treats ϱ and δ as constant endogenous variables. The values of ϱ and δ consistent with *hobs* and *iobs* are then found using `fsolve` calls **fun_RBC_3.m**.
- **RBC_4.mod** : calibrates hours worked, the investment share and the interest rate (*hobs*, *iobs*, *Rss*) and treats ϱ , δ and β as constant endogenous variables. The values of ϱ , δ and β consistent with *hobs*, *iobs* and *Rss* are then found using `fsolve` calls **fun_RBC_4.m**.

¹ Although Dynare can be solved without setting up the steady state by guessing initial values under `initval`, this approach is not robust and we do not advise it unless the model is linearized, in which case the steady state values are simply zero.

² Note that Dynare 4.6 treats the steady state differently from earlier versions. The Course codes are only compatible with 4.6.

Note that these examples in dynare are chosen only to *illustrate* the use of an external steady state. As we have seen above, the exercise can be carried out analytically, but in more developed models such as a NK one with financial frictions an analytical solution is not possible. Then it is *essential* to use the external steady state and fsolve facilities.

4.3 EXERCISE

Rework the examples above using the RBC model with external habit from the exercise in Section 3.

5

HOW TO LINEARIZE DYNAMIC MODELS

To solve the model it is usual to linearize about the steady state. In fact this is the easiest way to set up a model in Dynare for estimation, simulation and policy analysis. It is also useful as the linearized form can be used for some partial analysis providing insights into the workings of the model. That said, in Dynare it is not necessary to go through this step which becomes more arduous as models become larger.

5.1 LINEARIZATION TECHNIQUE

The technique of linearization uses a Taylor series expansion. Consider a general function of two variables $F(X_t, Y_t)$. Then up to first-order terms we have

$$F(X_t, Y_t) \approx F(X, Y) + \frac{\partial F}{\partial X_t}(X_t - X) + \frac{\partial F}{\partial Y_t}(Y_t - Y) \quad (5.1)$$

where partial derivatives are evaluated at the steady-state values X, Y . Now use the following notation

$$\frac{X_t - X}{X} = x_t \quad (5.2)$$

Then (5.1) becomes

$$f \approx \frac{X}{F} \frac{\partial F}{\partial X_t} x_t + \frac{Y}{F} \frac{\partial F}{\partial Y_t} y_t \quad (5.3)$$

Note that

$$\log \left[\frac{X_t}{X} \right] = \log \left[1 + \frac{X_t - X}{X} \right] \approx x_t \quad (5.4)$$

which is why this technique is referred to as ‘log-linearization’.

Define lower case variables $x_t = \log \frac{X_t}{\bar{X}_t}$ if X_t has a long-run trend, $\bar{X}_t$, or $x_t = \log \frac{X_t}{X}$ where X is the steady state value of a non-trended variable. For example, $r_t \equiv \log \left(\frac{R_t}{R} \right)$ and $y_t \equiv \log \left(\frac{Y_t}{Y} \right)$ or $y_t \equiv \log \left(\frac{Y_t}{Y} \right)$. Appendix 3 illustrates this method for the RBC Euler equation.

5.2 EXAMPLE AND WHY LINEARIZATION IS USEFUL

Full paper and pen stability analysis possible for linear models and optimal policy for linear-quadratic problems (see Woodford (2003) and Galí (2015)). Consider the **Euler Equation**. Useful analysis can be done in the case of a logarithmic utility function ($\sigma_c = 1$). In this case $u_{C,t} \approx c_t$ and the the linear Euler equation becomes:

$$\mathbb{E}_t[u_{C,t+1}] = u_{C,t} - r_t \Rightarrow c_t = -r_t + \mathbb{E}_t[c_{t+1}]$$

Solving this forward in time gives

$$c_t = -r_t - \mathbb{E}_t[-r_{t+1} + \mathbb{E}_{t+1}[c_{t+2}]] = -r_t - \mathbb{E}_t[r_{t+1}] + \mathbb{E}_t[\mathbb{E}_{t+1}[c_{t+2}]]$$

Then using $\mathbb{E}_t[\mathbb{E}_{t+1}[c_{t+2}]] = \mathbb{E}_t[c_{t+2}]$ (the “law of iterated expectations”) and reiterating we arrive at:

$$c_t = -r_t - \mathbb{E}_t[r_{t+1}] - \mathbb{E}_t[r_{t+2}] - \dots = -\sum_{i=0}^{\infty} \mathbb{E}_t[r_{t+i}]$$

Thus consumption depends negatively on expectations of all future real interest rates.

Another example is the household labour supply choice> Gather the linearization of the labour supply choice and the marginal utility of leisure together:

$$\begin{aligned} w_t &= u_{L,t} - u_{C,t} \\ u_{L,t} &= u_{C,t} + \left(\frac{c_t - \chi c_{t-1}}{1 - \chi} \right) + \frac{H}{1 - H} h_t \end{aligned}$$

It follow that $w_t = \left(\frac{c_t - \chi c_{t-1}}{1 - \chi} \right) + \frac{H}{1 - H} h_t$

We now have an expression for the inverse elasticity of hours with respect to the wage keeping consumption fixed at its steady state. Then $c_t = c_{t-1}$ and the constant consumption inverse elasticity (the **Frisch elasticity**) is then $\frac{H}{1-H} = 1/2$ with $H = 1/3$. From empirical studies this is on the low side (estimates suggest an elasticity > 1). But JR rather than CD preferences used for the RBC model will enable us to target the Frisch elasticity

5.3 LINEARIZATION OF RBC MODEL

Using this method and substituting for the steady-state you should be able to show for yourself that the linearized form of the full RBC model without

investment costs about the zero-growth steady state then takes the state-space form

$$\begin{aligned}a_t &= \rho_A a_{t-1} + \varepsilon_{A,t} \\g_t &= \rho_G g_{t-1} + \varepsilon_{G,t} \\k_t &= (1 - \delta)k_{t-1} + \delta i_t \\ \mathbb{E}_t[u_{C,t+1}] &= u_{C,t} - r_t\end{aligned}$$

with further outputs defined in terms of the dynamic state variables by

$$\begin{aligned}u_{C,t} &= -(1 + (\sigma_c - 1)(1 - \varrho))c_t + (\sigma_c - 1)\varrho \frac{H}{1 - H}h_t \\u_{L,t} &= u_{C,t} + c_t + \frac{H}{1 - H}h_t \\w_t &= u_{L,t} - u_{C,t} \\y_t &= \alpha(a_t + h_t) + (1 - \alpha)k_{t-1} \\y_t &= c_y c_t + i_y i_t + g_y g_t \\g_t &= t_t \\ \mathbb{E}_t[y_{t+1}] - k_t &= \frac{R}{R - 1 + \delta}r_t \\w_t &= y_t - h_t\end{aligned} \tag{5.5}$$

With investment costs we replace (5.5) with

$$\begin{aligned}r_t &= E_t[x_{t+1}] - q_t \\Rx_t &\equiv (R - 1 + \delta)(y_t - k_{t-1}) + (1 - \delta)q_t \\ \left(1 + \frac{1}{R}\right)i_t &= \frac{1}{R}E_t i_{t+1} + i_{t-1} + \frac{1}{S''(1)}q_t\end{aligned}$$

Note as investment costs go to zero, $S'' \rightarrow 0$, $q_t \rightarrow 0$ and this reduces to (5.5) again.

5.4 DYNARE AND MATLAB FILES

Dynare and Matlab files to run the linearized simple RBC model (with no investment adjustment costs and no habit) and compare it with the non-linear set-up are in the sub-folder **linear** of the folder **RBC**. The graph-plotter is **graphs_irfs_compare_RBC.m**

6 | THE NEW KEYNESIAN MODEL

6.1 INTRODUCTION

A crucial distinction of the Keynesian perspective, though, is the view that real economies are not perfectly flexible nor perfectly competitive, thus providing a powerful cause for (inefficient) business cycle fluctuations and the non-neutrality of money. This, consequently, implies that economic policy has a significant role in minimizing welfare distortions caused by these imperfections and rigidities. Thus, much of the research from the mid 1980s onwards was devoted to developing mechanisms that provide a rationale for price stickiness. These include models of imperfect competition, imperfect information and nominal price rigidities, which eventually made their way into current NK DSGE models.¹ In fact, incorporating price stickiness in a rigorous way requires that firms are price setters, as opposed to price takers in a perfectly competitive world. This can be achieved in a tractable way by using the monopolistic competition approach of Dixit and Stiglitz (1977), in which firms produce (and set prices for) differentiated goods.

Nominal price inertia then requires some form of staggered pricing, so that only a proportion of firms adjusts prices at a given time. While different formulations are possible, the Calvo (1983b) formulation has become standard, as it greatly simplifies aggregation.² An extension with similar persistence mechanisms to wage setting was suggested by Erceg *et al.* (2000), which, while playing an important role in explaining output and inflation dynamics, has also important implications for optimal monetary policy, as now a welfare-maximizing central bank should be concerned with both price and wage stability.

In the quest for increased theoretical and empirical adequacy, additional frictions have been suggested in order to further capture the observed inertial behaviour of inflation and other aggregate variables. Carlstrom and Fuerst (1997) and Bernanke *et al.* (1999), for example, propose the introduction of financial frictions, while Christiano *et al.* (2005b) stress the role played by habit

¹ See Mankiw and Romer (1991), which contains a collection of the most influential articles in the early New Keynesian tradition.

² See Rotemberg (1982) for an alternative and equally convenient specification of price stickiness.

persistence in consumption preferences, adjustment costs in investment and variable capital utilization.

6.2 THE NON-LINEAR NK MODEL WITH STICKY PRICES AND FLEXIBLE WAGES

We proceed to a series of NK models leading to that of Smets and Wouters (2007) in stages. First we study a model with sticky-prices but flexible wages. This will be the model we estimate in the Estimation Chapter. Then in the next section we make the latter sticky as well and add two further features from Smets and Wouters (2007), namely capacity utilization and a fixed cost of converting the wholesale into a retail good.

6.2.1 RetailSector

The retail sector uses a homogeneous wholesale good to produce a basket of differentiated goods for consumption

$$C_t = \left(\int_0^1 C_t(m)^{(\zeta-1)/\zeta} dm \right)^{\zeta/(\zeta-1)} \quad (6.1)$$

where ζ is the elasticity of substitution. For each m , the consumer chooses $C_t(m)$ at a price $P_t(m)$ to maximize (6.1) given total expenditure $\int_0^1 P_t(m) C_t(m) dm$. This results in a set of consumption demand equations for each differentiated good m with price $P_t(m)$ of the form

$$C_t(m) = \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} C_t \quad (6.2)$$

where $P_t = \left[\int_0^1 P_t(m)^{1-\zeta} dm \right]^{\frac{1}{1-\zeta}}$. P_t is the aggregate price index. Note that C_t and P_t are Dixit-Stiglitz aggregators – see Dixit and Stiglitz (1977). Demand for investment and government services takes the same form (see Appendix 5) so in aggregate

$$Y_t(m) = \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} Y_t \quad (6.3)$$

where $Y_t(m)$ is the quantities of output needed in the wholesale sector to produce good m in the retail sector. Integrating over m we then have

$$\int_0^1 Y_t(m) dm = Y_t^W = \left(\int_0^1 \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} dm \right) Y_t = \Delta_t Y_t \quad (6.4)$$

where $\Delta_t \equiv \int_0^1 \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} dm$ is *price dispersion*.

Following Calvo (1983a), we now assume that there is a probability of $1 - \zeta$ at each period that the price of each retail good m is set optimally to $P_t^O(m)$. If the price is not re-optimized, then it is held fixed.³ For each retail producer m , given its real marginal cost $MC_t = \frac{P_t^W}{P_t}$, the objective is at time t to choose $\{P_t^O(m)\}$ to maximize discounted real profits

$$\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} Y_{t+k}(m) \left[P_t^O(m) - P_{t+k} MC_{t+k} \right] \quad (6.5)$$

subject to

$$Y_{t+k}(m) = \left(\frac{P_t^O(m)}{P_{t+k}} \right)^{-\zeta} Y_{t+k} \quad (6.6)$$

where $\Lambda_{t,t+k} \equiv \beta^k \frac{U_{C,t+k}}{U_{C,t}}$ is the stochastic discount factor over the interval $[t, t+k]$. The solution to this is

$$\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} Y_{t+k}(m) \left[P_t^O(m) - \frac{1}{(1 - 1/\zeta)} P_{t+k} MC_{t+k} \right] = 0 \quad (6.7)$$

Using (6.6) and rearranging this leads to

$$P_t^O = \frac{1}{(1 - 1/\zeta)} \frac{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} (P_{t+k})^{\zeta} Y_{t+k} MC_{t+k}}{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} (P_{t+k})^{\zeta} Y_{t+k}} \quad (6.8)$$

where the m index is dropped as all firms face the same marginal cost so the right-hand side of the equation is independent of firm size or price history.

By the law of large numbers the evolution of the price index is given by

$$P_t^{1-\zeta} = \zeta P_{t-1}^{1-\zeta} + (1 - \zeta) (P_t^O)^{1-\zeta} \quad (6.9)$$

³ Thus we can interpret $\frac{1}{1-\zeta}$ as the average duration for which prices are left unchanged.

Now define k periods ahead inflation as

$$\Pi_{t,t+k} \equiv \frac{P_{t+k}}{P_t}$$

To ease the notation in what follows we denote $\Pi_t = \Pi_{t-1,t}$ and $\Pi_{t+1} = \Pi_{t,t+1}$.

We can now write the fraction (6.8)

$$\frac{P_t^O}{P_t} = \frac{1}{(1 - 1/\zeta)} \frac{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \Lambda_{t,t+k} (\Pi_{t,t+k})^{\zeta} Y_{t+k} MC_{t+k}}{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta^k \Lambda_{t,t+k} (\Pi_{t,t+k})^{\zeta-1} Y_{t+k}} \quad (6.10)$$

and (6.9) as

$$1 = \zeta (\Pi_t)^{\zeta-1} + (1 - \zeta) \left(\frac{P_t^O}{P_t} \right)^{1-\zeta} \quad (6.11)$$

6.2.2 Price Dynamics

In order to set up the model in non-linear form as a set of difference equations, required for software packages such as Dynare, we need to represent the price dynamics as difference equations. Both numerator and denominator the first-order condition for pricing, (6.10), are of the form considered in Appendix 5. Using the Lemma, price dynamics are given by

$$\frac{P_t^O}{P_t} = \frac{J_t}{JJ_t} \quad (6.12)$$

$$JJ_t - \zeta \mathbb{E}_t [\Lambda_{t,t+1} \Pi_{t+1}^{\zeta-1} JJ_{t+1}] = Y_t \quad (6.13)$$

$$J_t - \zeta \mathbb{E}_t [\Lambda_{t,t+1} \Pi_{t+1}^{\zeta} J_{t+1}] = \left(\frac{1}{1 - \frac{1}{\zeta}} \right) Y_t MC_t MS_t \quad (6.14)$$

$$1 = \zeta \Pi_t^{\zeta-1} + (1 - \zeta) \left(\frac{J_t}{JJ_t} \right)^{1-\zeta} \quad (6.15)$$

$$MC_t = \frac{P_t^W}{P_t} = \frac{W_t}{F_{H,t}} \quad (6.16)$$

where (6.36) uses (3.15). We have introduced a mark-up shock MS_t . Notice that the real marginal cost, MC_t , is no longer fixed as it was in the RBC model.

6.2.3 Indexing

Prices now are indexed to last period's aggregate inflation, with a price indexation parameter γ_p . Then the price trajectory with no re-optimization is given by $P_t^O(j)$, $P_t^O(j) \left(\frac{P_t}{P_{t-1}}\right)^{\gamma_p}$, $P_t^O(j) \left(\frac{P_{t+1}}{P_{t-1}}\right)^{\gamma_p}$, $\dots$ where $Y_{t+k}(m)$ is given by (6.3) with indexing so that

$$Y_{t+k}(m) = \left(\frac{P_t^O(m)}{P_{t+k}} \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma_p} \right)^{-\zeta} Y_{t+k} \quad (6.17)$$

With indexing by an amount $\gamma_p \in [0, 1]$ and an exogenous mark-up shock MS_t as before, the optimal price-setting first-order condition for a firm j setting a new optimized price $P_t^O(j)$ is now given by

$$P_t^O = \frac{\frac{1}{(1-\zeta/\gamma_p)} \mathbb{E}_t \left[\sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} MC_{t+k} MS_{p,t+k} Y_{t+k} \right]}{\mathbb{E}_t \left[\sum_{k=0}^{\infty} \zeta^k \frac{\Lambda_{t,t+k}}{P_{t+k}} Y_{t+k}(j) \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma_p} \right]}$$

Price dynamics are now given by

$$\begin{aligned} \frac{P_t^O}{P_t} &= \frac{J_t}{JJ_t} \\ JJ_t - \zeta \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^{\zeta-1} JJ_{t+1}] &= Y_t \\ J_t - \zeta \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^{\zeta} J_{t+1}] &= \frac{1}{1 - \frac{1}{\zeta}} MC_t MS_{p,t} Y_t \\ \tilde{\Pi}_t &\equiv \frac{\Pi_t}{\Pi_{t-1}^{\gamma_p}} \\ 1 &= \zeta \tilde{\Pi}_t^{\zeta-1} + (1 - \zeta) \left(\frac{J_t}{JJ_t} \right)^{1-\zeta} \end{aligned}$$

An alternative model of indexing assumes that prices are indexed to a weighted average of last period and trend (steady state) inflation. If we denote the two weights by γ_p and $\bar{\gamma}_p$ then the previous dynamics replaces $\tilde{\Pi}_t$ above with

$$\tilde{\Pi}_t \equiv \frac{\Pi_t}{\Pi_{t-1}^{\gamma_p} \Pi^{1-\bar{\gamma}_p}} \quad (6.18)$$

In Smets and Wouters (2007) it is assumed that $\bar{\gamma}_p = \gamma_p$ so that $\tilde{\Pi} = 1$ in the steady state which eliminates the effect of state-state inflation in the equilibrium. In the coding of our model we allow for options $\bar{\gamma}_p = \gamma_p$ and $\bar{\gamma}_p = 0$.

6.2.4 Remaining Features of the Non-Linear NK Model

With a nominal side of the model we need to distinguish between the *ex ante* nominal gross interest rate $R_{n,t}$ set at time t and the *ex post* real interest rate, R_t . These are related by the Fischer equation

$$R_t = \frac{R_{n,t-1}}{\Pi_t} \quad (6.19)$$

The stochastic Euler equation must now take the form

$$U_{C,t} = \beta \mathbb{E}_t \left[\frac{R_{n,t}}{\Pi_{t+1}} U_{C,t+1} \right] = \beta \mathbb{E}_t [R_{t+1} U_{C,t+1}] = 1 \quad (6.20)$$

which can be written as

$$\mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}] = 1 \quad (6.21)$$

Note that whereas nominal bonds are riskless, the real return is not.

Demand for capital by the wholesale firm owned by households is now given by

$$1 = \mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}] = \frac{\mathbb{E}_t \left[\Lambda_{t,t+1} \left[(1 - \alpha) \frac{P_{t+1}^W}{P_{t+1}} \frac{Y_{t+1}^W}{K_t} + (1 - \delta) Q_{t+1} \right] \right]}{Q_t} = \mathbb{E}_t [\Lambda_{t,t+1} R_{K,t+1}]$$

The final change is that price dispersion Δ_t^p reduces output which, as shown in Appendix 5, is now given by

$$Y_t = \frac{(A_t H_t)^\alpha K_{t-1}^{1-\alpha}}{\Delta_t^p} \quad (6.22)$$

$$\Delta_t^p \equiv \frac{1}{n} \sum_{j=1}^n (P_t(j)/P_t)^{-\zeta} = \zeta \Pi_t^\zeta \Delta_{t-1}^p + (1 - \zeta) \left(\frac{J_t^p}{J J_t^p} \right)^{-\zeta} \quad (6.23)$$

The nominal interest rate is given by the following Taylor-type rules

$$\begin{aligned} \log \left(\frac{R_{n,t}}{R_n} \right) &= \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right. \\ &\quad \left. + \theta_y \log \left(\frac{Y_t}{Y} \right) + \theta_{dy} \log \left(\frac{Y_t}{Y_{t-1}} \right) \right] + \epsilon_{M,t} \end{aligned} \quad (6.24)$$

or

$$\log \left(\frac{R_{n,t}}{R_n} \right) = \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right]$$

$$+ \theta_y \log \left(\frac{Y_t}{Y_t^F} \right) + \theta_{dy} \log \left(\frac{Y_t/Y_t^F}{Y_{t-1}/Y_{t-1}^F} \right) \Big] + \epsilon_{M,t} \quad (6.25)$$

where Y_t^F is the flexi-price level of output and $\epsilon_{M,t}$ is a monetary policy shock process. (6.85) is an ‘implementable’ form of the Taylor rule which stabilizes output about its steady state.⁴ Then θ_π and θ_y are the long-run elasticities of the inflation and output respectively with respect to the interest rate. The “Taylor principle” requires $\theta_\pi > 1$. The conventional Taylor rule, (6.85), stabilizes output about its flexi-price level which is that found by solving the RBC core of this model or simply allowing the contract parameter ξ to tend to zero. Unlike the implementable form, this requires observations of the output gap $\frac{Y_t}{Y_t^F}$ to implement and monitor.⁵

Finally in a zero-growth steady state there are three exogenous AR1 shock processes to technology, government spending, the mark-up shock and an i.i.d monetary policy shock:

$$\begin{aligned} \log A_t - \log A &= \rho_A (\log A_{t-1} - \log A) + \epsilon_{A,t} \\ \log G_t - \log G &= \rho_G (\log G_{t-1} - \log G) + \epsilon_{G,t} \\ \log MS_t - \log MS &= \rho_{MS} (\log MS_{t-1} - \log MS) + \epsilon_{MS,t} \end{aligned}$$

Clearly other shocks are possible such as to investment and preferences and capital quality.

To summarize, the NK model consists of the RBC model of real variables with a time-varying real marginal cost (6.36) and price dynamics given by (E.4)–(E.4). The expected ex post real interest rate $\mathbb{E}_t[R_{t+1}]$ over the interval $[t, t+1]$ replaces R_t in the RBC model and price dispersion given by (6.23) reduces output as in (6.36). The interest rate $R_{n,t}$ is the exogenous instrument typically given by a Taylor-type rule. By proceeding from the RBC to the NK model in this fashion we see how price stickiness introduces an inefficiency through the price dispersion mechanism. Figure 6.1 illustrates the NK model.

⁴ In our Dynare setups for policy analysis it is coded as

$$\log \left(\frac{R_{n,t}}{R_n} \right) = \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + \alpha_\pi \log \left(\frac{\Pi_t}{\Pi} \right) + \alpha_y \log \left(\frac{Y_t}{Y} \right) + \epsilon_{M,t}$$

⁵ Technically this should pose no problems in a perfect information rational expectations equilibrium, but the rationale for ‘simple rules’ is to have policies that are easy to observe without relying on the perfect information solution.

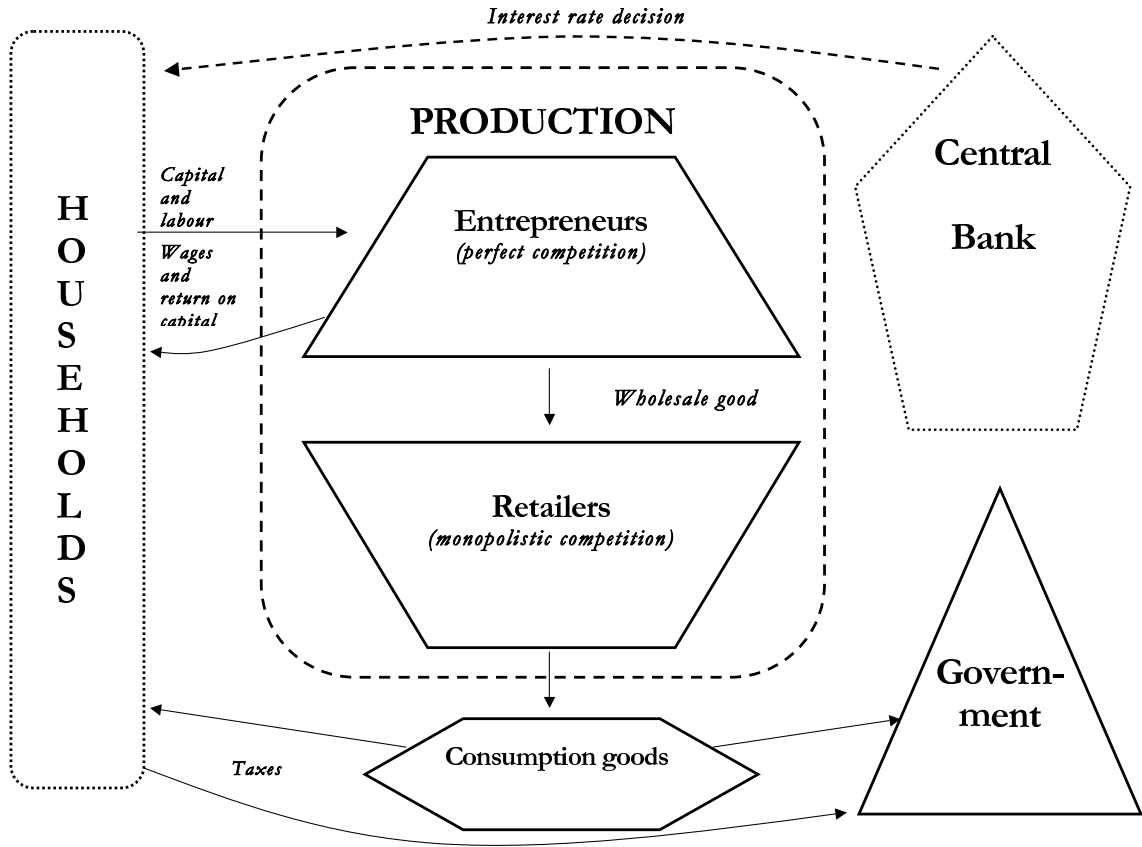


Figure 6.1: NK Model

6.2.5 Summary of Model 3

Household

$$\text{Non-separable CRRA} : U_t = \frac{(C_t^{(1-\varrho)}(1-H_t)^\varrho)^{1-\sigma_c} - 1}{1-\sigma_c}$$

$$\text{Euler Consumption} : 1 = R_t R S_t \mathbb{E}_t [\Lambda_{t,t+1}]$$

$$\text{Stochastic Discount Factor} : \Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t}$$

$$\text{where : } \lambda_t = U_{C,t}$$

$$\text{Labour Supply} : \frac{U_{H,t}}{\lambda_t} = -\frac{U_{L,t}}{\lambda_t} = -W_t$$

$$\text{Investment FOC} : IS_t Q_t (1 - S(\Xi_t) - \Xi_t S'(\Xi_t)) + \mathbb{E}_t [\Lambda_{t,t+1} IS_{t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2] = 1$$

$$\begin{aligned}
\text{Capital Supply : } \mathbb{E}_t \left[\Lambda_{t,t+1} R_{t+1}^K \right] &= 1 \\
\text{Capacity Utilization : } r_t^K &= a'(u_t) \\
\text{Capital Gross Return : } R_t^K &= \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}} \\
\text{Capital Accumulation : } K_t &= (1 - \delta)K_{t-1} + (1 - S(X_t))I_t I S_t
\end{aligned}$$

6.2.6 Price Dynamics

$$\begin{aligned}
\Pi_t &\equiv \frac{P_t}{P_{t-1}} \\
\tilde{\Pi}_t(\gamma) &\equiv \frac{\Pi_t^p}{(\Pi_{t-1}^p)^\gamma} \\
J_t - \xi \mathbb{E}_t[\Lambda_{t,t+1} \tilde{\Pi}_{t+1}(\gamma)^{\zeta-1} J_{t+1}] &= Y_t \\
J_t - \xi \mathbb{E}_t[\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^p(\gamma)^\zeta J_{t+1}] &= \frac{\zeta}{\zeta - 1} Y_t MC_t MCS_t \\
1 &= \xi \tilde{\Pi}_t(\gamma)^{\zeta-1} + (1 - \xi) \left(\frac{J_t}{J_t} \right)^{1-\zeta} \\
\frac{P_t^O}{P_t} &= \frac{J_t}{J_t}
\end{aligned}$$

and with price dispersion given by

$$\Delta_t = \xi \tilde{\Pi}_t \gamma^\zeta \Delta_{t-1} + (1 - \xi) \left(\frac{P_t^O}{P_t} \right)^{-\zeta}$$

Output and Labour Market Equilibrium

$$\begin{aligned}
Y_t &= \frac{Y_t^W}{\Delta_t} = C_t + I_t + G_t + K_{t-1} \\
Y_t^W &= (A_t H_t)^\alpha K_{t-1}^{1-\alpha}
\end{aligned}$$

Monetary Policy

The nominal interest rate is given by the following Taylor-type rules

$$\log \left(\frac{R_{n,t}}{R_n} \right) = \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right]$$

$$+ \theta_y \log \left(\frac{Y_t}{Y} \right) + \theta_{dy} \log \left(\frac{Y_t}{Y_{t-1}} \right) \Big] + \epsilon_{M,t}$$

or

$$\begin{aligned} \log \left(\frac{R_{n,t}}{R_n} \right) &= \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right. \\ &\quad \left. + \theta_y \log \left(\frac{Y_t}{Y_t^F} \right) + \theta_{dy} \log \left(\frac{Y_t/Y_t^F}{Y_{t-1}/Y_{t-1}^F} \right) \right] + \epsilon_{M,t} \end{aligned}$$

where Y_t^F is the flexi-price level of output and $\epsilon_{M,t}$ is a monetary policy shock process.

Shock Processes

AR(1): Risk Premium (RS_t), technology (A_t), investment (IS_t), and monetary policy ($\epsilon_{M,t}$).

i.i.d: monetary policy ($\epsilon_{M,t}$)

ARMA(1,1): marginal cost (MCS_t), marginal rate of substitution ($MRSS_t$)

Zero Growth and Positive Net Inflation Steady State

Consider first the case of a zero net inflation steady state ($\Pi = 1$). The *real* component of the steady state is that of the RBC model with investment adjustment costs and a wholesale-retail inverse of the mark-up $\frac{P^W}{P}$. Consider the case of no habit in consumption. The steady state is then given by

$$\begin{aligned} Q &= 1 \text{ (because } S(X) = S'(X) = S(0) = S'(0) = 0) \\ R &= \frac{1}{\beta} \\ \frac{P^W}{P} &= 1 - \frac{1}{\zeta} \\ \Delta &= 1 \\ \Lambda &= \beta \\ Y^W &= (AH)^\alpha K^{1-\alpha} \\ Y &= \frac{Y^W}{\Delta} \\ \frac{\varrho C}{(1 - \varrho)(1 - H)} &= W \end{aligned}$$

$$\begin{aligned}
\frac{\alpha P^W}{P} &= W \\
\frac{PK}{P^W Y^W} &= \frac{1-\alpha}{R-1+\delta} \\
I &= \delta K
\end{aligned}$$

which has a *recursive form*

$$\begin{aligned}
Q &= 1 \\
R &= \frac{1}{\beta} \\
\frac{P^W}{P} &= MC = 1 - \frac{1}{\zeta} \\
\Delta &= 1 \\
\Lambda &= \beta \\
\frac{K}{Y^W} &= \frac{(1-\alpha)}{R-1+\delta} \frac{P^W}{P} \\
\frac{I}{Y} &= \frac{\delta K}{Y^W} \frac{Y^W}{Y} = \frac{(1-\alpha)\delta}{R-1+\delta} \frac{P^W Y^W}{PY} = \frac{(1-\alpha)\delta}{R-1+\delta} \frac{P^W \Delta}{P} \\
\frac{C}{Y} &= 1 - \frac{I}{Y} - \frac{G}{Y} = 1 - \frac{I}{Y} - g_y \\
\frac{H\varrho}{(1-H)(1-\varrho)} &= \frac{WH/PY}{C/Y} = \frac{\alpha \frac{P^W Y^W}{PY}}{C/Y} \Rightarrow H = \frac{\alpha \frac{P^W \Delta^p}{P} (1-\varrho)}{\varrho C/Y + \alpha \frac{P^W \Delta}{P} (1-\varrho)} \\
Y^W &= (AH)^\alpha K^{1-\alpha} = (AH)^\alpha \left(\frac{K}{Y^W} \right)^{1-\alpha} (Y^W)^{1-\alpha} \Rightarrow Y^W = (AH)(K/Y^W)^{\frac{1-\alpha}{\alpha}} \\
Y &= \frac{Y^W}{\Delta} \\
G &= g_y Y \\
W &= \alpha \frac{P^W}{P} \frac{Y^W}{H} \\
I &= \frac{I}{Y} Y \\
C &= \frac{C}{Y} Y \\
K &= \frac{K}{Y^W} Y^W
\end{aligned}$$

For a steady state gross inflation rate $\Pi > 1$, the *nominal* NK features of the steady state are

$$R = \frac{R_n}{\Pi} \quad (6.26)$$

$$J^p(1 - \beta\zeta\Pi^\zeta) = \frac{YMC}{\left(1 - \frac{1}{\zeta}\right)} \quad (6.27)$$

$$JJ^p(1 - \beta\zeta\Pi^{\zeta-1}) = Y \quad (6.28)$$

$$\frac{J}{JJ} = \left(\frac{1 - \zeta\Pi^{\zeta-1}}{1 - \zeta}\right)^{\frac{1}{1-\zeta}} \quad (6.29)$$

$$MC = \left(1 - \frac{1}{\zeta}\right) \frac{J(1 - \beta\zeta\Pi^\zeta)}{JJ(1 - \beta\zeta\Pi^{\zeta-1})} \quad (6.30)$$

$$\Delta^p = \frac{(1 - \zeta)^{\frac{1}{1-\zeta}} (1 - \zeta\Pi^{\zeta-1})^{\frac{-\zeta}{1-\zeta}}}{1 - \zeta\Pi^\zeta} \quad (6.31)$$

For a zero-inflation steady state $\Pi = 1$ we arrive $\frac{J}{JJ} = \Delta = 1$ and $MC = \left(1 - \frac{1}{\zeta}\right)$ but in general there is a long-run inflation-output trade-off in the choice of the steady-state inflation rate. The implications of introducing a *non-zero* inflation steady state into the standard NK model are explored by Ascari and Ropele (2007).

As for the RBC model, the steady state can be used to calibrate the preference parameter ϱ to hit a particular target for the proportion of hours worked, $H = \bar{H}$. Then solving the steady state expression above giving H :

$$H = \frac{\alpha^{\frac{P^W\Delta}{P}}(1 - \varrho)}{\varrho^{\frac{C}{Y}} + \alpha^{\frac{P^W\Delta}{P}}(1 - \varrho)} = \bar{H} \quad (6.32)$$

for ϱ we have

$$\varrho = \frac{(1 - \bar{H})\alpha^{\frac{P^W\Delta}{P}}}{\alpha^{\frac{P^W\Delta}{P}} + \left(\frac{C}{Y} - \alpha^{\frac{P^W\Delta}{P}}\right)\bar{H}} \quad (6.33)$$

The case of habit in consumption now follows as for the RBC model replacing C/Y in (6.33) with $\frac{C}{Y}(1 - \chi)$ for external habit and with $\frac{C}{Y} \frac{(1-\chi)}{(1-\beta\chi)}$ for internal habit.

6.2.7 Linearized Model for Estimation

The linearized full NK model then adds to the RBC model:

$$\begin{aligned}\pi_t &= \beta \mathbb{E}_t \pi_{t+1} + \frac{(1 - \beta \tilde{\xi})(1 - \tilde{\xi})}{\tilde{\xi}} (mc_t + ms_t) \\ mc_t &= p_t^w - p_t = w_t + h_t - y_t \\ r_{n,t} &= \rho_r r_{n,t-1} + (1 - \rho_r)(\theta_\pi \pi_t + \theta_y y_t) + \epsilon_{M,t} \\ ms_t &= \rho_{ms} ms_{t-1} + \epsilon_{MS,t}\end{aligned}$$

The rest of the model is as above for the RBC model. The non-linear NK model steady state about which the model is linearized is given in the next section.

In **folder NK**, the mod file is **NKlinear.mod**. A version with both a flexi-price bloc and the option of non-zero inflation the the steady state is also included in this folder.

6.3 UNDERSTANDING THE IMPULSE RESPONSES

The linearized model can be used to understand the impulse response functions set out below. Consider the simpler version where $\sigma_c = 1$ (Cobb-Douglas utility) and $\chi = 0$ (no external habit). Then the supply of labour, h_t^s , is given by equating the real wage with the marginal rate of substitution between consumption and leisure to give

$$\begin{aligned}w_t &= c_t + \frac{H}{1-H} h_t^s \\ c_t &= \frac{1}{c_y} (y_t - i_y i_t - g_y g_t)\end{aligned}$$

Output is given by

$$y_t = a_t + \alpha h_t^d + (1 - \alpha) k_{t-1}$$

where labour demand, h_t^d , is given by equating the real wage with the marginal product of labour to give

$$w_t = y_t - h_t^d + mc_t = a_t + (1 - \alpha)(k_{t-1} - h_t^d) + mc_t$$

Finally the real marginal cost (the inverse of the retail mark-up), mc_t , is given by the Phillip's curve as

$$mc_t = \frac{\pi_t - \beta \mathbb{E}_t \pi_{t+1}}{\lambda}$$

which implies that a temporary increase in inflation (a looser monetary policy) will see marginal costs increase and the mark-up decrease thus increasing retail output.

In the labour market equilibrium $h_t^s = h_t^d = h_t$. In the steady state at $t - 1$, $k_{t-1} = 0$ and a little algebra sees equilibrium hours as

$$h_t = \frac{\left(1 - \frac{1}{c_y}\right) a_t + i_t + \left(\frac{g_y}{c_y}\right) g_t + mc_t}{\frac{\alpha}{c_y} + \frac{H}{1-H} + 1 - \alpha} \quad (6.34)$$

Thus hours *rise* on impact with a positive technology shock (a_t rises) keeping $g_t = 0$ iff

$$\left(1 - \frac{1}{c_y}\right) a_t + i_t + mc_t > 0 \quad (6.35)$$

which since $c_y < 1$, requires as a necessary condition $i_t + mc_t > 0$. For the RBC model $mc_t = 0$ and from the irfs below i_t rises with an increase in a_t . Then provided that adjustment costs ϕ_X are absent or small, hours will then increase. With the NK model by contrast a positive technology shock leads to an immediate fall in inflation π_t (a tightening of monetary policy) and $mc_t < 0$, so the possibility of a drop in hours appears.

For the real wage rate w_t we have

$$\begin{aligned} w_t &= -(1 - \alpha)h_t + a_t + mc_t \\ &= \left(1 - \frac{(1 - \alpha)(1 - \frac{1}{c_y})}{\frac{\alpha}{c_y} + \frac{H}{1-H} + 1 - \alpha}\right) a_t - \frac{(1 - \alpha)(mc_t + i_t + \frac{g_y}{c_y}g_t)}{\frac{\alpha}{c_y} + \frac{H}{1-H} + 1 - \alpha} + mc_t \\ &= \underbrace{\left(1 - \frac{(1 - \alpha)(1 - \frac{1}{c_y})}{\frac{\alpha}{c_y} + \frac{H}{1-H} + 1 - \alpha}\right)}_{\text{positive}} (a_t + mc_t) - \frac{(1 - \alpha)(i_t + \frac{g_y}{c_y}g_t)}{\frac{\alpha}{c_y} + \frac{H}{1-H} + 1 - \alpha} \end{aligned}$$

In a similar manner to hours, this equation can be used to explain the response of the real wage to impulses a_t and g_t in the irfs below. In particular in response to a positive technology shock the real wage will *fall* in the NK model iff $a_t + mc_t < 0$ or, in other words, the effect of a tightening of monetary policy where $mc_t < 0$ outweighs the productivity effect.

6.4 THE WORKHORSE SIMPLE NK MODEL

The workhorse NK model is the above model with no capital, no habit and no indexing. It is summarized as follows.

$$\begin{aligned}
Y_t &= C_t + G_t = \frac{Y_t^W}{D_t} \\
Y_t^W &= F(A_t, N_t) = (A_t N_t)^\alpha \\
\mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}] &= 1 \\
\Lambda_{t,t+1} &= \beta \frac{U_{C,t+1}}{U_{C,t}} \\
\frac{U_{N,t}}{U_{C,t}} &= -W_t \\
R_t &= \frac{R_{n,t-1}}{\Pi_t} \\
W_t &= \alpha \frac{P_t^W}{P_t} A_t^\alpha N_t^{\alpha-1} = \alpha \frac{P_t^W}{P_t} \frac{Y_t^W}{N_t} \\
\frac{P_t^0}{P_t} &= \frac{J_t}{JJ_t} \\
JJ_t - \xi \mathbb{E}_t [\Lambda_{t,t+1} \Pi_{t+1}^{\zeta-1} JJ_{t+1}] &= Y_t \\
J_t - \xi \mathbb{E}_t [\Lambda_{t,t+1} \Pi_{t+1}^\zeta J_{t+1}] &= \left(\frac{1}{1 - \frac{1}{\zeta}} \right) Y_t MC_t MS_t \\
1 &= \xi \Pi_t^{\zeta-1} + (1 - \xi) \left(\frac{P_t^0}{P_t} \right)^{1-\zeta} \\
MC_t &= \frac{P_t^W}{P_t} = \frac{W_t}{F_{H,t}} = \frac{W_t}{\alpha A_t^\alpha N_t^{\alpha-1}} \\
D_t &= \xi \Pi_t^\zeta D_{t-1} + (1 - \xi) \left(\frac{P_t^0}{P_t} \right)^{-\zeta}
\end{aligned}$$

With $R_{n,t}$ and A_t exogenous (noting that W_t is the real wage in consumption units). The model is then closed with AR 1 processes for A_t , G_t , MS_t and a Taylor-type rule for $R_{n,t}$

$$R_{n,t} = R_{n,t-1}^{\rho_r} (\Pi_t^{\theta_\pi} Y_t^{\theta_y})^{1-\rho_r} \exp(\epsilon_{MPS,t})$$

Note that Production Function in (6.36) is slightly different in Galí (2008) who has $Y_t^W = A_t N_t^{1-\alpha}$.

To close the model instead of a non-separable utility function up to now:
 $U_t = \frac{(C_t^{(1-\sigma_c)}(1-H_t)^{\psi})^{1-\sigma_c}-1}{1-\sigma_c}$, the workhorse model assumes a separable utility function:

$$U(C_t, H_t) = \frac{C_t^{1-\sigma_c}}{1-\sigma_c} - \frac{H_t^{1+\psi}}{1+\psi}$$

where ψ is the Frisch parameter. Then

$$\begin{aligned} U_{C,t} &= C_t^{-\sigma_c} \\ U_{H,t} &= -H_t^{\psi} \end{aligned}$$

6.4.1 Linearized Workhorse Model

First we log-linearize (6.36)–(6.36) using the notation for any variable X_t

$$x_t \equiv \log(X_t/X) = \log(X_t) - \log(X) = \log\left(1 + \frac{X_t - X}{X}\right) \approx \frac{X_t - X}{X}$$

up to first order. X is the deterministic steady state of X_t which can easily be generalized to a trend. The following is a linearization about a zero-growth and zero net inflation ($\Pi = 1$) steady state

$$\begin{aligned} \text{Technology shock} &: a_t = \rho_A a_{t-1} + \epsilon_{A,t} \\ \text{Gov Spending shock} &: g_t = \rho_G g_{t-1} + \epsilon_{G,t} \\ \text{Mark-up Shock} &: ms_t = \rho_{ms} ms_{t-1} + \epsilon_{MS,t} \\ \text{Euler Equation} &: \mathbb{E}_t[u_{C,t+1}] = u_{C,t} - \mathbb{E}_t(r_{n,t} - \pi_{t+1}) \\ \text{Labour Supply} &: w_t = u_{H,t} - u_{C,t} \\ \text{Real Marginal Cost} &: mc_t = p_t^w - p_t \\ \text{Labour Demand} &: w_t = mc_t + y_t - h_t \\ \text{Production Function} &: y_t = \alpha(a_t + h_t) \\ \text{Equilibrium} &: y_t = c_y c_t + g_y g_t \\ \text{Philips Curve} &: \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \frac{(1 - \beta \xi)(1 - \xi)}{\xi} (mc_t + ms_t) \\ \text{Monetary Rule} &: r_{n,t} = \rho_r r_{n,t-1} + (1 - \rho_r)(\theta_\pi \pi_t + \theta_y y_t) + \epsilon_{M,t} \\ \text{Marginal utility of } C_t &: u_{C,t} = -\sigma c_t \\ \text{Marginal utility of } H_t &: u_{H,t} = \psi h_t \end{aligned}$$

6.4.2 The Flexi-Price Economy

This is the limit as $\xi \rightarrow 0$ in which case $mc_t = mc_t^F = 0$ denoting the flexi-price model with a superscript F . It is given by

$$\begin{aligned} \text{Euler Equation : } \mathbb{E}_t[u_{C,t+1}^F] &= -\sigma_c \mathbb{E}_t[c_{t+1}^F] = u_{C,t}^F - r_t^F \\ &= -\sigma_c c_t^F - r_t^F \end{aligned} \quad (6.36)$$

$$\text{Labour Supply : } w_t = u_{H,t} - u_{C,t} = \psi h_t^F + \sigma_c c_t^F \quad (6.37)$$

$$\text{Labour Demand : } w_t^F = y_t^F - h_t^F \quad (6.38)$$

$$\text{Production Function : } y_t^F = \alpha(a_t + h_t^F) \quad (6.39)$$

$$\text{Equilibrium : } y_t^F = c_y c_t^F + g_y g_t \quad (6.40)$$

with exogenous shock a_t and g_t processes as before. r_t^F in what Woodford calls the Wicksellian natural rate of interest which is the equilibrium real rate of interest in the case of fully flexible prices. Equations (6.37) – (6.40) define an equilibrium in n_t^F , w_t^F , y_t^F and c_t^F given exogenous processes for a_t and g_t . Then (6.36) gives the natural rate of interest as:

$$r_t^F = \sigma_c (\mathbb{E}_t[c_{t+1}^F] - c_t^F) \quad (6.41)$$

Note that the flexi-price model still differs from its RBC counterpart in the steady state which has a price mark-up. But as the price elasticity $\xi \rightarrow \infty$ the mark-up disappears and the flexi-price NK and RBC models are same.

6.4.3 Relationship to Galí and Woodford

Our notation is close to Galí (2008), Chapter 3 which differs from Woodford (2003), Chapter 4. Differences from both are that (i) we consistently express the linearized equations as deviations about the steady state and so there are no intercept term. (ii) we have a persistence term in the interest rate rule (6.36) and (iii) they have $G_t = g_t = 0$ so the output equilibrium is simply $y_t = c_t$. However in Section 4.1 of Woodford (2003) he introduces an a different exogenous shock which only brings about a temporary “shift in the relationship between real income and its marginal utility due to either preference shocks or variations in government purchases” which is still referred to as g_t but does not affect the steady state. Then $g_y = 0$ in our set-up and becomes (in our notation)

$$r_t^F = \sigma_c \mathbb{E}_t[y_{t+1}^F - g_{t+1}] - (y_t^F - g_t) \quad (6.42)$$

and for the NK model we have

$$y_t - g_t = (\mathbb{E}_t[y_{t+1} - g_{t+1}] - \frac{1}{\sigma_c}(r_{n,t} - \mathbb{E}_t[\pi_{t+1}]) \quad (6.43)$$

Note σ in Woodford in $1/\sigma_c$ here and in Gali.

To arrive at the Philips Curve of Woodford and Gali requires some further algebraic steps. From (6.36), (6.36) and (6.37) we have (putting $c_t = y_t$)

$$w_t = \psi n_t + \sigma_c c_t = \psi n_t + \sigma_c y_t \quad (6.44)$$

Hence from (6.36) we have

$$mc_t = (\psi + 1)n_t + (\sigma_c - 1)y_t \quad (6.45)$$

Finally substituting for n_t from (6.36) we arrive at

$$mc_t = \left(\frac{\psi + 1}{\alpha} + \sigma_c - 1 \right) y_t - (1 + \psi)a_t \quad (6.46)$$

Putting $mc_t^F = 0$ for the flexi-price case we then have

$$y_t^F = \frac{\alpha(1 + \psi)}{1 + \psi + \alpha(\sigma_c - 1)} a_t \quad (6.47)$$

$$mc_t = \frac{\alpha(1 + \psi)}{1 + \psi + \alpha(\sigma_c - 1)} (y_t - y_t^F) \quad (6.48)$$

Hence the RE NK Philips Curve becomes

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa (y_t - y_t^F) + \frac{(1 - \beta \xi)(1 - \xi)}{\xi} ms_t \quad (6.49)$$

where

$$\kappa \equiv \frac{(1 - \beta \xi)(1 - \xi)}{\xi} \frac{\alpha(1 + \psi)}{1 + \psi + \alpha(\sigma_c - 1)} \quad (6.50)$$

By rescaling the shock process ms_t we can write (6.49) more simply as

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa (y_t - y_t^F + ms_t) \quad (6.51)$$

Given that we now have an expression for y_t^F we can consider a conventional monetary rule that responds to the output gap $y_t - y_t^F$ rather than its ‘implementable’ form (6.36)

$$r_{n,t} = \rho_r r_{n,t-1} + (1 - \rho_r)(\theta_\pi \pi_t + \theta_y (y_t - y_t^F) + \epsilon_{M,t}) \quad (6.52)$$

Indeed subtracting (6.42) from (6.43) we can then write the model in terms of the output gap $x_t = y_t - y_t^F$ as

$$x_t - g_t = (\mathbb{E}_t[x_{t+1} - g_{t+1}] - \frac{1}{\sigma_c}(r_{n,t} - \mathbb{E}_t[\pi_{t+1}] - r_t^F) \quad (6.53)$$

$$r_{n,t} = r r_{n,t-1} + (1 - \rho_r)(\theta_\pi \pi_t + \theta_y x_t) + \epsilon_{M,t} \quad (6.54)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa(x_t + ms_t) \quad (6.55)$$

$$a_t = \rho_A a_{t-1} + \epsilon_{A,t} \quad (6.56)$$

$$g_t = \rho_G g_{t-1} + \epsilon_{G,t} \quad (6.57)$$

$$ms_t = \rho_{MS} ms_{t-1} + \epsilon_{MS,t} \quad (6.58)$$

6.5 THE FULL SMETS-WOUTERS NON-LINEAR MODEL

To introduce wage stickiness we now assume that each household supplies homogeneous labour at a nominal wage rate $W_{h,t}$ to a monopolistic trade-union who differentiates the labour and sells type $H_t(j)$ at a nominal wage $W_{n,t}(j) > W_{h,t}$ to a labour packer in a sequence of Calvo staggered nominal wage contracts. The real wage is then defined as $W_t \equiv \frac{W_{n,t}}{P_t}$. We now have to distinguish between *price inflation* which now uses the notation $\Pi_t^p \equiv \frac{P_t}{P_{t-1}}$ and *wage inflation*. $\Pi_t^w \equiv \frac{W_{n,t}}{W_{n,t-1}}$

As with price contracts we employ Dixit-Stiglitz quantity and price aggregators. Calvo probabilities are now ξ_p and ξ_w for price and wage contracts respectively. The competitive labour packer forms a composite labour service according to $H_t = \left(\int_0^1 H_t(j)^{(\mu-1)/\mu} dj \right)^{\mu/(\mu-1)}$ and sells onto the intermediate firm. where μ is the elasticity of substitution. For each j , the labour packer chooses $H_t(j)$ at a wage $W_{n,t}(j)$ to maximize H_t given total expenditure $\int_0^1 W_{n,t}(j) H_t(j) dj$. This results in a set of labour demand equations for each differentiated labour type j with wage $W_{n,t}(j)$ of the form

$$H_t(j) = \left(\frac{W_{n,t}(j)}{W_{n,t}} \right)^{-\mu} H_t^d \quad (6.59)$$

where $W_{n,t} = \left[\int_0^1 W_{n,t}(j)^{1-\mu} dj \right]^{\frac{1}{1-\mu}}$ is the aggregate wage index. H_t and $W_{n,t}$ are Dixit-Stiglitz aggregators for the labour market and $H_t(j)$ is the quantity of

homogeneous labour supplied by households needed to produce labour type j . Integrating over j we then have

$$\int_0^1 H_t(j) dj = H_t = \left(\int_0^1 \left(\frac{W_{n,t}(j)}{W_{n,t}} \right)^{-\mu} dj \right) H_t^d = \Delta_t^w H_t^d \quad (6.60)$$

where $\Delta_t^w \equiv \int_0^1 \left(\frac{W_{n,t}(j)}{W_{n,t}} \right)^{-\mu} dj$ is *wage dispersion*.

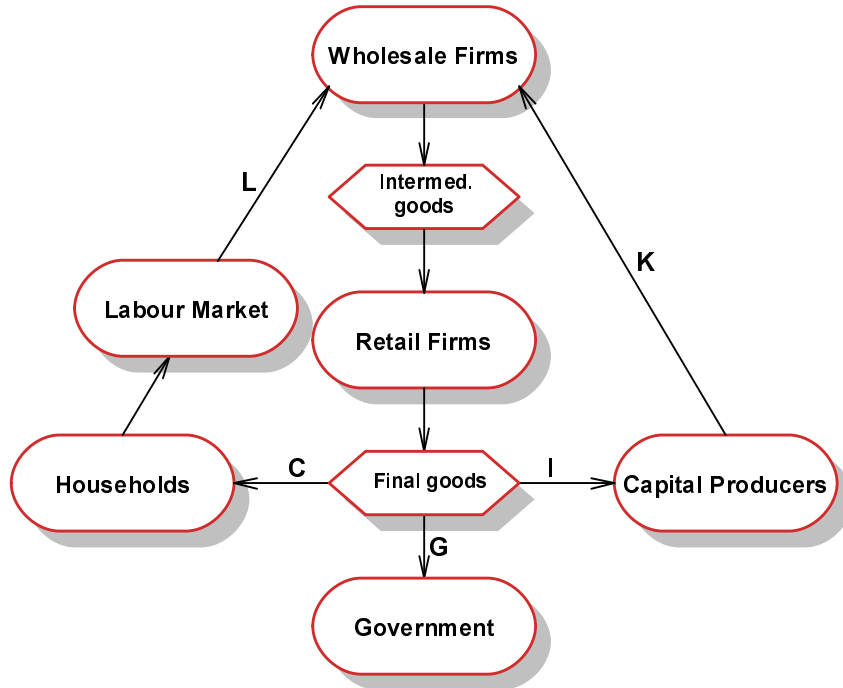


Figure 6.2: Model Structure

Wage setting by the trade-union again follows the standard Calvo framework supplemented with indexation. At each period there is a probability $1 - \xi_w$ that the wage is set optimally. The optimal wage derives from maximizing discounted profits. For those trade-unions unable to reset, wages are indexed to last period's aggregate inflation, with wage indexation parameter γ_w . Then as for price contracts the wage rate trajectory with no re-optimization is given by $W_{n,t}^O(j)$, $W_{n,t}^O(j) \left(\frac{P_t}{P_{t-1}} \right)^{\gamma_w}$, $W_{n,t}^O(j) \left(\frac{P_{t+1}}{P_{t-1}} \right)^{\gamma_w}$, $\dots$. The trade union then buys homogeneous labour at a nominal price $W_{h,t}$ and converts it into a differenti-

ated labour service of type j . The trade union time t then chooses $W_{n,t}^O(j)$ to maximize real profits

$$\mathbb{E}_t \sum_{k=0}^{\infty} \zeta_w^k \frac{\Lambda_{t,t+k}}{P_{t+k}} H_{t+k}(j) \left[W_{n,t}^O(j) \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma_w} - W_{h,t+k} \right] \quad (6.61)$$

where using (6.59) with indexing $H_{t+k}^d(j)$ is given by

$$H_{t+k}(j) = \left(\frac{W_{n,t}^O(j)}{W_{n,t+k}} \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma_w} \right)^{-\mu} H_{t+k}^d \quad (6.62)$$

and μ is the elasticity of substitution across labour varieties.

By analogy with (6.7) this leads to the following first-order condition

$$\mathbb{E}_t \sum_{k=0}^{\infty} \zeta_w^k \frac{\Lambda_{t,t+k}}{P_{t+k}} H_{t+k}^d(j) \left[W_{n,t}^O(j) \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma_w} - \frac{1}{(1 - 1/\mu)} W_{h,t+k} \right] = 0 \quad (6.63)$$

and hence by analogy with (6.10) this leads to the optimal real wage

$$\frac{W_{n,t}^O}{P_t} \equiv W_t^O = \frac{1}{(1 - 1/\mu)} \frac{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta_w^k \Lambda_{t,t+k} \left(\Pi_{t,t+k}^w \right)^{\zeta} H_{t+k}^d \frac{W_{h,t+k}}{P_{t+k}}}{\mathbb{E}_t \sum_{k=0}^{\infty} \zeta_w^k \Lambda_{t,t+k} \left(\Pi_{t,t+k}^w \right)^{\zeta} \left(\Pi_{t,t+k}^p \right)^{-1} H_{t+k}^d} \quad (6.64)$$

Then by the law of large numbers the evolution of the wage index is given by

$$W_{n,t}^{1-\mu} = \zeta_w \left(W_{n,t-1} \left(\frac{P_t}{P_{t-1}} \right)^{\gamma_w} \right)^{1-\mu} + (1 - \zeta_w) (W_{n,t}^0(j))^{1-\mu} \quad (6.65)$$

6.5.1 Dynare Price and Wage Dynamics

$$\begin{aligned} \Pi_t^p &\equiv \frac{P_t}{P_{t-1}} \\ \tilde{\Pi}_t^p(\gamma) &\equiv \frac{\Pi_t^p}{\Pi_{p,t-1}^\gamma} \\ JJ_t^p - \zeta_p \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^p(\gamma_p)^{\zeta-1} JJ_{t+1}^p] &= Y_t \\ J_t^p - \zeta_p \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^p(\gamma_p)^\zeta J_{t+1}^p] &= \frac{\zeta}{\zeta - 1} Y_t MC_t MS_{p,t} \end{aligned}$$

$$1 = \xi_p \tilde{\Pi}_t^p (\gamma_p)^{\zeta-1} + (1 - \xi_p) \left(\frac{J_t^p}{JJ_t^p} \right)^{1-\zeta}$$

$$\frac{P_t^O}{P_t} = \frac{J_t^p}{JJ_t^p}$$

$$\Pi_t^w \equiv \frac{W_{n,t}}{W_{n,t-1}} \quad (6.66)$$

$$\tilde{\Pi}_t^w \equiv \frac{\Pi_t^w}{(\Pi_{t-1}^p)^{\gamma_w}} \quad (6.67)$$

$$MRS_t = -\frac{U_{H,t}}{U_{C,t}} = \frac{W_{h,t}}{P_t} \quad (6.68)$$

$$JJ_t^w - \xi_w \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{(\tilde{\Pi}_{t,t+1}^w)^\mu}{\tilde{\Pi}_{t,t+1}^p (\gamma_w)} JJ_{t+1}^w \right] = H_t^d \quad (6.69)$$

$$J_t^w - \xi_w \mathbb{E}_t \left[\Lambda_{t,t+1} \tilde{\Pi}_{w,t+1}^\mu J_{t+1}^w \right] = -\frac{\mu}{\mu-1} MRS_t MS_{w,t} H_t^d \quad (6.70)$$

$$(W_{n,t})^{1-\mu} = \xi_w \left((W_{n,t-1}) \frac{1}{\tilde{\Pi}_t^p (\gamma_w)} \right)^{1-\mu} + (1 - \xi_w) (W_{n,t}^O)^{1-\mu} \Rightarrow$$

$$1 = \xi_w \left(\frac{\Pi_t^w \tilde{\Pi}_{p,t} (\gamma_w)}{\Pi_t^p} \right)^{\mu-1} + (1 - \xi_w) \left(\frac{W_t^O(j)}{W_{n,t}} \right)^{1-\mu} \quad (6.71)$$

$$\frac{W_t^O}{W_{n,t}} = \frac{J_t^w}{W_t JJ_t^w} \quad (6.72)$$

$$W_t \equiv \frac{W_{n,t}}{P_t} \quad (6.73)$$

The set-up is completed with

$$W_t = \frac{\Pi_t^w}{\Pi_t^p} W_{t-1} \quad (6.74)$$

6.5.2 Capacity Utilization and Fixed Costs of Production

We now add two remaining features to the model. As in Christiano *et al.* (2005a) and Smets and Wouters (2007) we assume that using the stock of capital with

intensity u_t produces a cost of $a(u_t)K_t$ units of the composite final good. The functional form is chosen consistent with the literature:

$$a(u_t) = \gamma_1(u_t - 1) + \frac{\gamma_2}{2}(u_t - 1)^2 \quad (6.75)$$

and satisfies $a(1) = 0$ and $a'(1), a''(1) > 0$. Then we must add a term $(r_t^K - a(u_t)K_t)$ to the household budget constraint on the income side leading to the following first-order condition determines capacity utilization:

$$r_t^K = a'(u_t) \quad (6.76)$$

Capital now enters the production function as $u_t K_{t-1}$. The final change is to add fixed costs, necessary to transform homogeneous wholesale goods into differentiated retail goods, to the Cobb-Douglas production function which becomes

$$F(A_t, H_t^d, u_t K_{t-1}) = (A_t H_t^d)^\alpha (u_t K_{t-1})^{1-\alpha} + F \equiv Y_t^W + F \quad (6.77)$$

Smets and Wouters (2007) has one more feature: *Kimball preferences* as in Kimball (1995) and Klenow and Willis (2016) generalize the Dixit-Stiglitz aggregator. But we do not include this feature

6.5.3 Price and Wage Dispersion

The output and labour market clearing conditions must take into account relative price dispersion across varieties and wage dispersion across firms. Integrating across all firms, taking into account that the capital-labour ratio is common across firms and that the wholesale sector is separated from the retail sector we obtain aggregate demand for intermediate (wholesale) goods necessary to produce final retail goods as

$$Y_t^W - F = (C_t + I_t + G_t + a(u_t)K_{t-1}) \int_0^1 \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} dm \quad (6.78)$$

where labour market clearing gives total demand for labour, H_t^d , as

$$H_t = \int_0^1 H_t(j) dj = \int_0^1 \left(\frac{W_{n,t}(j)}{W_{n,t}} \right)^{-\mu} dj H_t^d = \Delta_t^w H_t^d \quad (6.79)$$

where the price dispersion is given by $\Delta_t^p = \int_0^1 \left(\frac{P_t(f)}{P_t} \right)^{-\zeta} df$ and wage dispersion is given by $\Delta_t^w = \int_0^1 \left(\frac{W_{n,t}(j)}{W_{n,t}} \right)^{-\mu} dj$. By analogy with (6.23)

$$\Delta_t^p = \zeta_p + \tilde{\Pi}_t^\zeta \Delta_{t-1}^p + (1 - \zeta_p) \left(\frac{P_t^O}{P_t} \right)^{-\zeta} \quad (6.80)$$

$$\Delta_t^w = \zeta_w \tilde{\Pi}_{w,t}^\mu \Delta_{t-1}^w + (1 - \zeta_w) \left(\frac{W_{n,t}^O}{W_{n,t}} \right)^{-\mu} \quad (6.81)$$

$$(6.82)$$

6.5.4 Internal and External Habit

We can distinguish between *external habit* and *internal habit* in consumption by rewriting the utility function of household j , the single-period utility, marginal utility of consumption and labour supply as respectively⁶:

$$\begin{aligned} U_t^j &= U(C_t^j, C_{t-1}, H_t^j) = \frac{((C_t^j - \chi C_{t-1})^{(1-\varrho)}(1 - H_t^j)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\ U_{C,t}^j &= (1 - \varrho)(C_t^j - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)-1} (1 - H_t^j)^{\varrho(1-\sigma_c)} \\ U_{H,t}^j &= -\varrho(C_t^j - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)} (1 - H_t^j)^{\varrho(1-\sigma_c)-1} \end{aligned}$$

Then in evaluating the marginal utility of consumption, $U_{C,t}^j$, the household takes external habit χC_{t-1} as *given* where C_t is the *aggregate* per capita consumption. In an equilibrium of *identical households* $C_t^j = C_t$ and $H_t^j = H_t$ (as in the notes previously).

By contrast with internal habit household j enjoys consumption *relative to her own previous consumption*. Then

$$\begin{aligned} U_t^j &= U(C_t^j, H_t^j) = \frac{((C_t^j - \chi C_{t-1}^j)^{(1-\varrho)}(1 - H_t^j)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\ U_{C,t}^j &= (1 - \varrho)(C_t^j - \chi C_{t-1}^j)^{(1-\varrho)(1-\sigma_c)-1} (1 - H_t^j)^{\varrho(1-\sigma_c)} \\ &\quad - \beta \chi (1 - \varrho) \mathbb{E}_t \left[(C_{t+1}^j - \chi C_t^j)^{(1-\varrho)(1-\sigma_c)-1} (1 - H_{t+1}^j)^{\varrho(1-\sigma_c)} \right] \\ U_{H,t}^j &= -\varrho(C_t^j - \chi C_{t-1}^j)^{(1-\varrho)(1-\sigma_c)} (1 - H_t^j)^{\varrho(1-\sigma_c)-1} \end{aligned}$$

As for external habit in an equilibrium of identical household $C_t^j = C_t$ the average consumption and $H_t^j = H_t$.

⁶ In the codes and slides we adopt a different form of the utility function, but the Cobb-Douglas form here serves to illustrate the internal-external habit features and the stationarization.

6.5.5 Unemployment

Unemployment can easily be introduced into the set-up in a manner similar to Galí *et al.* (2015). Let $H_t > H_t^d$ be the total supply households would be willing to supply at the prevailing real wage W_t in the absence of the intervention of the trade union. Let $U(C_t, H_t)$ be the utility from supplying H_t and $U_N(C_t, N_t)$ the corresponding marginal utility. Then H_t and unemployment rate UN_t is given by

$$\frac{U_{H,t}}{U_{C,t}} = -MRS_t \equiv W_t \quad (6.83)$$

$$UN_t = \frac{H_t - H_t^d}{H_t} \quad (6.84)$$

6.5.6 Summary of SW07 Model

Households

$$\text{Euler Consumption : } 1 = R_t R S_t \mathbb{E}_t [\Lambda_{t,t+1}]$$

$$\text{Stochastic Discount Factor : } \Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t}$$

$$\text{where : } \lambda_t = U_{C,t}$$

$$\text{Labour Supply : } \frac{U_{H,t}}{\lambda_t} = -\frac{U_{L,t}}{\lambda_t} = -W_t$$

$$\text{Investment FOC : } IS_t Q_t (1 - S(\Xi_t) - \Xi_t S'(\Xi_t)) + \mathbb{E}_t [\Lambda_{t,t+1} IS_{t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2] = 1$$

$$\text{Capital Supply : } \mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}^K] = 1$$

$$\text{Capacity Utilization : } r_t^K = a'(u_t)$$

$$\text{Capital Gross Return : } R_t^K = \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}}$$

$$\text{Capital Accumulation : } K_t = (1 - \delta)K_{t-1} + (1 - S(X_t))I_t IS_t$$

Price and Wage Dynamics

$$\Pi_t^p \equiv \frac{P_t}{P_{t-1}}$$

$$\tilde{\Pi}_t^p(\gamma) \equiv \frac{\Pi_t^p}{(\Pi_{t-1}^p)^\gamma}$$

$$\begin{aligned}
JJ_t^p - \xi_p \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^p (\gamma_p)^{\zeta-1} JJ_{t+1}^p] &= Y_t \\
J_t^p - \xi_p \mathbb{E}_t [\Lambda_{t,t+1} \tilde{\Pi}_{t+1}^p (\gamma_p)^\zeta J_{t+1}^p] &= \frac{\zeta}{\zeta-1} Y_t MC_t MCS_{p,t} \\
1 &= \xi_p \tilde{\Pi}_t^p (\gamma_p)^{\zeta-1} + (1 - \xi_p) \left(\frac{J_t^p}{JJ_t^p} \right)^{1-\zeta} \\
\frac{P_t^O}{P_t} &= \frac{J_t^p}{JJ_t^p}
\end{aligned}$$

$$\begin{aligned}
\Pi_t^w &\equiv \frac{W_{n,t}}{W_{n,t-1}} \\
\tilde{\Pi}_t^w &\equiv \frac{\Pi_t^w}{(\Pi_{t-1}^p)^{\gamma_w}} \\
MRS_t &= -\frac{U_{H,t}}{U_{C,t}} = \frac{W_{h,t}}{P_t} \\
JJ_t^w - \xi_w \mathbb{E}_t \left[\Lambda_{t,t+1} \frac{(\tilde{\Pi}_{t+1}^w)^\mu}{\tilde{\Pi}_{t,t+1}^p (\gamma_w)} JJ_{t+1}^w \right] &= H_t^d \\
J_t^w - \xi_w \mathbb{E}_t [\Lambda_{t,t+1} (\tilde{\Pi}_{t+1}^w)^\mu J_{t+1}^w] &= -\frac{\mu}{\mu-1} MRS_t MRSS_{w,t} H_t
\end{aligned}$$

$$\begin{aligned}
(W_{n,t})^{1-\mu} &= \xi_w \left((W_{n,t-1}) \frac{1}{\tilde{\Pi}_t^p (\gamma_w)} \right)^{1-\mu} + (1 - \xi_w) (W_{n,t}^O)^{1-\mu} \Rightarrow \\
1 &= \xi_w \left(\frac{\Pi_t^w \tilde{\Pi}_{p,t} (\gamma_w)}{\Pi_t^p} \right)^{\mu-1} + (1 - \xi_w) \left(\frac{W_t^O(j)}{W_{n,t}} \right)^{1-\mu} \\
\frac{W_t^O}{W_{n,t}} &= \frac{J_t^w}{W_t JJ_t^w} \\
W_t &\equiv \frac{W_{n,t}}{P_t}
\end{aligned}$$

The real wage is given by

$$W_t = \frac{\Pi_t^w}{\Pi_t^p} W_{t-1}$$

and with price and wage dispersion given by

$$\Delta_t^p = \xi_p \tilde{\Pi}_t^p (\gamma_p)^\zeta \Delta_{t-1}^p + (1 - \xi_p) \left(\frac{P_t^O}{P_t} \right)^{-\zeta}$$

$$\Delta_t^w = \xi_w (\tilde{\Pi}_t^w)^\mu \Delta_{t-1}^w + (1 - \xi_w) \left(\frac{W_t^O}{W_t} \right)^{-\mu}$$

Output and Labour Market Equilibrium

$$\begin{aligned} Y_t &= \frac{Y_t^W - F}{\Delta_{p,t}} = C_t + I_t + G_t + a(u_t)K_{t-1} \\ Y_t^W &= (A_t H_t^d)^\alpha (u_t K_{t-1})^{1-\alpha} \\ H_t^d &= \frac{H_t}{\Delta_t^w} \end{aligned}$$

Monetary Policy

The nominal interest rate is given by the following Taylor-type rules

$$\begin{aligned} \log \left(\frac{R_{n,t}}{R_n} \right) &= \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right. \\ &\quad \left. + \theta_y \log \left(\frac{Y_t}{Y} \right) + \theta_{dy} \log \left(\frac{Y_t}{Y_{t-1}} \right) \right] + \epsilon_{M,t} \end{aligned}$$

or

$$\begin{aligned} \log \left(\frac{R_{n,t}}{R_n} \right) &= \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + (1 - \rho_r) \left[\theta_\pi \log \left(\frac{\Pi_t}{\Pi} \right) \right. \\ &\quad \left. + \theta_y \log \left(\frac{Y_t}{Y_t^F} \right) + \theta_{dy} \log \left(\frac{Y_t/Y_t^F}{Y_{t-1}/Y_{t-1}^F} \right) \right] + \epsilon_{M,t} \end{aligned}$$

where Y_t^F is the flexi-price level of output and $\epsilon_{M,t}$ is a monetary policy shock process.

Unemployment

$$\begin{aligned} \frac{U_{H,t}}{U_{C,t}} &= -MRS_t \equiv W_t \\ UN_t &= \frac{H_t - H_t^d}{H_t} \end{aligned}$$

Shock Processes

AR(1): Risk Premium (RS_t), technology (A_t), investment (IS_t), and monetary policy ($\epsilon_{M,t}$).

i.i.d: monetary policy ($\epsilon_{M,t}$)

ARMA(1,1): marginal cost (MCS_t), marginal rate of substitution ($MRSS_t$)

Steady State

In recursive form the zero-growth zero-inflation ($\Pi_p = \Pi_w = 1$) steady state of can be written

$$\begin{aligned}
 Q &= 1 \\
 R &= \frac{1}{\beta} \\
 \frac{P^W}{P} &= 1 - \frac{1}{\zeta} = \frac{1}{MS_p} \\
 \frac{W_h/P}{W/P} &= 1 - \frac{1}{\mu} = \frac{1}{MS_w} \\
 \Delta_p = \Delta_w &= 1 \\
 \frac{K}{Y^W} &= \frac{(1-\alpha)\delta}{R-1+\delta} \frac{P^W}{P} \\
 \frac{I}{Y} &= \frac{\delta K}{Y^W} \frac{Y^W}{Y} = \frac{(1-\alpha)\delta}{R-1+\delta} \frac{P^W Y^W}{PY} = \frac{(1-\alpha)\delta}{R-1+\delta} \frac{P^W}{P} \\
 \frac{C}{Y} &= 1 - \frac{I}{Y} - \frac{G}{Y} = 1 - \frac{I}{Y} - g_y \\
 \frac{WH^d}{P^W Y^W} &= \alpha \\
 \frac{H\varrho}{(1-H)(1-\varrho)} &= \frac{W_h H / PY}{C/Y} = \frac{\alpha \frac{P^W Y^W}{PY} \frac{W_h}{W} \Delta^w}{C/Y} \Rightarrow N = \frac{\alpha \frac{W_h}{W} \Delta^w (1-\varrho)}{\varrho C/Y + \alpha \frac{W_h}{W} \Delta^w (1-\varrho)} \\
 Y^W &= (AH^d)^\alpha K^{1-\alpha} = (AH^d)^\alpha \left(\frac{K}{Y^W} \right)^{1-\alpha} (Y^W)^{1-\alpha} \Rightarrow Y^W = (AH^d) (K/Y^W)^{\frac{1-\alpha}{\alpha}} \\
 H^d &= \frac{H}{\Delta_w} \\
 Y &= \frac{Y^W - F}{\Delta_p} \\
 G &= g_y Y
 \end{aligned}$$

$$\begin{aligned}
W &= \alpha \frac{P^W}{P} \frac{Y^W}{H} \\
I &= \frac{I}{Y} Y \\
C &= \frac{C}{Y} Y \\
K &= \frac{K}{Y^W} Y^W
\end{aligned}$$

where we have imposed a zero-profit condition $PY = P^W Y^W$.

For a particular steady state inflation rate $\Pi_p = \Pi_w > 1$ the NK features of the steady state become for price dynamics:

$$\begin{aligned}
\tilde{\Pi}_p(\gamma) &\equiv \Pi^{1-\gamma} \\
\frac{P^O}{P} = \frac{J^p}{H_p} &= \left(\frac{1 - \xi_p \tilde{\Pi}_p(\gamma_p)^{\zeta-1}}{1 - \xi_p} \right)^{\frac{1}{1-\zeta}} \\
MC = \frac{P^W}{P} &= \left(1 - \frac{1}{\zeta} \right) \frac{J^p (1 - \beta \xi_p \tilde{\Pi}_p(\gamma_p)^\zeta)}{H_p (1 - \beta \xi_p \tilde{\Pi}_p(\gamma_p)^{\zeta-1})} \\
&= \text{Inverse of price mark-up} \\
\Delta_p &= \frac{1 - \xi_p}{1 - \xi_p \tilde{\Pi}_p(\gamma_p)^\zeta} \left(\frac{J^p}{H_p} \right)^{-\zeta}
\end{aligned}$$

and for wage dynamics

$$\begin{aligned}
\frac{W_h}{P} &= -\frac{U_N}{U_C} \\
\frac{W^O}{W} = \frac{\frac{J^w}{H_w}}{\frac{W}{P}} &= \left(\frac{1 - \xi_w \tilde{\Pi}_p(\gamma_w)^{\mu-1}}{1 - \xi_w} \right)^{\frac{1}{1-\mu}} \\
\frac{J^w}{JJ^w} &= MS_w \frac{W_h}{P} \frac{(1 - \beta \xi_w \tilde{\Pi}_p(\gamma_w)^{\mu-1})}{(1 - \beta \xi_w \tilde{\Pi}_p(\gamma_w)^\mu)} \\
\text{i.e., } \frac{\frac{W_h}{P}}{\frac{W}{P}} &= \left(1 - \frac{1}{\mu} \right) \frac{\frac{J^w}{H_w}}{\frac{W}{P}} \frac{(1 - \beta \xi_w \tilde{\Pi}_p(\gamma_w)^\mu)}{(1 - \beta \xi_w \tilde{\Pi}_p(\gamma_w)^{\mu-1})} \\
&= \text{Inverse of wage mark-up} \\
\Delta_w &= \frac{1 - \xi_w}{1 - \xi_w \tilde{\Pi}_p(\gamma_w)^\mu} \left(\frac{\frac{J^w}{H_w}}{\frac{W}{P}} \right)^{-\mu}
\end{aligned}$$

Note that these expressions for dispersion in the steady state imply the following conditions must apply

$$\tilde{\Pi}^p(\gamma_p) < \left(\frac{1}{\bar{\xi}_p} \right)^{\frac{1}{\xi_p}} \quad (6.85)$$

$$\tilde{\Pi}^p(\gamma_w) < \left(\frac{1}{\bar{\xi}_w} \right)^{\frac{1}{\xi_w}} \quad (6.86)$$

Given Δ^p and Δ^w , for CD production we now can write

$$Y = \frac{(Y^W - F)}{\Delta^p} \quad (6.87)$$

To pin down F we make the assumption that entry occurs until retail profits are eliminated in the steady state, i.e., $P^W Y^W = P Y$. It follows that

$$\frac{P^W}{P} = MC = \frac{Y}{Y^W} = \frac{\left(1 - \frac{F}{Y^W}\right)}{\Delta_p} \quad (6.88)$$

It follow that

$$\frac{F}{Y^W} = 1 - \Delta_p MC \quad (6.89)$$

For the zero inflation, $MC = 1 - \frac{1}{\xi}$ and $\Delta_p = \Delta_w = 1$ and therefore $\frac{F}{Y^W} = \frac{1}{\xi}$. Finally we still have

$$H = \frac{\alpha \frac{W_h}{W} \Delta^w (1 - \varrho)}{\varrho C / Y + \alpha \frac{W_h}{W} \Delta^w (1 - \varrho)} \quad (6.90)$$

which can be used to calibrate the preference parameter ϱ to hit a particular target for the proportion of hours worked, $N = N_{ss}$.

6.5.7 A Balanced-Non-Zero-Growth Steady State

We can easily set up the model with a balanced-exogenous-growth steady state. Now the process for A_t is replaced with

$$\begin{aligned} A_t &= \bar{A}_t A_t^c \\ \bar{A}_t &= (1 + g_t) \bar{A}_{t-1} \\ \log(1 + g_t) &= \log(1 + g) + \epsilon_{A,t} \\ \log A_t^c - \log A^c &= \rho_A (\log A_{t-1}^c - \log A^c) + \epsilon_{A,t} \end{aligned}$$

A_t is a labour-augmenting technical progress parameter which is now decomposed into a cyclical component, A_t^c , modelled as a temporary AR1 process, a stochastic trend, a random walk with drift, $\bar{A}_t$. Thus the balanced growth deterministic steady state path (bgp) is driven by labour-augmenting technical change growing at a net rate g . If we put $g = \epsilon_{trend,t} = 0$ and $\bar{A}_t = 1$, we arrive at our previous formulation with $A_t^c = A_t$.

$$\begin{aligned} U_t^c &= \frac{((C_t^c - \chi C_{t-1}^c)/(1 + g_t))^{(1-\varrho)}(1 - H_t)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\ U_{C,t}^c &\equiv \frac{U_{C,t}}{\bar{A}_t^{(1-\varrho)(1-\sigma_c)-1}} = (1 - \varrho)(C_t^c - \chi C_{t-1}^c/(1 + g_t))^{(1-\varrho)(1-\sigma_c)-1}(1 - H_t)^{\varrho(1-\sigma_c)} \\ \Lambda_{t,t+1} &= \beta \frac{U_{C,t+1}^c}{U_{C,t}^c} = \beta(1 + g_{t+1})^{(1-\varrho)(1-\sigma_c)-1} \frac{U_{C,t+1}^c}{U_{C,t}^c} \equiv \beta_{g,t+1} \frac{U_{C,t+1}^c}{U_{C,t}^c} \end{aligned}$$

where the growth-adjusted discount rate is defined as

$$\beta_{g,t} \equiv \beta(1 + g_t)^{(1-\varrho)(1-\sigma_c)-1},$$

the Euler equation is still

$$E_t [\Lambda_{t,t+1} R_{t+1}]$$

Now stationarize remaining variables by defining cyclical components:

$$\begin{aligned} Y_t^c &\equiv \frac{Y_t}{\bar{A}_t} = \frac{(A_t H_t^d)^\alpha \left(\frac{K_{t-1}^c}{\bar{A}_t}\right)^{1-\alpha} - F}{\Delta_t^p} = \frac{(A_t H_t^d)^\alpha \left(\frac{K_{t-1}^c}{(1+g_t)}\right)^{1-\alpha} - F}{\Delta_t^p} \\ K_t^c &\equiv \frac{K_t}{\bar{A}_t} \\ K_t^c &= (1 - \delta) \frac{K_{t-1}^c}{1 + g_t} + (1 - S(X_t^c)) I_t^c \\ X_t^c &= (1 + g_t) \frac{I_t^c}{I_{t-1}^c} \\ S(X_t^c) &= \phi_X(X_t^c - 1 - g_t)^2 \\ S'(X_t^c) &= 2\phi_X(X_t^c - 1 - g_t) \\ C_t^c &\equiv \frac{C_t}{\bar{A}_t} \\ I_t^c &\equiv \frac{I_t}{\bar{A}_t} \end{aligned}$$

$$W_t^c \equiv \frac{W_t}{\bar{A}_t}$$

With non-zero steady state growth, the steady state for the rest of the system is the same as the zero-growth one except for the following relationships:

$$\begin{aligned} R &= \frac{(1+g)^{1+(\sigma_c-1)(1-\varrho)}}{\beta} \equiv \frac{1}{\beta_g} \\ R_n &= \Pi R \\ \Lambda &= \frac{1}{R} \\ I^c &= \frac{(\delta+g)K^c}{1+g} \\ X^c &= 1+g \\ \frac{(J^p)^c}{(JJ^p)^c} &= \left(\frac{1 - \xi_p(\Pi^p)^{\zeta-1}}{1 - \xi_p} \right)^{\frac{1}{1-\zeta}} \\ MC &= \left(1 - \frac{1}{\zeta} \right) \frac{J^p(1 - \beta_g(1+g)\xi_p\Pi^\zeta)}{JJ^p(1 - \beta_g(1+g)\xi(\Pi^p)^{\zeta-1})} \\ \frac{\frac{W_h}{P}}{\frac{W}{P}} &= \left(1 - \frac{1}{\mu} \right) \frac{\frac{J^w}{JJ^w}}{\frac{W}{P}} \frac{(1 - \beta_g\xi_w\tilde{\Pi}^p(\gamma_w)^\mu)}{(1 - \beta_g\xi_w\tilde{\Pi}_p(\gamma_w)^{\mu-1})} \\ &= \text{Inverse of wage mark-up} \end{aligned}$$

where R and R_n are the real and nominal steady state interest rates and Π is inflation.

6.6 IMPULSE RESPONSES TO A TECHNOLOGY SHOCK

We are now in a position to compare the RBC and NK models. For the RBC model we adopt the calibration set out in Table 1. The additional parameters in the NK model are set as $\xi = 0.75$, $\chi = 0$ and $\gamma = 0$, $\rho = 0.75$, $\theta_\pi = 1.5$ and $\theta_y = 0.15$ so we leave out habit and indexation for this comparison. $\xi = 0.75$ corresponds to an average price contract length of 4 quarters.

We compare the impulse response functions (IRFs), as percentage deviations about the steady state, for two AR(1) shocks with persistence parameters 0.7 (as with the policy rule). The first is a 1% supply-side shock to total factor productivity, $\epsilon_{A,t}$ and the second is a 1% demand-side shock to government

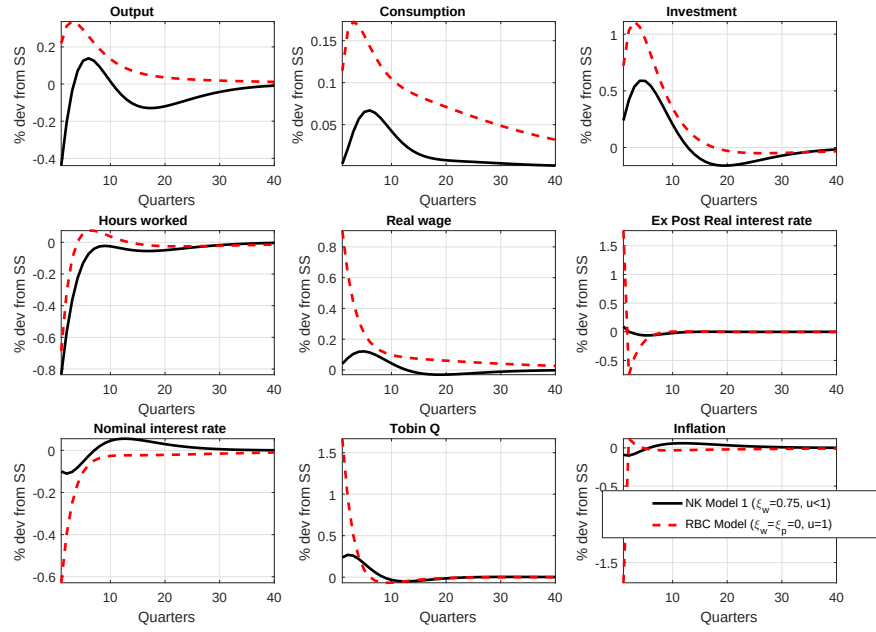
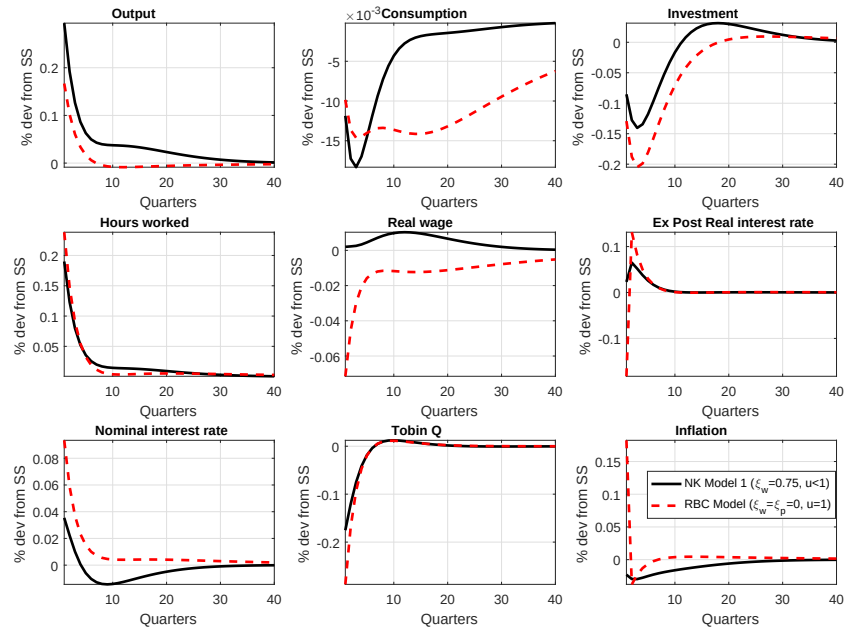
expenditure $\epsilon_{G,t}$. Figures 3 and 4 show the IRFs as proportional deviations about the steady state for RBC variables: real output, consumption, investment, hours, the real wage, the real interest rate and the real price of capital and the same for the NK model with, in addition, inflation and the nominal interest rate (set to zero in the RBC model). In the NK model the ex post real interest rate replaces the ex ante rate of the RBC model.

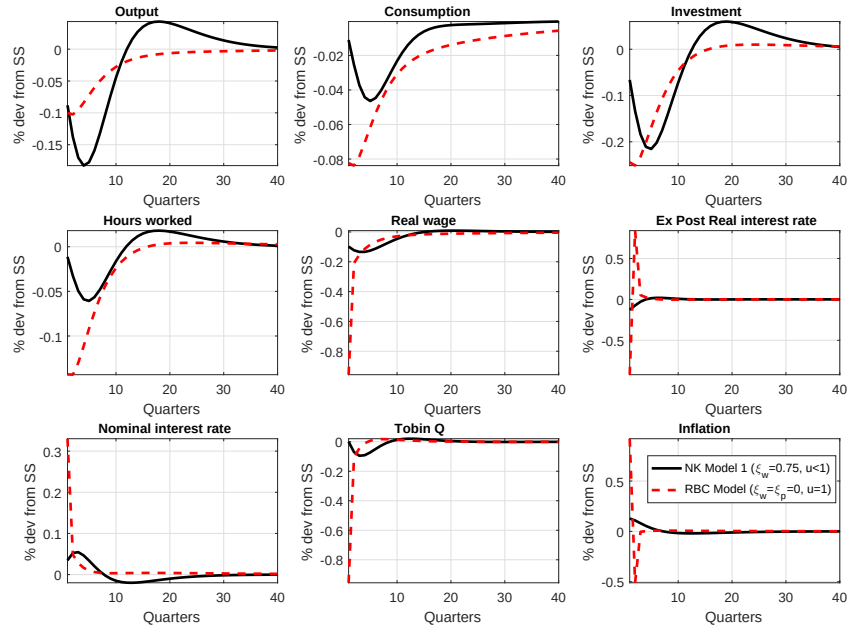
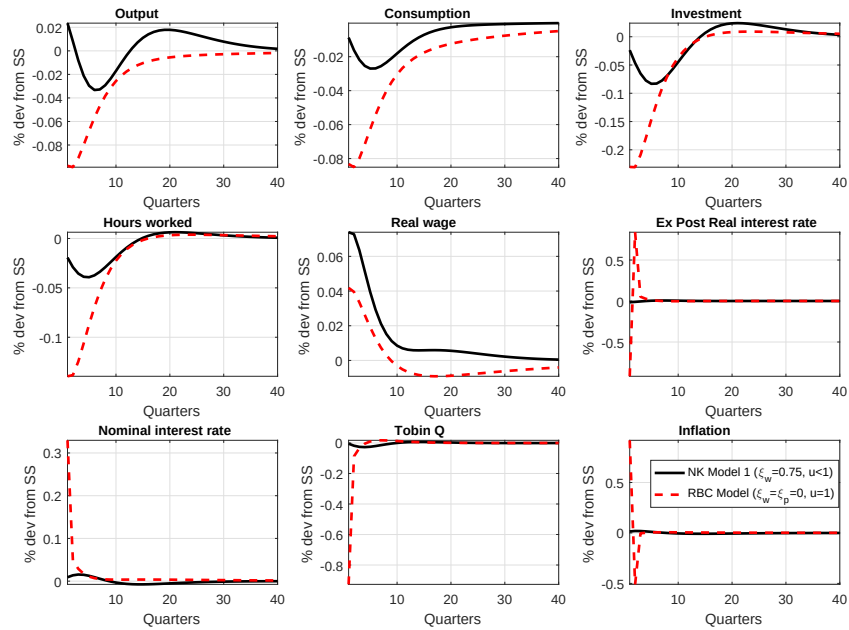
Consider first a positive technology shock. For both RBC and NK models output, consumption and investment rise and then falls back to the steady state as the shock fades away. The increase in the supply of capital reduces its cost, the real interest rate. Thus the discounted sum of future profits rises bringing about a rise in the price of capital. Real wage increases because labour supply increases less than proportionally with output, the substitution effect outweighs the income effect and hours worked increase. In the NK model inflation falls sharply at first then quickly returns and slightly overshoots the steady state. With our chosen Taylor rule this calls for a reduction in the nominal interest rate a lower immediate increase in output and consumption and an initial decline in the real wage that is now affected by the increase in marginal costs and the reduction of labour supply. In the long-run the real wage increases as soon as hours worked overshoots its steady state value. There are rather small differences between the IRFs of real variables except for hours worked and the real wage in the short-run. Whereas in the short-run, hours and the real wage increase in the RBC model⁷ the opposite is the case in the NK model. For the latter case sticky prices results in hours-technology debate is probably one of the most controversial issue in business cycle theory and, as shown by our simulations, the major point of distinction between these two models. The reason being that the sign and response of hours to a productivity shock can have important consequences for policy analysis. For comprehensive reviews about the hours-technology debate see Galí and Rabanal (2005) and Whelan (2009).

Turning to the demand shock (in effect a fiscal stimulus), we now observe significant differences between the two models. The response of a shock the government spending is to crowd out consumption and investment in both RBC and NK worlds, the latter occurring through the an increase in the real interest rate. The fall in consumption increases the marginal utility of consumption and induces the household to switch from consumption and leisure.

⁷ The actual reduction in hours on impact for the RBC model showed in the simulations is due to the presence of investment adjustment costs. In this case is costly to adjust capital so firms respond to a TFP shock by reducing hours worked in the first period. From the second quarter on the usual result that hours increase in RBC models still applies.

Hours therefore increase and with capital only able to gradually change we see output increase in both the RBC and NK models. However in the NK model with sticky prices firms respond to an increase in demand by raising output and there is an additional demand effect on aggregate output. In both cases there is an increase in hours supplied which in the RBC model leads to a reduction in the real wage. In the NK model demand for labour increases offsetting the decrease in the real wage which in fact now increases.

Figure 6.3: RBC-NK SW Model Comparison: IRFs to $\epsilon_{A,t}$ shockFigure 6.4: RBC-NK SW Model Comparison: IRFs to $\epsilon_{G,t}$ shock

Figure 6.5: RBC-NK SW Model Comparison: IRFs to $\epsilon_{MCS,t}$ shockFigure 6.6: RBC-NK SW Model Comparison: IRFs to $\epsilon_{MRSS,t}$ shock

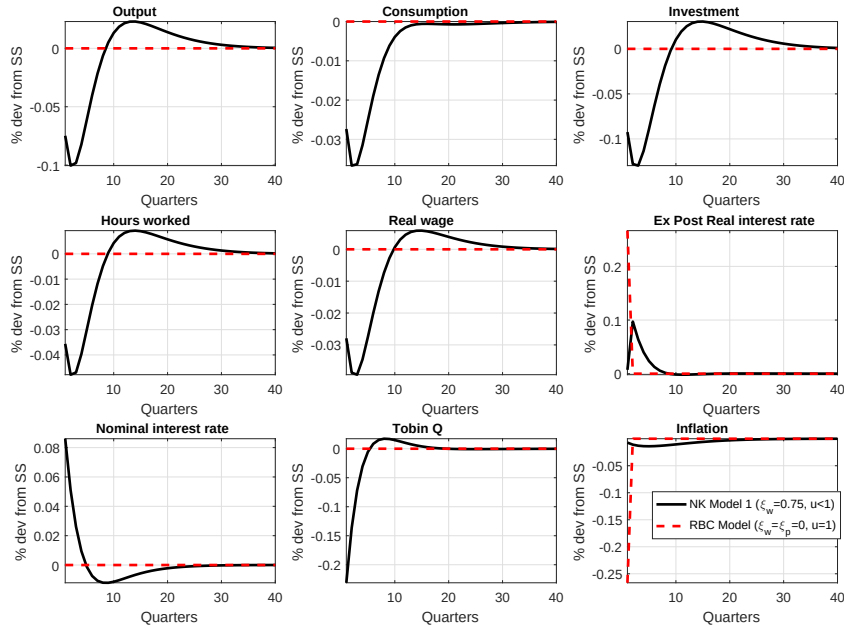
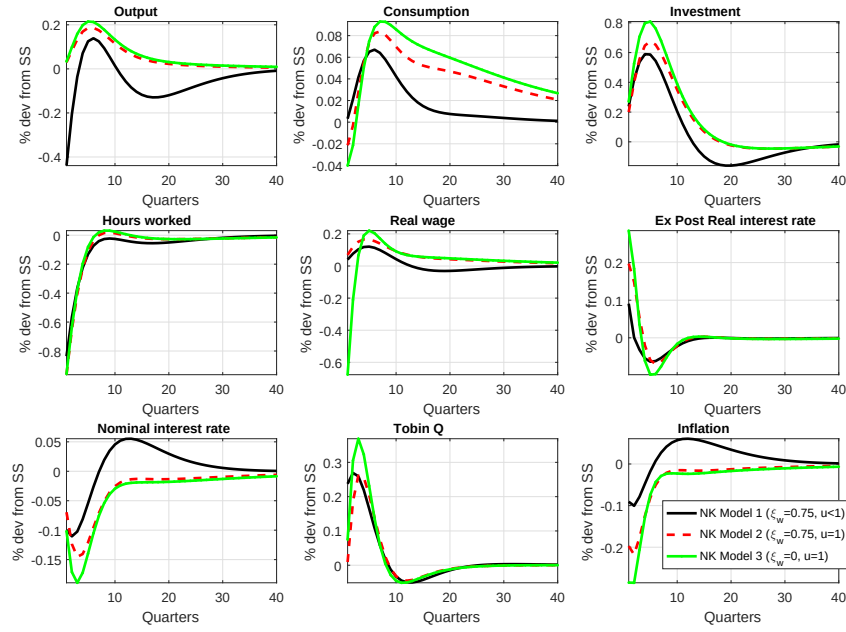
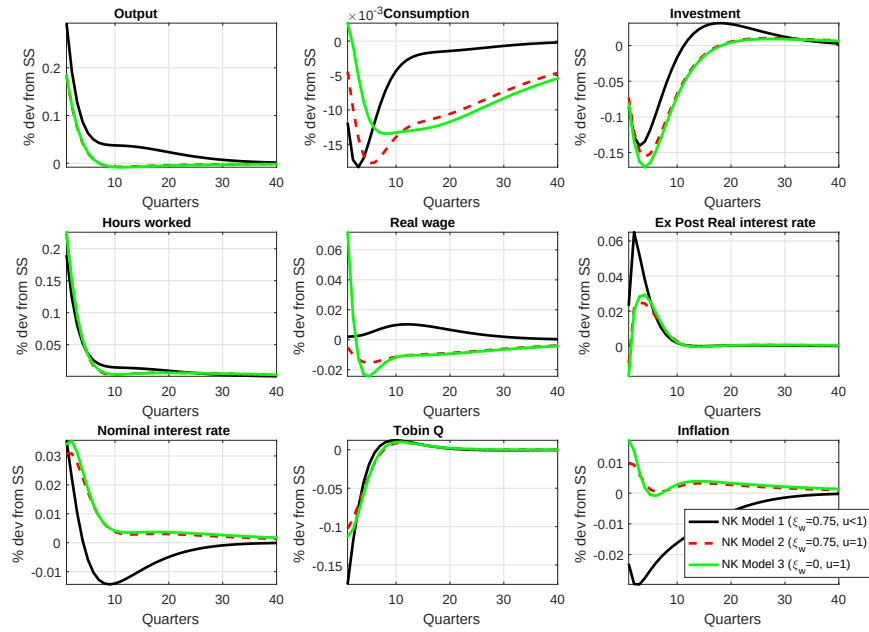
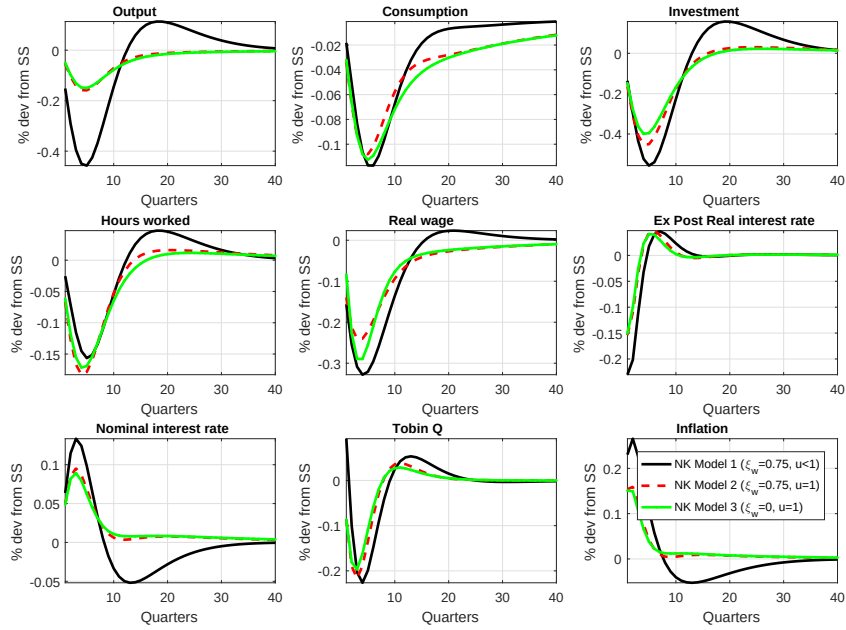
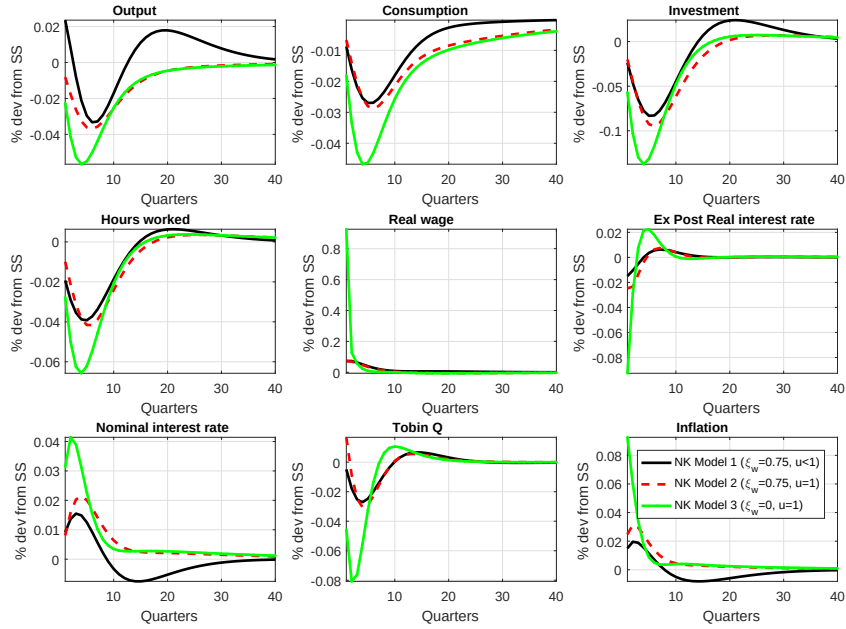
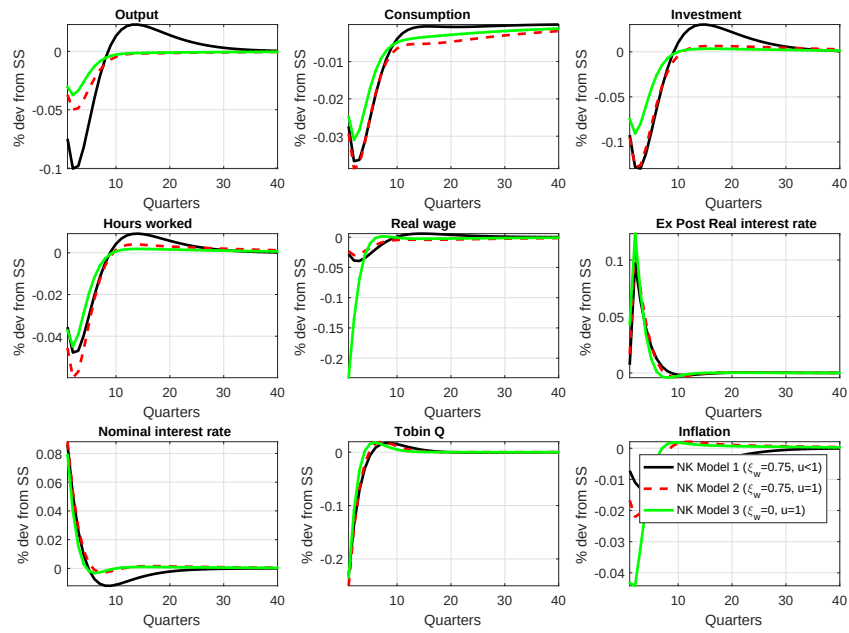


Figure 6.7: RBC-NK SW Model Comparison: IRFs to $\epsilon_{MS,t}$ shock

So sticky prices in the NK model leads to important differences in the response to both supply but especially demand shocks. It is interesting to compare the impact fiscal multipliers at $t = 0$ in the two models. The IRFs are elasticities which for output is $\frac{G\Delta Y}{Y\Delta G} = g_y \frac{\Delta Y}{\Delta G}$. With $g_y = 0.2$, the multiplier $\frac{\Delta Y}{\Delta G}$ as a percentage is therefore $5 \times 100 \times$ the values shown; so for the RBC model the fiscal multiplier is around a 0.38 and rises to about twice that in the NK model.

Figure 6.8: SW Model: IRFs to $\epsilon_{A,t}$ shockFigure 6.9: SW Model: IRFs to $\epsilon_{G,t}$ shock

Figure 6.10: SW Model: IRFs to $\epsilon_{MCS,t}$ shockFigure 6.11: SW Model: IRFs to $\epsilon_{MRSS,t}$ shock

Figure 6.12: SW Model: IRFs to $\epsilon_{MS,t}$ shock

6.7 DYNARE AND MATLAB FILES

In folder NK

- **NKlinear.mod: Linear NK model with habit and indexing options.** With a flexi-price option and therefore set up for both an implementable rule and conventional Taylor rule.
- **NK.mod:** Non-linear NK model with habit, indexing options and price dispersion. No flexi-price option and therefore only set up for an implementable rule.
- **NK_SW.mod:** Full SW07 Non-linear NK model with habit, indexing, capacity utilization, wage/price stickiness options. Indexing can take place in both the steady state and out of the steady state. Full indexing in the steady state eliminates the steady state welfare cost of inflation. No flexi-price option and therefore only set up for an implementable rule.
- **graphs_irfs_compare_NK** Graph plotter for irs of non-linear or linear NK model

6.8 DYNARE EXERCISES

1. Examine the role of habit and indexing in the NKlinear model by comparing impulse responses for zero, medium and high parameter settings of χ and γ .
2. Compare the impulse response functions and the variance decomposition of RBC with external habit and NK models, both with investment costs, for low, medium and high values of the investment cost parameter $\phi_X = 0.1, 2.0, 4.0$.⁸ The easiest way to do this is to first run the NK model with perfect competition in the retail sector ($\zeta = 10000$, a large number) and $\xi = 0$. This is in effect the “RBC model”. Then reset ζ and ξ as before for the NK model. Does the monetary shock have real effects in the RBC case?
3. Compare the impulse response functions of NKlinear with implementable and conventional Taylor rules.

⁸ The comparison of responses of hours to a technology shock relates to the hours-technology debate (see, for example, Galí and Rabanal (2005), Whelan (2009) and Cantore *et al.* (2010))

4. Modify the code **NK.mod** by introducing a preference shock (PS_t) in utility, and an investment specific technology shock (IS_t). Modified equilibrium conditions are:

$$\begin{aligned}
 U_t &= PS_t \frac{((C_t - \chi C_{t-1})^{(1-\varrho)}(1 - H_t)^\varrho)^{1-\sigma_c} - 1}{1 - \sigma_c} \\
 U_{C,t} &= PS_t(1 - \varrho)(C_t - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)-1}((1 - H_t)^\varrho)^{(1-\sigma_c)} \\
 U_{H,t} &= -PS_t\varrho(C_t - \chi C_{t-1})^{(1-\varrho)(1-\sigma_c)}(1 - H_t)^{\varrho(1-\sigma_c)-1} \\
 K_t &= (1 - \delta)K_{t-1} + (1 - S(X_t))I_t IS_t \\
 1 &= Q_t IS_t(1 - S(X_t) - X_t S'(X_t)) \\
 &+ E_t \left[\Lambda_{t,t+1} Q_{t+1} S'(X_{t+1}) X_{t+1}^2 IS_{t+1} \right]
 \end{aligned}$$

5. Discuss the irfs for these extra shocks.
6. Use the options in the code **NK_SW.mod** to explore the effect of introducing wage stickiness into the NK model. Compare impulse responses, volatility, co-movement and the contribution of shocks to the output variance for models with and without wage flexibility.
7. Do the same for external and internal habit.
8. Do the same for models with and without variable capacity utilization.

7

SOLUTION, STABILITY AND INDETERMINACY

All our DSGE models are all special cases of the following general setup recognized by Dynare in non-linear form

$$Z_t = J(Z_{t-1}, X_t) + \nu \epsilon_{t+1} \quad (7.1)$$

$$\mathbb{E}_t X_{t+1} = K(Z_t, X_t) \quad (7.2)$$

where Z_{t-1}, X_t are $(n - m) \times 1$ and $m \times 1$ vectors of backward and forward-looking variables, respectively, and ϵ_t is a $\ell \times 1$ shock variable and ν is an $(n - m) \times \ell$ matrix.

The two methods Dynare uses to solve models are perturbation methods for **stochastic simulations** when the command *stoch_simul* is used and Newton-type methods for **deterministic simulations** when the command *simul* is used. On the Course we only use the former.

7.1 SOLUTION METHOD

To see how the perturbation solution method works define

$$y_t \equiv \begin{bmatrix} Z_t \\ X_t \end{bmatrix} \quad (7.3)$$

Then, as in Dynare User Guide, (7.1) and (7.2) can be written

$$\mathbb{E}_t[f(y_{t+1}, y_t, y_{t-1}, \epsilon_{t+1})] = 0 \quad (7.4)$$

$$\mathbb{E}_{t-1}[\epsilon_t] = 0$$

$$\mathbb{E}_{t-1}[\epsilon_t \epsilon_t'] = \Sigma_\epsilon$$

Solution in Dynare is by *Undetermined Coefficients*. We look for a solution of the form

$$y_t = g(y_{t-1}, \epsilon_t) = 0 \quad (7.5)$$

called the *policy or decision function*. Then we have $y_{t+1} = g(y_t, \epsilon_{t+1}) = g(g(y_{t-1}, \epsilon_t), \epsilon_{t+1})$. Hence (7.4) can be rewritten:

$$\mathbb{E}_t[f(y_{t+1}, y_t, y_{t-1}, \epsilon_{t+1})] = \mathbb{E}_t[F(y_{t-1}, \epsilon_t, \epsilon_{t+1})] = 0 \quad (7.6)$$

Now expand $F(\cdot)$ as a *Taylor series expansion* about a deterministic (possibly trended) steady state defined from (7.4) by

$$f(y, y, y, 0) = 0 \quad (7.7)$$

Define a deviation $\hat{y}_t \equiv y_t - y$ and denote partial derivatives $f_{y+} \equiv \frac{\partial f}{\partial y_{t+1}}$, $g_y \equiv \frac{\partial g}{\partial y_{t-1}}$ etc evaluated at the steady state. Then using $\mathbb{E}_t[\epsilon_{t+1}] = 0$, the *first order Taylor Series expansion* of $F(\cdot)$ yields

$$(f_{y+}g_yg_y + f_yg_y + f_{y-})\hat{y}_{t-1} + (f_{y+}g_yg_\epsilon + f_yg_\epsilon + f_\epsilon)\epsilon_t = 0 \quad (7.8)$$

Now the crucial step! Since (7.8) must hold for any random shock, each term in the brackets $(\cdot)$ above must be equal to zero. Therefore can solve for g_y and g_ϵ yielding the policy function in linearized form

$$y_t = \bar{y} + g_y\hat{y}_{t-1} + g_\epsilon\epsilon_t \quad (7.9)$$

We have now obtained the *first-order perturbation method solution*. Second and higher order approximations can be obtained by higher order Taylor series and are available in Dynare up to order 3. Default is order 2. When you linearize by hand and solve you arrive at the first-order perturbation solution from the non-linear model

The Blanchard-Kahn minimal state-space form considered next can be obtained from the general set-up using the algorithm of Levine and Pearlman (2011). One can express linearized forms of the models in Blanchard-Kahn linear state-space form:

$$\begin{bmatrix} z_{t+1} \\ \mathbb{E}_t x_{t+1} \end{bmatrix} = A \begin{bmatrix} z_t \\ x_t \end{bmatrix} + B r_{n,t} + C \epsilon_{t+1}; \quad o_t = E \begin{bmatrix} z_t \\ x_t \end{bmatrix} \quad (7.10)$$

where z_t is a $(n - m) \times 1$ vector of predetermined variables at time t with z_0 given, x_t is a $m \times 1$ vector of non-predetermined variables and o_t is a vector of outputs. All variables are expressed as absolute or proportional deviations about a steady state. A , B , C and E are fixed matrices and ffl_t as a vector of random zero-mean shocks. Rational expectations are formed assuming a full information set $\{z_s, x_s, \epsilon_s\}$, $s \leq t$, the model and the monetary rule. For example the linearized NK model can be expressed in this form where z_t consists of exogenous shocks, lags in non-predetermined and output variables and capital stock; x_t consists of current inflation, investment, Tobin's Q , consumption and flexi-price outcomes for the latter two variables, and outputs o_t consist

of marginal costs, the marginal rate of substitution between consumption and leisure, the cost of capital, labour supply, output, flexi-price outcomes, the output gap and other target variables for the monetary authority.

Rules so far are of the form

$$r_{n,t} = Dy_t = D \begin{bmatrix} z_t \\ x_t \end{bmatrix} \quad (7.11)$$

where D is constrained to be sparse in some specified way.

7.2 BLANCHARD-KAHN SADDLEPATH STABILITY CONDITIONS

Substituting (7.11) into (7.10), the condition for a stable and unique equilibrium depends on the magnitude of the eigenvalues of the matrix $A + BD$. If the number of eigenvalues outside the unit circle is equal to the number of non-predetermined variables, the system has a unique equilibrium which is also stable with saddle-path $x_t = -Nz_t$ where $N = N(D)$. (See Blanchard and Kahn (1980); Currie and Levine (1993)). Instability occurs when the number of eigenvalues of $A + BD$ outside the unit circle is larger than the number of non-predetermined variables. This implies that when the economy is pushed off its steady state following a shock, it cannot ever converge back to it, but rather finishes up with explosive inflation dynamics (hyperinflation or hyperdeflation).

By contrast, indeterminacy occurs when the number of eigenvalues of $A + BD$ outside the unit circle is smaller than the number of non-predetermined variables. Put simply, this implies that when a shock displaces the economy from its steady state, there are many possible paths leading back to equilibrium, i.e. there are multiple well-behaved rational expectations solutions to the model economy. With forward-looking rules this can happen when policymakers respond to private sector's inflation expectations and these in turn are driven by non-fundamental exogenous random shocks (i.e. not based on preferences or technology), usually referred to as 'sunspots'. If policymakers set the coefficients of the rule so that this accommodates such expectations, the latter become self-fulfilling. Then the rule is unable to uniquely pin down the behavior of one or more real and/or nominal variables, making many different paths compatible with equilibrium (see Chari *et al.* (1998); CGG (2000); Carlstrom and Fuerst (2001) and Carlstrom and Fuerst (2000); Benhabib *et al.*

(2001) and Woodford (2003)). The fact that the rule itself may introduce indeterminacy and generate so called ‘sunspot equilibria’ is of interest because sunspot fluctuations – i.e., persistent movements in inflation and output that materialize even in the absence of shocks to preferences or technology – are typically welfare-reducing and can potentially be quite large.

The Matlab program **indeter.m** loops a .mod file that reports and plots the results of the Blanchard-Kahn determinacy and stability condition for a given model. The file provided call the NK models and examines the condition for different values of coefficients in the Taylor rule.

7.3 DYNARE AND MATLAB FILES

The folder **Indeterminacy** provides an example of exploring the Blanchard-Kahn (BK) conditions our linear NK model in the parameter space of (ρ_r, θ_π) . To carry out this exercise run the Matlab file **indeter** which in turn runs the dynare mod file **NK_indeter.mod** which has been amended to pick out the parameters to be changed and only do the BK check.

7.4 DYNARE EXERCISE

1. Rework the stability analysis with a one-period ahead inflation targeting rule ($j = 1$).
2. Extend this case to $j = 2, 3, 4$. What do you notice?¹

¹ Your results should demonstrate that forward-looking rules introduce indeterminacy unless there is sufficient persistence - see Batini and Pearlman (2002) and Batini *et al.* (2006).

8

BAYESIAN ESTIMATION AND VALIDATION OF THE NK MODEL

Up to now we have used *calibration* to pin down parameters in the model. We have used the deterministic steady state to solve for parameter values that result in observed long-run outcomes for macro-variables such as hours worked, the great ratios (consumption, investment shares given a government spending share) and the real interest rate. This *first-moment matching* can be extended to a second-moment matching of variances, correlations and autocorrelations to calibrate shock processes. Current practice in empirical macroeconomics is to replace this informal moment matching with formal *systems estimation*. Maximum Likelihood (ML), General Method of Moments (GMM) and Bayesian estimation are three widely used systems estimation methods for DSGE models.

We use a Bayesian approach for several reasons. First, these procedures, unlike full information maximum likelihood, for example, allow us to use prior information to identify key structural parameters. In addition, the Bayesian methods employed here utilize all the cross-equation restrictions implied by the general equilibrium set-up, which makes estimation more efficient when compared to the partial equilibrium approaches. Moreover, Bayesian estimation and model comparison are consistent even when the models are misspecified, as shown by Fernandez-Villaverde and Rubio-Ramirez (2004). Finally, this framework provides a straightforward method of evaluating the ability of the models in capturing the cyclical features of the data, while allowing for a fully structural approach to analyze the sources of fluctuations.¹

8.1 MODELS FOR ESTIMATION

For the most part we use the intermediate model with sticky prices and flexible wages in non-linear form **NKlinear_Est.mod**. The steady state of this model has a zero growth and net inflation steady state. Then at the end of this topic

¹ See Fernandez-Villaverde (2009) for an excellent survey.

we show results for the estimation of the full Smets and Wouters (2007) model `SW_NKEst.mod`.

8.2 BAYESIAN METHODOLOGY

Bayesian analysis requires:

- Initial information $\Rightarrow$ Prior distribution
- Data $\Rightarrow$ Likelihood density or the probability of observing the data given the model and parameters
- Prior and Likelihood $\Rightarrow$ Bayes theorem $\Rightarrow$ Posterior distribution
- Posterior distribution used for confidence intervals for parameters and impulse responses.
- The posterior distribution Also provides information regarding identification of parameters -how much information does the data provide on parameters?

Bayesian estimation is a full information systems estimation method (much like ML). It can be thought of as ‘hybrid’ approach between informal calibration and ML. In the absence of prior information it converges to ML and if we are sure the priors are correct we are back to calibration. It uses prior information to identify key structure parameters enabling the utilization of additional sources of information. A major problem with likelihood approaches is that the likelihood surface can be flat (or almost flat) in some directions. Priors then add ‘curvature’ to likelihood. The Bayesian approach allows straightforward facilities for the construction of confidence intervals for parameter estimates and impulse, forecasting and model comparison.

In practice, the Bayesian approach uses the log-linear approximation of the original model’s non-linear optimality conditions around a non-stochastic steady state, obtaining a linear rational expectations system, which is then solved for the state-space form in its predetermined variables. Subsequently, standard Kalman recursions are applied to compute the likelihood function² which, combined with the prior assumptions about model parameters to be estimated, allows us to evaluate their posterior probability.

² The likelihood function is computed under the assumption of normally distributed disturbances.

8.2.1 The Bayesian Approach to Econometrics

Bayesian analysis is based on a few simple rules of probability. First **Some notation**: Suppose A, B are random variables (or events), then

$$\begin{aligned}\text{probability of event A} &\equiv p(A) \\ \text{probability of A and B} &\equiv p(A, B) \text{ or } p(A \cap B) \\ \text{probability of A given B} &\equiv p(A|B) = P(A) \text{ if A, B are independent}\end{aligned}$$

Then by definition of conditional probability

$$p(A|B) \equiv \frac{p(A, B)}{p(B)}$$

Reversing the roles of events A and B, we also have

$$p(B|A) \equiv \frac{p(A, B)}{p(A)}$$

Equating these expressions and rearranging, we arrive at the famous **Bayes' rule**:

$$p(B|A) = \frac{p(A|B)p(B)}{p(A)}$$

In Bayesian econometrics, we want to use data (say T data points $y \equiv (y_1, y_2, \dots, y_T)$) to learn about the model's parameters (say n variables, $\theta \equiv (\theta_1, \theta_2, \dots, \theta_n)$) A Bayesian approach does just that: replacing B by θ and A by y

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

Our focus is on $p(\theta|y)$: given the data, what can we tell about θ ? Whereas classical (frequentist) econometrics treats θ as some unknown fixed value(s), Bayesian econometrics assumes that, if θ is unknown, then it should be expressed using rules of probability (i.e., θ is effectively a random object).

Noting that we're interested in θ , we can drop $p(y)$, so

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

Whereas ML maximizes $p(y|\theta)$ wrt θ ; Bayesian estimation maximizes $p(\theta|y)$. Note that the prior $p(\theta)$ gives the surface more curvature,

The posterior density $p(\theta|y)$ which summarises what we know about θ after (hence *posterior*) seeing the data. $p(y|\theta)$ is the likelihood density given

the model parameters - also denoted as $L(\theta; y)$. $p(\theta)$ is the prior density $p(\theta|y) \propto p(y|\theta)p(\theta)$ is like an updating rule: the data allow us to update our priors about θ , resulting in the posterior, which combines data and non-data information. The likelihood density is computed using the *Kalman Filter* which is a recursive forecasting procedure for the unobserved states given the observables in the linear state space form seen in Topic I (see Miao, chapters 10 and 15).

8.3 COMPUTATION

Our focus is on the posterior distribution of $p(\theta|y)$ that summarise what we know about θ , such as (posterior) means, medians, modes, etc (and respective standard deviations). Knowing this allows Bayesian inference expressed as $E[g(\theta)|y]$, where $g(\theta)$ is a function of interest:

$$E[g(\theta)|y] = \int g(\theta)p(\theta|y)d\theta$$

Bar a few exceptions, it is often impossible to evaluate the integral analytically $\Rightarrow$ simulation methods (Monte Carlo), drawing from the posterior density $p(\theta|y)$. Then as the number of draws (N) increases, then we can invoke the Law of Large Numbers and the Central Limit Theorem

Dynare proceeds through the following steps to arrive at the estimated posterior distribution:

1. Solves the model for a particular parameter vector θ . Currently this is a first-order (linear) solution
2. Evaluates the likelihood density $p(y | \theta)$ using the linear Kalman filter and assuming Gaussian shocks
3. Maximizes $p(y | \theta)$ numerically to arrive at the mode of θ (repeating 1 and 2 each time)
4. Computes an estimate of covariance matrix of the parameters, $\hat{\Sigma}_\theta$ using the result

$$\hat{\Sigma}_\theta = \left(-\frac{\partial^2 \log(p(y | \theta))}{\partial \theta \partial \theta'} \right)^{-1} \quad (8.1)$$

evaluated at the mode (see De Jong and Dave, chapter 15, page 403). The term in the brackets is the Hessian.

5. Output is reported at this stage (the prior mean, the estimated mode, its standard deviation and a t-test). The user can stop here.
6. Proceeds to the computation of the posterior distribution using MCMC

8.3.1 Markov Chain Monte Carlo (MCMC)

The MCMC methods sample from θ wandering over the posterior, taking most draws from high probability areas. The "Markov Chain" bit is as follows: a given draw $\theta^{(s)}$ usually depends on $\theta^{(s-1)}$. It is not easy to draw directly from $p(\theta|y)$ - we need methods that work well for any case $\Rightarrow$ MH MCMC, drawing from a candidate ("transition") distribution. From a starting value, θ_0 , and thus $p(\theta_0)p(y|\theta_0)$ can be evaluated to generate draws from the posterior distribution. The idea here is to specify a transition density for a MC such that, starting from some initial value and iterating a number of times, we produce a limiting distribution which is the target distribution from which we need to sample. The intuition here is that we want to sample from the region with highest posterior probability, but we also want to visit the whole parameter space as much as possible. given that there is a discrepancy between the candidate and target densities, the MCMC will not take the correct draws. The *Metropolis-Hasting* (MH) algorithm corrects this calculating an acceptance probability and eventually discarding some draws.

Because it is difficult to find a good candidate density, we usually employ a *Random Walk* Chain MH algorithm:

$$\theta^* = \theta^{(s-1)} + z$$

The sampler wanders in random directions, thus visiting most of the parameter space. z has a distribution $\alpha(\theta^*|\theta_{s-1})$ that is usually multivariate Normal and here the key choice is its covariance matrix. For each draw i , $\hat{\theta}_i = \theta_{i-1}$ with probability $1 - r$ and $\hat{\theta}_i = \theta_i^*$ with probability r . The acceptance probability of each new draw is then defined by:

$$r = \min \left[\frac{p(\theta_i^*)p(y|\theta_i^*)}{p(\theta_{i-1})p(y|\theta_{i-1})}, 1 \right] \quad (8.2)$$

The **acceptance rate** is dependent on the choice of α . One chooses α to obtain 'reasonable' acceptance rate (by adjusting σ_v^2).

- where $\alpha(\theta_i^*|\theta_{i-1}) \sim N(\theta_{i-1}, \sigma_v^2)$ and $\sigma_v^2 = c * \sigma^2 \Rightarrow$ choose the **scaling parameter** c

- Ideally 20-40% $\Rightarrow$ each move goes a reasonable distance in parameter space, but not so low as to reject too frequently

8.3.2 Illustration of the MCMC Algorithm

To illustrate this, consider a simple example where the parameter may take one of three values $\theta^A, \theta^B, \theta^C$. The likelihood could for example be the result of examining the number r of 'heads' from tossing a biased coin n times, so that $L(r|\theta) = {}_n C_r \theta^r (1 - \theta)^{n-r}$. Clearly the posterior distribution is very simple to calculate, with the three probabilities each proportional to $L(r|\theta^j)$ for $j = A, B, C$.

To examine how the MCMC algorithm works, we assume for simplicity that $\alpha(\theta_i^*|\theta_{i-1}) = 1/2$ i.e. there is an equal probability of jumping to either one of the other θ values from θ_{i-1} . Also assume that $L(r|\theta^A), L(r|\theta^B), L(r|\theta^C)$ are in the ratio $\alpha : \beta : 1 - \alpha - \beta$ where $\alpha < \beta < 1 - \alpha - \beta$. What we will show is that the frequency of the draws of the $\{\theta^j\}$ has the same distribution.

If $\theta_{i-1} = \theta^A$, then since $L(r|\theta^A)$ is the lowest, we will always move so that θ_i will be one of the other θ^j s, and since we sample them with equal probability, the probability that θ_i is one or the other is $1/2$. Similarly if $\theta_{i-1} = \theta^B$, then the probability that $\theta_i = \theta^C$ is $1/2$; however since $\alpha/\beta < 1$, the probability that $\theta_i = \theta^A$ is only $\frac{\alpha}{2\beta}$. Similar complications arise when $\theta_{i-1} = \theta^C$. The Markov chain for the transitions between these 3 choices of θ^j is now easily shown to be

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{\alpha}{2\beta} & \frac{1}{2} - \frac{\alpha}{2\beta} & \frac{1}{2} \\ \frac{\alpha}{2(1-\alpha-\beta)} & \frac{\beta}{2(1-\alpha-\beta)} & 1 - \frac{\alpha+\beta}{2(1-\alpha-\beta)} \end{bmatrix} \quad (8.3)$$

It is now straightforward to show that the steady state distribution of this Markov chain, which satisfies $\pi^T P = \pi^T$ is given by $\pi^T = [\alpha \ \beta \ 1 - \alpha - \beta]$.

8.3.3 Testing for Convergence of MCMC

Testing for convergence of the posterior distribution is notoriously difficult, and Dynare utilises some indicative statistics, summarised by diagrams, as recommended by Brooks and Gelman (1998b). These diagrams are made up of

- 3 multivariate figures, representing convergence indicators for all parameters considered together

- 3 figures for each parameter, representing univariate convergence indicators

The diagnostics in figure 8.1 are generated by the estimation command if `mh_replic` is larger than 2000, `mh_nblocks`=2 and if option `nodiagnostics` is not used.

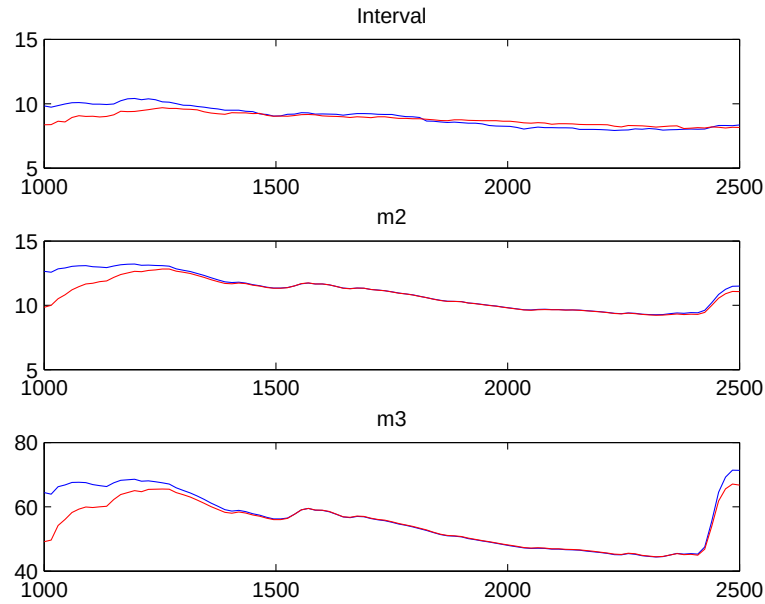


Figure 8.1: Multivariate Convergence Check

As an example, consider the diagnostics for the NK model estimated later. The multivariate diagnostics shown below indicate that the chains converge to similar means and distributions - *Interval* refers to the interval measure, and m_2 , m_3 refer to second and third order multivariate moment measures. However, if one examines the individual parameters, the picture is somewhat mixed. Only for a subset of the parameters is there clear evidence of convergence in mean and distribution. As a minimum requirement, the multivariate diagnostics should be seen to converge to the same values. To ensure convergence of individual parameters is not straightforward as there are no analytic results that ensure convergence; the only analytic result is that if convergence is achieved under MCMC then it is to the correct posterior distribution. Recommended actions to achieve convergence are to increase the number of draws or else to

increase the value of the *scale* parameter in the *estimation* command. The latter ensures that more of the parameter region is searched more regularly, but at the expense of reducing the acceptance ratio. The bottom line is that one often one has to accept that some of the convergence indicators will not always be satisfied.

8.3.4 Summary

Finally, below is a summary of estimation/evaluation algorithm:

1. Compute the steady state (when dealing with the non-linear solutions of the DSGE model, i.e. FOCs, other equilibrium conditions)
2. Construct a linear representation of the model and transform it into a state space framework
3. Specify prior distribution (for Bayesian method): typically the mean of prior is centred around calibrated value. Std. error reflect subjective or objective (to cover the range of existing estimated)
4. Transform the actual data to fit properties of the model (e.g. first-differencing or detrending)
5. Compute likelihood numerically via Kalman filter
6. Compute marginal likelihood for competitors. Bayes factors computed via laplace approximation in large scale system
7. Draw posterior sequences using MH: compute posterior kernel, choose a transition and use a rejection rule.
8. Check for Convergence of MCMC.
9. Construct statistics of interest from the draws (after burned in period): point estimates, credible sets, impulse responses, variance decomposition, forecast, etc.
10. Examine sensitivity of the results to choice of prior.

8.4 IDENTIFICATION ISSUES AND DIAGNOSTICS IN DSGE MODELS

It is necessary to confront the question of parameter identifiability in DSGE models before taking them to the data, as model or parameter identification is a prerequisite for the informativeness of different estimators, and their effectiveness when one uses the models to address policy questions. The sources of identification failure could be marginalisation (from the model structure: i.e. mapping the deep parameters to the reduced form coefficients of the solution and mapping the solution to the population objective function) or lack of information (from the data: i.e. mapping the population to the sample objective) and these notes provide an overview that discusses about the former. Although it is not often addressed in the empirical DSGE literature, that parameter identification is a potentially serious issue for DSGE models is not a new concern. Among the authors who have made this point are Rothenberg (1971), Sargent (1987) and Pesaran (1989). More recently Beyer and Farmer (2004), Iskrev (2008a) and Iskrev (2010b), Canova and Sala (2009a), and Komunjer and Ng (2011) exhibit several identification issues in practice. Using some commonly used models that are unidentifiable or weakly and partially identifiable and are responsible for leading to objective functions with flat surfaces in the economically reasonable portion of the parameter space, they argue that the problem is likely to be pertinent to DSGE models and needs a more careful assessment. Detecting and tackling identification issues is difficult because it is often the case in DSGE models that the mapping from structural parameters into the model reduced form solution is highly nonlinear and the problem is compounded in large-scale models. The current approaches in the literature can mainly be summarised by three branches: 1. classical rank condition approaches that check rank deficiency of gradient matrices based on model-implied moments (e.g. Iskrev (2010b)); 2. objective function approaches focusing on population moments of the data (e.g. Canova and Sala (2009a) and Iskrev (2008a)) and 3. observational equivalence approaches focusing on (non-linear) system matrices in the solution model (e.g. Komunjer and Ng (2011)).

The standard remedy suggested by most empirical DSGE literature is to fix some (potentially non-identifiable) parameters and re-maximise with the parameters that are well-identified. However, this can be problematic if parameters are not fixed at a consistent estimate. Canova and Sala (2009a) argue that flat objective functions lead to serious biases in estimates and that fixing some

of the troublesome parameters at arbitrary values may induce distortions in the distribution of parameter estimates. An alternative is to always use all possible information adding steady states or the covariance matrix of the shocks information, as pointed out by Canova and Sala (2009a), a partial information procedure that uses distance function is likely to have worse identification properties. However a commonly discussed issues using the full information techniques is that since DSGE models are frequently estimated using Bayesian methods, even adding a weakly informative prior can increase the curvature of the likelihood surface. Potential under-identification can remain hidden due to the improper use of priors. The posterior distribution can be well defined as long as the joint prior distribution is proper and the only symptom of poor identification is a posterior distribution looking like the prior. As a result, it can often be unclear to what extent the reported estimates reflect information in the data instead of subjective beliefs indicated by the prior distribution. Another possible solution is to work with higher order approximations instead of linearized models. McManus (1992) shows that identification failures are much rarer in non-linear than in linear set-ups, and argues that using linear approximations is a major cause for poor parameter identifiability in econometrics in general. Though the computational cost of working and estimating non-linear models is higher.

In what follows these notes set out a brief survey on the key literature with regard to the latest advances on identification issues and diagnostics in DSGE models. The review summarises the works done by Canova and Sala (2009a), Komunjer and Ng (2011), Iskrev (2010b) and Ratto and Iskrev (2011)). The main focus is on Iskrev (2010b) and Ratto and Iskrev (2011) who develop a diagnostic tool to detect identification failures that are inherent in the structure of the DSGE model, and not caused by data deficiencies, by evaluating analytically the information matrix of the reduced-form model (Iskrev (2008a)) and checking for rank deficiency of gradient matrix (Iskrev (2010b)). Iskrev's approach has some important advantages compared to the other studies, in particular, to the important contribution made by Canova and Sala (2009a) which exclusively deal with the issues by using numerical approximation of the Hessian from the perspective of the impulse response matching estimation (i.e. a limited information approach) and depend only on the population moments of the data. Evaluating the analytical information matrix, if possible, should be more precise than the numerical differentiation when the model involves highly non-linear functions. The application of Iskrev's approach also aims to characterise parameter identification in a global sense as the analytical evaluation of local identification can be performed over a large number of points

randomly drawn from the parameter space (e.g. prior distribution). Moreover, practical implementation of identification strengthen of DSGE model parameters can be carried out using the Global Sensitivity Analysis (GSA) techniques developed by Ratto (2008a).

8.4.1 Canova and Sala (2009)

Canova and Sala (2009a) focus on the methods that match model and empirical impulse responses and the distance is measured by the objective function and therefore depend on the population moments of the data. The paper makes a clear distinction between *observational equivalence* (2 structural models generate the same IRFs), *under-identification* or *non-identification* (structural parameters may disappear from the log-linearized solution), *weak identification* (insufficient curvature of the likelihood) and *partial identification* (parameters cannot be recovered separately when they enter the objective function).

To formalise the definitions above we use Canova and Sala (2009a)'s notations. If the objective function characterises a DSGE model as a likelihood function $L(y|\theta)$ where $\theta \in \Theta$ is a vector of structural parameters, identification is simply defined as the concept that θ_1 is (locally) identified if $L(y|\theta_1) = L(y|\theta_2)$, for all $y \Rightarrow \theta_1 = \theta_2$ (in Canova and Sala (2009a) the objective is to minimise the weighted distance function between the data-based and model-based impulse responses). In particular, for the objective function that minimises (minus) $L(y|\theta)$, identification is related to the shape and the rank of the information matrix which measures the curvature of the likelihood function. Any curvature deficiencies imply identification failures in θ . In general, assuming an objective function of interest $g(y, \theta)$, observing the data y , the following summarises the various identification difficulties:

- *Observational equivalence*: suppose there are two models m_1 and m_2 for θ_1 and θ_2 , given y , the data, and some θ_1^*, θ_2^* , if $g(y, \theta_1^*) = g(y, \theta_2^*)$ or $\frac{\partial g(y, \theta)}{\partial \theta_1} \big|_{\theta_1^*} = \frac{\partial g(y, \theta)}{\partial \theta_2} \big|_{\theta_2^*} = 0$ and both $\frac{\partial^2 g(y, \theta)}{\partial \theta_1 \partial \theta_1'} \big|_{\theta_1^*}$ and $\frac{\partial^2 g(y, \theta)}{\partial \theta_2 \partial \theta_2'} \big|_{\theta_2^*}$ are positive definite, then the two economic models are observationally equivalent.
- *Under-identification*: let $\theta = [\theta_1, \theta_2]$ and $m_1 = m_2 = m$, if $g(y, \theta_2) = g(y, \theta_1, \theta_2) \forall \theta_1 \in \Theta_1$ where Θ_1 is a subset of the parameter space for θ_1 , in other words, if $\frac{\partial g(y, \theta_1)}{\partial \theta_1} \big|_{\theta_1} = 0$ and $\frac{\partial^2 g(y, \theta_1)}{\partial \theta_1 \partial \theta_1'} \big|_{\theta_1}$ is rank deficient, then θ_1 is (locally) under-identifiable and disappears from the reduced form solution and the mapping to the objective function.

- *Weak identification*: if there exist a unique θ^* such that $g(y, \theta^*) < g(y, \theta), \forall \theta \in \Theta$, then θ is weakly identified which also implies that $\frac{\partial g(y, \theta)}{\partial \theta} |_{\theta^*} = 0$ and that $\frac{\partial^2 g(y, \theta)}{\partial \theta \partial \theta'} |_{\theta^*}$ is positive definite. Even if all the parameters enter the objective function independently and there is a unique optimum, there may be insufficient curvature in a neighborhood of θ^* .
- *Partial identification*: let $\theta = [\theta_1, \theta_2]$, if $g(y, \theta) = g(y, \theta_1 f(\theta_2)) \forall \theta_1 \in \Theta_1, \theta_2 \in \Theta_2$ where $\Theta_{1,2}$ are subsets of the parameter space for $\theta_{1,2}$, in other words, if $\frac{\partial g(y, \theta)}{\partial \theta} |_{\theta} = 0$ and $\frac{\partial^2 g(y, \theta)}{\partial \theta \partial \theta'} |_{\theta}$ is rank deficient then θ is (locally) partially identifiable when the two structure parameters $\theta_{1,2}$ enter the objective function proportionally, making them separately unrecoverable.

The analysis of the paper summarises the following main messages: poor identification leads to serious biases in the search for the optimal from the objective function when the algorithm has difficulties to explore the problem surfaces depending on the choice of minimisation algorithm and/or initial condition. Difficulties in identification combined with estimation often lead to very biased estimates of structure parameters which are inconsistent when estimation and inference is based on the conventional asymptotic distribution and theory and this problem for inference can be magnified with small samples that we typically use in macroeconomics. Calibration may induce distortions in the distribution of parameter estimates so the location of objective function may shift away from the true maximum. All identification difficulties lead to objective functions with large flat surfaces and the problems depend on the objective function used. So as mentioned full information methods are likely to have better identification properties than the limited information ones (e.g. maximum likelihood vs. distance functions). Bayesian methods can help detect or improve identifiability if properly used but may cover up the identification problems as priors convexify the objective function. As Poirier (1998) points out, if the parameter space is not variation free, the prior on the under-identified parameters may be updated because of the parameter constraints imposed by the model.

What can one do to solve the problems? Their main recommendations on detecting potential identification problems: i) less formally but should be costlessly and routinely implemented (before the estimation) is that one can plot the shape of the objective function fixing parameters in turn so that one checks the (local) curvature of the objective one dimension at the time, conditional on the other parameters being fixed at the 'true' values. Then to proceed to a more 'global' test, it is useful to look at the concentration statistics:

$\mathcal{C}_{\theta_0}(i) = \int_{i \neq j} \frac{g(\theta) - g(\theta_0) d\theta}{(\theta - \theta_0) d\theta}$, where i is the number of parameters, over a reasonable range of θ_0 . The interpretation is straightforward, more curvature there is in the objective function less it changes $\mathcal{C}_{\theta_0}(i)$ changing the value of θ_0 and $\mathcal{C}_{\theta_0}(i)$ is closer to 1 for each θ_0 throughout the range for θ_0 (in other words the objective function is not flat in any dimension of θ_0). So a large $\mathcal{C}_{\theta_0}(i)$, exceeding the number of draws taken for θ_0 in the grid range, indicates good curvature in the objective function in a more global sense. This approach is costless and useful to point out parameters responsible for the problems; ii) more formally check the rank of the Hessian of the objective function at or around the estimated parameters, using appropriate tests (Cragg and Donald (1997), Klaibergen and Paap (2006)) that examine whether, when scaled properly, the smallest of the eigenvalues of the Hessian is zero. Hessians (or the sample analogs) that are rank deficient or the magnitude of their smallest eigenvalues are induced by under, partial and weak identification and provide formal statistical evidence of the presence of information deficiencies but are not likely to pinpoint the where particularly the problem is; iii) a constructive way is to work separately with portions/equations of the model to understand source of identification failures. Reasonable estimates could be obtained because of a-priori restrictions (even if the data is uninformative on these parameters), but then one could also reparameterise the model to incorporate the long-run restrictions in the form of a-priori restrictions if the some problem parameters are not separably identified (conditional identification). Again as noted Bayesian Methods can help detect the problem if the priors are properly marginalised, and by implications, can also provide a general diagnostic by checking the sensitivity of varying the prior selections.

8.4.2 Komunjer and Ng (2011)

Canova and Sala (2009a) recommend a simple diagnostic tool that checks the Hessian of the likelihood (objective function) which depends on parameters that may not be consistently estimated. The approach does not provide formal conditions for identification and assumes the gradient of the mapping from the deep to the reduced form parameters to be identified. Komunjer and Ng (2011) focus on formally testing necessary and sufficient conditions for (non-)identification of the reduced form parameters which provide an upper bound for how many free parameters can be estimated. The paper shifts the attention to treating non-identifiability as a population problem that needs to be checked before estimation. Their paper exploits restrictions on observation-

ally equivalent transformations of the model that generates all information in the autocovariances (mean, long-run and other nonlinear restrictions) of the observed variables. The analysis depends only on the system matrices in the solution equations and the rank and the order conditions of the system function form provide the necessary and sufficient conditions for identification of the deep parameters. The following briefly summarises their evaluation procedure and the main findings: Consider the general state space solution form (the canonical $ABCD$ form where all matrices are functions of the parameter vector of interest θ):

$$X_{t+1} = AX_t + B\varepsilon_{t+1} \quad (8.4)$$

$$Y_{t+1} = CX_t + D\varepsilon_{t+1} \quad (8.5)$$

where X_t is the potentially unobservable state vector and Y_t is the vector of the observables. ε_t is the vector of economic shocks and measurement errors impinging on the states and observables. In this case where the D matrix in (1) is square (i.e., the number of observable variables is equal to the number of structural shocks) and D^{-1} exists, the impulse responses from the structural shocks ε_t to Y_t are given by the following VMA representation if the eigenvalues of A in (1) are inside the unit-circle:

$$Y_t = \mathcal{D}(L)\varepsilon_t = \sum_{j=0}^{\infty} d_j L^j \varepsilon_{t-j} \quad (8.6)$$

where L is the lag operator, $\sum_{j=0}^{\infty} \text{trace}(d_j d_j') < +\infty$, $d_0 = D$ and $d_j = CA^{j-1}B$ for $j \geq 1$. The transfer (impulse response) function is:

$$\mathcal{D}(z) = D + C(I_{nX} - A)^{-1}B = \sum_{j=0}^{\infty} d_j L^j z^{-j} \quad (8.7)$$

And the spectral densities:

$$\Omega(z) \equiv \Gamma(0) + \sum_{j=1}^{\infty} \Gamma(j)z^{-j} + \sum_{j=1}^{\infty} \Gamma(-j)z^{-j} = \mathcal{D}(z)\Sigma_{\varepsilon}\mathcal{D}(z_{-1})' \quad (8.8)$$

where nX is the size of X_t , $\Gamma(j) = \Gamma(-j)$ is the autocovariance matrix at j , Σ_{ε} variance-covariance matrix and $\Omega(z)$ is of size $nY \times nY$ and only depends on the reduced form parameters: $\Lambda(\theta) = (A, B, C, D, \Sigma_{\varepsilon})$. The *identification question* is that given $\Omega(z, \theta)$, can we recover the θ that generates $\Omega(z, \theta)$? Based on the state space solution of the model, θ_0 and θ_1 are observational equivalent

if $\Omega(z, \theta_0) = \Omega(z, \theta_1)$ or if $\Gamma(j, \theta_0) = \Gamma(j, \theta_1)$ (equivalence of spectral densities), mathematically, θ_0 is *identifiable* (locally) from the autocovariances of Y_t at the point θ_0 . In other words, using the restrictions implied by equivalent spectral densities (the second moment), θ is identifiable from this spectrum, if the values of θ in a neighborhood are able to be uncovered from a locally unique solution that generate the same model-implied moments for Y_t (local uniqueness).

The main results in this paper are:

- The reduced form parameters of DSGE models $\Lambda(\theta)$ are generally not identifiable (only a subset of $\Lambda(\theta)$ is identifiable from the spectral densities)
- Rank and order conditions for identifying the structural parameters from the observable spectrum of Y_t : let $\Delta(\theta_0)$ be the matrix of partial derivatives of the finite system of nonlinear equations that admits a unique solutions in terms of $\Lambda(\theta)$ evaluated at θ_0 , if the rank of $\Delta(\theta_0)$ remains constant in a neighborhood of θ_0 , then a necessary and sufficient *rank* condition for θ_0 to be locally identified from Γ or Ω is:

$$\text{rank}\Delta(\theta_0) = n_\theta + n_X^2 + n_\varepsilon^2$$

And a necessary order condition is:

$$n_\theta \leq n_Y n_X + n_\varepsilon (n_X + n_Y - n_\varepsilon) + \frac{n_\varepsilon (n_\varepsilon + 1)}{2}$$

where n represents the size of the structure specified in (1) and (2). The paper uses An and Schorfheide (2007)'s model as an example and finds the rank condition fails for the unrestricted model without fixing/restricting any parameters and that has a full-dimensional θ but finds conditional identification under a priori restrictions with a reparameterised model that has an explicit expression for the slope of the Phillips Curve (it is found that the individual components are not separately identifiable).

8.4.3 Iskrev (2010), Ratto and Iskrev (2011)

Komunjer and Ng (2011) focus their analysis on all information in the autocovariances (and other non-linear restrictions) of the observed variables derived from the system of the reduced form solution equations and provide formal

rank and order conditions for identifying the structural parameters from the observable spectrum. The most complete analysis to date in this area is Iskrev (2010b) which also performs formal identification checks on the reduced form parameters and structure or deep parameters, provide a computational toolbox for local and global identification tests as well as for testing the strength of identification that is useful for detecting parameters that are weakly identified. The procedures put forward in Iskrev (2010a), Iskrev (2010b) and Ratto and Iskrev (2011) are the focus of these notes.

DSGE models consist of a system of non-linear equations involving a vector $\mathbf{z}_t$ of endogenous variables, a $\mathbf{u}_t$ vector of random structural shocks and a $\theta \in \Theta$ k -dimensional vector of deep parameters. Most applications use a linear approximation of the original model, i.e. expressing the model in terms of stationary variables linearized around their steady-states. The unique solution (if it exists), can be expressed as

$$\mathbf{z}_t = \mathbf{A}(\theta)\mathbf{z}_{t-1} + \mathbf{B}(\theta)\mathbf{u}_t \quad (8.9)$$

where $\mathbf{A}$ and $\mathbf{B}$ are functions of θ . For later use, and using Iskrev (2010b) notation, denote τ as the vector collecting all the reduced-form coefficients from the DGSE model, i.e. the elements in $\mathbf{A}$, $\Omega = \mathbf{B}\mathbf{B}'$ and the steady-state of $\mathbf{z}_t$ that depend on θ , respectively $\tau = [\tau_z', \tau_A', \tau_\Omega']$.

A condition for identification is that distinct values of θ imply distinct values of the probability density function of the data, as the latter contains all available sample information about the value of the parameter vector of interest θ . Usually, the distribution of the data $\mathbf{X}$ is unknown or assumed to be Gaussian and estimation of θ is based on the first two moments of the data. If $\mathbf{X}$ is not normally distributed, higher-order moments may provide additional information about θ , not contained in the first two moments. Define $\mathbf{m}_T := [\mu', \sigma_T']$ as the vector collecting the first and second order moments of the observable variables. Identification based on the mean and the variance of $\mathbf{X}$ is only sufficient but not necessary for identification with the complete distribution, so that the mapping from the population moments of the sample, $\mathbf{m}_T$, to θ is unique.

Global identification cannot, in general, be established for unique solutions of systems of non-linear equations, but local identification can be verified by means of a rank condition of the Jacobian matrix $J_T = \frac{\delta \mathbf{m}_T}{\delta \theta'}$. Indeed, θ_0 is said to be locally identified if J_T has full column rank when evaluated at θ_0 , although this does not guarantee that the model is locally identified everywhere in the parameter space. However, studying the rank of J_T is helpful, as local identifiability ensures consistent estimation of the parameters of interest and, moreover,

it will help to pinpoint the parameters that cause identification problems. For example, the column of J_T corresponding to an unidentified parameter θ_j will be a vector of zeros and the rank condition will fail. Another possibility may occur when the columns of J_T are not linearly independent, due to parameters entering the solution in a way that makes them indistinguishable (partial identification).

While numerical differentiation could be used to obtain J_T , this is unreliable when the model is highly non-linear. Instead, Iskrev (2010b) proposes using analytical derivatives, employing implicit derivation, breaking down the mapping from θ to $\mathbf{m}_T$ in two steps: i) a transformation of θ to τ , ii) a transformation from τ to $\mathbf{m}_T$. The Jacobian can then be expressed as

$$J_T = \frac{\delta m_T}{\delta \tau'} \frac{\delta \tau'}{\delta \theta'} \quad (8.10)$$

The first term $J_1 = \frac{\delta m_T}{\delta \tau'}$ may be obtained by direct differentiation. Regarding the second term, Iskrev (2010b) establishes a necessary condition for identification: the point θ_0 is locally identifiable only if the rank of $H_T = \frac{\delta \tau'}{\delta \theta'}$ evaluated at θ_0 is equal to k . The second term can be split into the three components of τ (Ratto and Iskrev (2011) provide details on the computation and implementation of these analytical derivatives in Dynare).³ Note that H_T does not depend on the data, thus implying that it is possible to detect lack of identification, inherent to the structure of the DSGE model, *before* taking the model to the data. On the other hand, the rank of J_T will provide information on the identification θ given the set of observable variables and the sample size.

The above suggests a procedure based on Monte Carlo exploration of the parameter space Θ of model parameters. The local analysis is performed in turn using the Monte Carlo realisations. This provides a ‘global’ prior exploration of point identification properties of DSGE models. One starts by constructing a sample drawn from Θ (discarding values that do not ensure stability and determinacy). This step can be guided by use of priors, specifying a theoretically admissible range and/or a particular distribution for θ . The identifiability of each draw for θ_j is then established by studying the rank of J_T and H_T , resorting to the necessary and sufficient conditions enumerated in Iskrev (2010b):

- if H_T is rank-deficient at θ_j , this particular point is unidentifiable (full rank of H_T is necessary for identification)

³ This includes the case of non-linear DSGE models, for which the derivative of τ_z may not be obtained by direct differentiation, unlike linear models.

- if H_T has full rank but J_T does not, then θ_j cannot be identified for the particular set of observables and contemporaneous and lagged moments under consideration, i.e. given X and T
- weak identification issues when the columns of J_T and H_T are nearly linearly dependent (multicollinearity analysis of the re-scaled Jacobians)

Thus, a given parameter θ_j may be poorly identified because it has little impact on the reduced-form coefficients of the model ($\frac{\delta \tau}{\delta \theta_j} \approx 0$ or because its impact on the reduced-form coefficients is approximately to a linear combination of $\frac{\delta \tau}{\delta \theta_i}$ of other parameters.

Performing identification analysis in Dynare along the lines of Iskrev (2010b) and Ratto and Iskrev (2011) can be carried out in ‘standard’ or ‘advanced’ modes (providing a more detailed identification analysis, which however is beyond the scope of this discussion). The ‘point estimate’ mode is the default option in Dynare, in which local identification checks are done for the whole set, or only a subset, of the parameters in the model at a chosen central tendency measure, either at the defined prior or the estimated posterior, or at the calibrated value if no prior is declared. If priors have been defined, a Monte Carlo exploration is also available, in which the identification checks are based on samples from the prior distribution. The routines are triggered by simply using the command *identification*, which runs a check on the rank of J_T and H_T based on the priors and a list of observables defined by the user. Note that this can be carried out even before model estimation. A list of (*< options >= < values >*) can be used to control Monte Carlo replications, see Ratto and Iskrev (2011) for more details. The output of this procedure then reveals whether or not there are identification problems, stemming from J_T and/or H_T , as illustrated below for the benchmark NK model we estimate in this Course Note:

Our model has 4 shocks and 3 observables. The RE solution cannot be invertible (see Fernandez-Villaverde *et al.* (2007)) and the standard perfect information assumption on the part of agents is inconsistent with the information of the econometrician. (See Levine *et al.* (2019)). this results in a warning in the first Komunjer and Ng test. We return to this issue when we come to the Exercises.

```
==== Identification analysis ====
```

```
Testing prior mean
```

```
WARNING: Komunjer and Ng (2011) failed:
```

```
There are more shocks and measurement errors
than observables this is not implemented (yet).
Skip identification analysis based on minimal state space system.
The number of moments with non-zero derivative is smaller
than the number of parameters
Try increasing ar = 2
```

```
REDUCED-FORM:
```

```
All parameters are identified in the Jacobian of steady state
and reduced-form solution matrices
(rank(Tau) is full with tol = robust).
```

```
SPECTRUM (QU AND TKACHENKO, 2012):
```

```
All parameters are identified in the Jacobian of mean and
spectrum (rank(Gbar) is full with tol = robust).
```

```
MOMENTS (ISKREV, 2010):
```

```
All parameters are identified in the Jacobian of first two moments
(rank(J) is full with tol = robust).
```

```
Testing prior mean
```

```
All parameters are identified in the model (rank of H).
```

```
All parameters are identified by J moments (rank of J)
```

The upper block shows the local point identification check at the prior mean and the second is a Monte Carlo testing using 250 draws from the prior space (*identification(prior_mc=250)*). If the rank condition fails in any of these two procedures, the procedure indicates which parameters are responsible for identification problems. In particular, it reports the associated parameter(s) that are responsible for failing the column rank condition in the J_T and/or H_T and the

parameter(s) with pair-wise and multi-correlation coefficients for each column of the Jacobian matrix equal to unity.

A further output of the *identification* routine is the analysis of identification ‘strength’, i.e., focusing on weak identification, summarized in two additional plots. The procedures are based on either the asymptotic or a moment information matrix. The first can be obtained given a sample of size T , whereas the second can be computed based on Monte Carlo simulations for samples of size T , from which sample moments of the observed variables are computed, forming a sample of N replicas of simulated moments. The corresponding information matrix is then obtained as $I_T(\theta|\mathbf{m}_T) = H_T \Sigma_{\mathbf{m}_T} H_T$, where $\Sigma_{\mathbf{m}_T}$ is the covariance matrix of simulated moments.

The ‘strength’ of identification for θ_i is computed as

$$s_i = \sqrt{\theta_i^2 / I_T(\theta)_{(i,i)}^{-1}} \quad (8.11)$$

which works like a t-test for θ_i . This can be decomposed into a ‘sensitivity’ and ‘correlation’ parts, the first referring to the case when weak identification arises when the moments do not change with θ_i and the second when collinearity dampens the effect of θ_i . The first referring to the case when weak identification arises when the moments do not change with θ_i and the second when collinearity dampens the effect of θ_i . The former is defined as

$$\Delta_i = \sqrt{\theta_i^2 \cdot I_T(\theta)_{(i,i)}} \quad (8.12)$$

which can also be normalised relative to the prior standard deviation for θ_i : $\sigma(\theta_i)$, weighting the information matrix using the prior uncertainty:

$$\Delta_i^{prior} = \sigma(\theta_i) \cdot \sqrt{I_T(\theta)_{(i,i)}} \quad (8.13)$$

This is particularly useful to distinguish the cases where (8) and (9) are singular because $\theta_i \approx 0$. It is possible to show the standard error of a parameter:

$$s.e.(\theta_i) = \frac{1}{\Delta_i} \frac{1}{\sqrt{1 - \varrho_i^2}} \quad (8.14)$$

where ϱ_i denotes collinearity between the effects of different parameters so that lack of identification and a flat likelihood may be due to either $\Delta_i = 0$ or $\varrho_i = 1$.

Thus, the *identification* command plots the measures described above (also normalised relative to the prior standard deviation for θ_i), where large bars

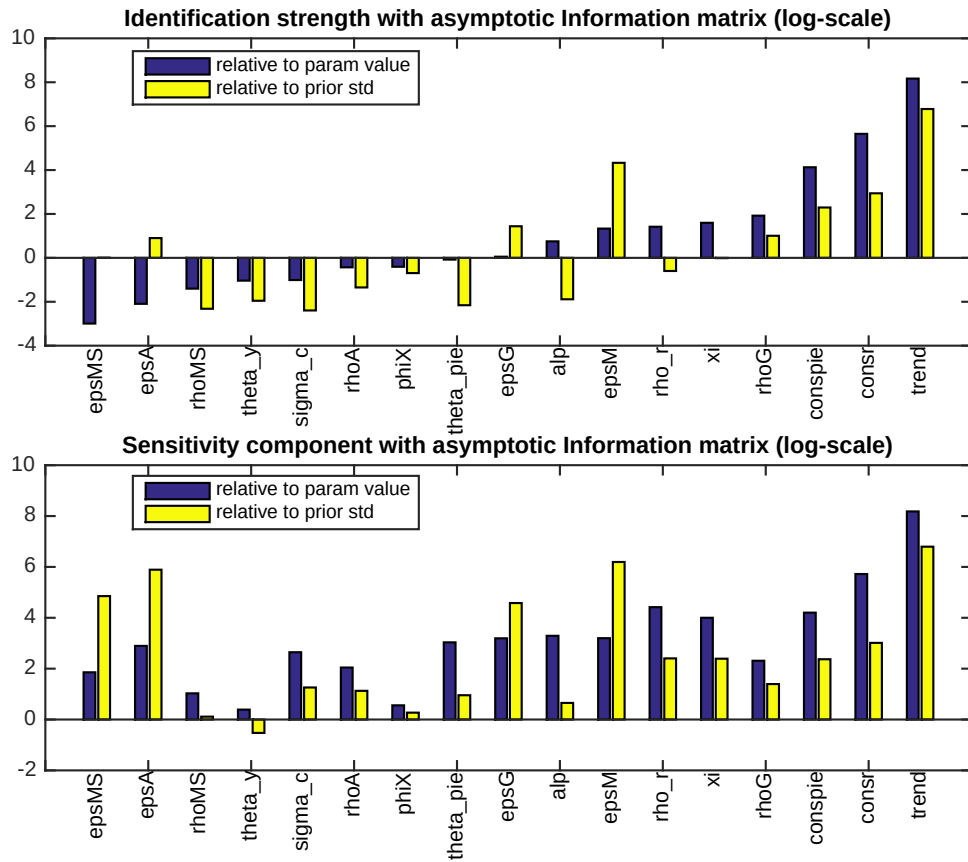


Figure 8.2: Identification Strength in NK Model

in absolute value (in the log-scale plot) imply strong identification, for the respective θ_i . The following Figures 8.2 shows the identification strength and sensitivity component in the moments for the above NK model.

In the upper panel the bars depict the identification strength of the parameters based in the Fisher information matrix normalized by either the parameter at the prior mean (blue bars) or by the standard deviation at the prior mean (yellow bars).⁴ Intuitively, the bars represent the normalized curvature of the log likelihood function at the prior mean in the direction of the parameter. If the strength is 0 (for both bars) the parameter is not identified as the likelihood function is flat in this direction. The larger the absolute value of the bars,

⁴ The weighting with the prior standard deviation is particularly useful for cases where the prior mean is 0. In this case the weighting with the prior mean would falsely signal an identification strength of 0.

the stronger the identification. The lower panel decomposes further the effect shown above. A weak identification can be due to either other parameters having exactly the same effect on the likelihood or that the likelihood does not change at all with the respective parameter. This latter effect is the sensitivity effect (see Ratto and Iskrev (2011)) and is plotted in the bottom panel. Again, the weighting that place either with the prior mean (blue) or the prior standard deviation (yellow).

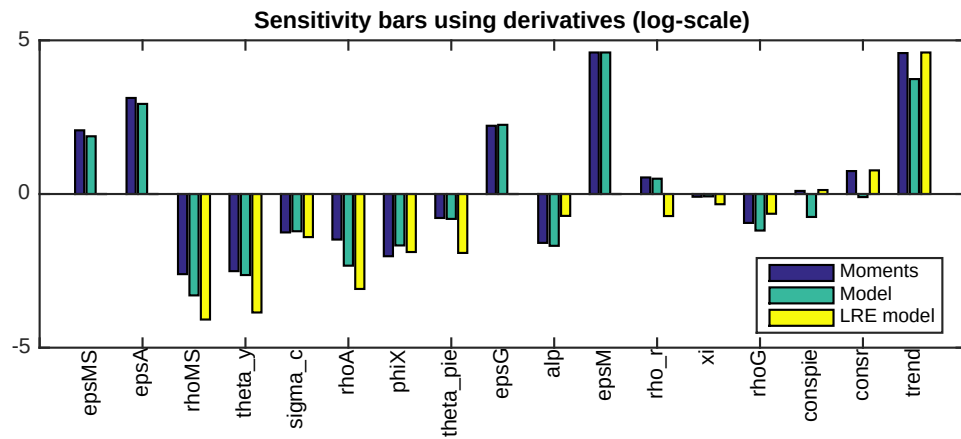


Figure 8.3: Identification Strength in NK Model

Figure 8.3 shows an aggregate measure of how changes in the elements of the parameter vector θ impact on the model moments. The impact is measured locally using the Jacobian. The problem is that the derivatives are not scale invariant so not easily comparable. For this reason here Dynare uses a normalization/standardization procedure described in Ratto and Iskrev (2011). Dynare plots here three different measure of sensitivity. The bars depict the

norm of the columns of three different standardized Jacobian matrices for the respective parameter shown on the x-axis. The respective Jacobian refer to i) the moments matrix, indicating how well a parameter can be identified due to the strength of its impact on moments, ii) the solution matrices, indicating how well a parameter could in principle be identified if all state variables were observed, and iii) the Linear Rational Expectation (LRE) model, indicating trivial cases of non-identifiability due for example to the fact that some parameters always show up as a product in the model equations.⁵

To completely rule out a flat likelihood at the local point one can also check collinearity between the effects of different parameters on the likelihood. If there exists an exact linear dependence between a pair and among all possible combinations their effects on the moments are not distinct and the violation of this condition must indicate a flat likelihood and lack of identification. From high correlations to near-exact collinearity one may suspect some weak identification (due to partial identification). The details of collinearity analysis require the advanced analysis option (*advanced* = 1)) which prints the results of the brute force search for the groups of parameters whose columns in the Jacobian matrix best explain each column of the Jacobian (i.e. best reproducing the behaviour of each single parameter). The default maximum dimension of the group is 2 and the dimensionality is controlled by the option (*max_dim_cova_group* = *INTEGER*). For more information on parameter similarities and high correlation Figure 8.4 plots collinearity patterns for the critical collinearities (i.e. dark yellow spots for higher correlation).

Collinearity patterns with 1 parameter(s)

Parameter	[Expl. params]	cosn
epsA	[epsMS]	0.9725474
epsG	[rhoG]	0.9541539
epsM	[theta_pie]	0.9863971
epsMS	[epsA]	0.9725474
rhoA	[rhoMS]	0.9886306
rhoG	[epsG]	0.9541539
rhoMS	[rhoA]	0.9886306
phiX	[theta_y]	0.8243693
sigma_c	[theta_y]	0.7693756
xi	[alp]	0.7911396

⁵ This does not apply to the estimation of exogenous shocks.

The following box prints the syntax for the default identification procedures in DYNARE: point identification at the prior mean, the MC exploration using the draws from the prior distribution and identification strength measured at the mean and weighted by the prior standard deviation.

```

estimated_params;
stderr epsA,      INV_GAMMA_PDF,0.10, 2.00;
stderr epsG,      INV_GAMMA_PDF,0.50, 2.00;
stderr epsM,      INV_GAMMA_PDF,0.10, 2.00;
stderr epsMS,     INV_GAMMA_PDF,0.10, 2.00;
rhoA,             BETA_PDF,      0.50, 0.20;
rhoG,             BETA_PDF,      0.50, 0.20;
rhoMS,           BETA_PDF,      0.50, 0.20;
phiX,            NORMAL_PDF,    2.00, 1.50;
sigma_c,         NORMAL_PDF,    1.50, 0.375;
chi,             BETA_PDF,      chi, 0.10;
xi,             BETA_PDF,      0.50, 0.10;
gammap,         BETA_PDF,      gammap, 0.15;
alp,            BETA_PDF,      0.70, 0.05;
theta_pie,      NORMAL_PDF,    2.00, 0.25;
rho_r,          BETA_PDF,      0.75, 0.10;
theta_y,        NORMAL_PDF,    0.125,0.05;
conspie,        GAMMA_PDF,     0.625,0.10;
trend,          NORMAL_PDF,    0.40, 0.10;
consr,          NORMAL_PDF,     1.5,0.1;
end;

varobs dy pinfobs robs;
identification;
//identification(advanced=1);

```

8.5 ESTIMATION OF NK MODEL

We estimate the NK model set out in chapter 6 around zero steady state inflation using the Bayesian methodology. The mod file for this analysis is

NKlinear_Est.mod which provides results from posterior optimization and provides posterior distribution from posterior simulation – MCMC. In Dynare, observed variables are declared after *varobs* (i.e. *varobs VARIABLE_NAME...*;) and must be available in the data file. Estimated parameters are declared in the *estim_params; ... end;* block. For each estimated parameter, declare the initial value and, optionally, a lower and upper bound for the ML estimation. If their prior distributions are further declared, Dynare chooses to perform the Bayesian estimation. Computing the estimation is triggered by the keyword *estimation* and the only required option in brackets after *estimation* is *datafile=FILENAME*.

We use the same data set as in Smets and Wouters (2007) in first difference at quarterly frequency. Namely, these observable variables are the log difference of real GDP, the log difference of the GDP deflator and the federal funds rate. All series are seasonally adjusted. Since the variables in the model state space are measured as deviations from a constant steady state, we take the first difference of the real GDP in order to obtain DSP.

The raw data are stored in **raw_data.mat** in which for real variables we take the log of the original data. Inflation and nominal interest rates are used as they are in percentage terms. The original data are taken from the FRED Database available through the Federal Reserve Bank of St.Louis. The sample period is 1984:1-2008:2 which starts at observation 144 in the data file. There is a pre-sample period of 4 quarters so the observations actually used for the estimation go from 148:245. A full description of the data used can be found in Appendix 6 and Smets and Wouters (2007).

We estimate the linear NK model using SW data with output in first-differences. Observables are: output, inflation and interest rate. The corresponding measurement equations for the 3 observables are:

$$\begin{bmatrix} D(\log GDP_t) * 100 \\ \log(GDPDEF_t / GDPDEF_{t-1}) * 100 \\ FEDFUNDS_t / 4 * 100 \end{bmatrix} = \begin{bmatrix} Y_t - Y_{t-1} + \text{trend growth} \\ \Pi_t + \text{constant}_\Pi \\ R_{n,t} + \text{constant}_{R_n} \end{bmatrix}$$

Note that the quarterly trend growth rate in real GDP; the quarterly steady-state inflation rate and the steady-state nominal interest rate are estimated together with the other parameters.

It is possible to estimated the non-linear model directly and allow Dynare to perform the linearization. The corresponding measurement equations for the 3 observables are then:⁶

$$\begin{bmatrix} D(\log GDP_t) \\ \log(GDPDEF_t/GDPDEF_{t-1}) \\ FEDFUNDS_t/4 \end{bmatrix} = \begin{bmatrix} \log\left(\frac{Y_t}{\bar{Y}_t}\right) - \log\left(\frac{Y_{t-1}}{\bar{Y}_{t-1}}\right) + \text{trend growth} \\ \log\left(\frac{\Pi_t}{\bar{\Pi}}\right) + \text{constant}_{\Pi} \\ \log\left(\frac{R_{n,t}}{\bar{R}_n}\right) + \text{constant}_{R_n} \end{bmatrix} \quad (8.15)$$

In a stochastic framework, Dynare computes a Taylor approximation around the computed steady state of order one or two (for a non-linear model). In particular, Dynare computes approximated (Taylor) decision rules and transition equations (the equations listing current values of the endogenous variables of the model as a function of the previous state of the model and current shocks) for the model by a perturbation method (i.e. perturbation method: recovering a Taylor expansion of the solution function from a Taylor expansion of the original model.). For a log-linear model, a first order approximation is nothing else than a standard solution through linearization and a first order approximation in terms of the logarithm of the variables provides standard log-linearization.

A few structural parameters are kept fixed or calibrated based on some parameters being estimated in the estimation procedure, in accordance with the usual practice in the literature (see Table 8.1). The choice of priors for the estimated parameters is usually determined by the theoretical implications of the model and evidence from previous studies. In general, inverse gamma distributions are used as priors when non-negativity constraints are necessary, and beta distributions for fractions or probabilities. Normal distributions are used when more informative priors seem to be necessary.

Note that in non-linear estimation Dynare solves and updates the steady state at each MCMC realization given the parameter draws. In consequence, the implied steady state relationships reported in Table 8.1 are no longer kept fixed in this setup (in a special case where χ and H are instead imposed in the (log-)linear setup, it is straightforward to show that the linear estimates and the non-linear estimates are completely consistent).

In order to avoid stochastic singularity when evaluating the likelihood function, Dynare requires at least as many shocks or measurement errors in the models as observable variables (i.e. requires that the covariance matrix of endogenous variables is nonsingular). In this estimation an additional structural

⁶ $Y_t = GDP_t$, $\bar{Y}_t = \text{trend}$ and $\text{trend growth} = \log \bar{Y}_t - \log \bar{Y}_{t-1} = \text{constant}$.

Calibrated parameter	Symbol	Value
Discount factor	β	0.99
Depreciation rate	δ	0.025
Growth rate	g	0
Substitution elasticity of goods	ζ	7
Fixed cost	c	$\frac{1}{\zeta} = 0.1429$
Preference parameter	ϱ	$\frac{(1-H)\alpha}{(1-H)\alpha + c_y H(1-\chi)}$
Implied steady state relationship		
Government expenditure-output ratio	g_y	0.2
Investment-output ratio	i_y	$\frac{(1-\alpha)\delta}{(1/\beta - 1 + \delta)}$
Consumption-output ratio	c_y	$1 - g_y - i_y$

Table 8.1: Calibrated Parameters

shock is included to capture to some extent aggregation effects (e.g. monetary policy shock) and there is no measurement error in the data set without facing stochastic singularity.

In the process of parameter estimation, the mode of the posterior is first estimated using Chris Sim's `csmnwel`⁷ after the models' log-prior densities and log-likelihood functions have been obtained by running the Kalman recursion and maximized (this is triggered by the option `mode_compute=4`). Then a sample from the posterior distribution is obtained with the Metropolis-Hastings (MH) algorithm using the inverse Hessian at the estimated posterior mode as the covariance matrix of the jumping distribution. The covariance matrix needs to be adjusted in order to obtain reasonable acceptance rates. Thus the scale used for the jumping distribution in the MH is set to 0.40 (option `mh_jscale=0.40`), allowing good acceptance rates (around 20%-30%). Two parallel Markov chains of 100000 runs each are run from the posterior kernel for the MH (`mh_replic=INTEGER`: sets the number of draws - longer chains are more likely to have converged⁸). The first 25% of iterations (initial burn-in period) are discarded in order to remove any dependence of the chain from its starting values (use `mh_drop`: sets the percentage of discarded draws).

The following list summarizes the options used in the estimated models based on the Dynare user manual (e.g. **NKlinear_Est.mod**):

⁷ See, for more details, Chris Sim's homepage: <http://www.princeton.edu/~sims/>

⁸ In this example, the univariate diagnostic statistics produced by Dynare indicate convergence by comparing between and within moments of multiple chains (Brooks and Gelman (1998a)).

- *prefilter*: a value of 1 means that the estimation procedure will demean the data
- *first_obs*: specifies the first observation to be used
- *mode_compute*: specifies the optimizer. For example: 0: switch mode computation off; 1: fmincon; 4: csminwel; 6: Monte-Carlo based optimization; 9: CMA-ES (Covariance Matrix Adaptation Evolution Strategy). On the Course we mostly use mode compute 4 using an initial from a closely related model. But the first estimation of NKlinear.mod uses mode compute 6 which always constructs a negative definite Hessian. This takes a lot time and the file is provided.
- *mh_replic*: sets the number of replications for MH algorithm - longer chains are more likely to have converged
- *bayesian_irf*: computes posterior distribution of IRFs
- *mh_drop*: sets the percentage of discarded draws
- *mh_jscale*: specifies the scale to be used for the jumping distribution in MH algorithm
- *presample*: number of initial periods that do not enter into likelihood computation (to initialize Kalman)
- *mode_file=FILENAME_mode*: reloads the computed posterior mode (stored in FILENAME_mode.mat in the working dir.)
- *load_mh_file*: recovers the estimates using the existing MH replications
- *mode_check*: plots the objective function around the computed mode
- *forecast = INTEGER*: posterior distribution of out-of-sample forecast

The estimation outputs report the Bayesian inference which summarizes the prior distribution - posterior mean and 95% confident interval. The marginal data density of the model is computed using Geweke (1999) modified harmonic-mean estimator. The outputs are displayed on the MATLAB Command Window:

ESTIMATION RESULTS

Log data density is -67.215505.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
rhoA	0.500	0.8231	0.7320	0.9147	beta	0.2000
rhoG	0.500	0.3889	0.0537	0.7197	beta	0.2000
rhoMS	0.500	0.9739	0.9482	0.9985	beta	0.2000
hss	0.400	0.4035	0.1714	0.6582	beta	0.1000
phiX	2.000	1.9283	0.1240	3.3399	norm	1.5000
sigma_c	2.000	1.8315	1.0560	2.5972	gamm	0.5000
chi	0.700	0.8413	0.7292	0.9559	beta	0.1000
xi	0.500	0.7879	0.7160	0.8608	beta	0.1000
gammap	0.500	0.2305	0.0715	0.3780	beta	0.1500
alp	0.700	0.6772	0.5931	0.7605	beta	0.0500
theta_pie	2.000	2.1258	1.7597	2.4636	norm	0.2500
rho_r	0.750	0.8177	0.7754	0.8614	beta	0.1000
theta_y	0.125	0.0087	-0.0280	0.0411	norm	0.0500
conspie	0.625	0.6772	0.5448	0.8145	gamm	0.1000
trend	0.400	0.5163	0.3617	0.6623	norm	0.1000
consr	1.500	1.3760	1.2323	1.5112	norm	0.1000

standard deviation of shocks

	prior mean	post. mean	90% HPD interval		prior	pstdev
epsA	0.100	1.3531	0.7778	1.8989	invga	2.0000
epsG	0.100	1.1763	0.6156	1.9139	invga	2.0000
epsM	0.100	0.1571	0.1338	0.1794	invga	2.0000
epsMS	0.100	2.7650	1.4163	4.3266	invga	2.0000

8.6 BAYESIAN MODEL COMPARISONS

First let us go back to Bayes Rule

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} \propto p(y|\theta)p(\theta)$$

So far we have not needed the unconditional density $p(y)$ to maximize $p(\theta|y)$ wrt θ . This is computed by integrating over the prior distribution to obtain

$$p(y) = \int_{\Theta} p(y|\theta)p(\theta)d\theta$$

For a particular model i from a number of alternatives, say m_i , we can define a density conditional on this model

$$p(y|m_i) = \int_{\Theta} p(y|\theta, m_i)p(\theta, m_i)d\theta$$

where $p(\theta, m_i)$ is the prior for that model. We refer to $p(y|m_i)$ as the *marginal likelihood* associated with model m_i .

Bayesian inference also allows a framework for comparing alternative and potentially misspecified models based on their marginal likelihood. For a given model $m_i \in M$ and common dataset, the latter is obtained by integrating out vector θ ,

$$p(y|m_i) = \int_{\Theta} p(y|\theta, m_i) p(\theta|m_i) d\theta$$

where $p_i(\theta|m_i)$ is the prior density for model m_i , and $p(y|m_i)$ is the data density for model m_i given parameter vector θ . To compare models (say, m_i and m_j) we calculate the posterior odds ratio which is the ratio of their posterior model probabilities (or Bayes Factor when the prior odds ratio, $\frac{p(m_i)}{p(m_j)}$, is set to unity):

$$PO_{i,j} = \frac{p(m_i|y)}{p(m_j|y)} = \frac{p(y|m_i)p(m_i)}{p(y|m_j)p(m_j)} \quad (8.16)$$

$$BF_{i,j} = \frac{p(y|m_i)}{p(y|m_j)} = \frac{\exp(LL(y|m_i))}{\exp(LL(y|m_j))} \quad (8.17)$$

defining the log-likelihood

$$LL(y|m_i) \equiv \log(p(y|m_i))$$

noting that $x = \exp(\log x)$). Components (8.16) and (8.17) provide a framework for comparing alternative and potentially misspecified models based on their marginal likelihood. Such comparisons are important in the assessment of rival models.

Given Bayes factors we can easily compute the model probabilities $p_1, p_2, \dots, p_n$ for n models. Since $\sum_{i=1}^n p_i = 1$ we have that

$$\frac{1}{p_1} = \sum_{i=2}^n BF_{i,1}$$

from which p_1 is obtained. Then $p_i = p_1 BF(i, 1)$ gives the remaining model probabilities. The MATLAB programme, **model_odds.m**, computes these probabilities given the log-likelihood values from the competing models.

8.7 DYNARE AND MATLAB FILES

- **NKlinear_Est_All.mod** estimates model NKlinear with both habit and indexing. Includes Brook-Gelman convergence diagnostics, second-order stochastic simulation and historical variance decomposition.
- **NKlinear_Est_Habit.mod** estimates model NKlinear with habit, but no indexing.
- **NKlinear_Est_Index.mod** estimates NKlinear with indexing, but no habit
- **NKlinear_Est_None.mod** estimates NKlinear with neither habit nor indexing (no persistence mechanisms)
- Matlab program **model_odds.m** computes model odd from log-likelihood values
- Matlab program **acfs_plot.m** plots the sample and estimated **Auto-Correlation Functions (ACFs)**. Requires subfunctions **acfcomp.m** and **autocov.m**.
- Matlab program **irfs_plot.m** plots the impulse response functions for the estimated model.

The following table provides a formal Bayesian comparison of the four models.

	Model All	Model Habit	Model Index	Model None
LLs(mode)	-80.14	-73.52	-93.89	-85.91
LLs(mcmc)	-72.29	-68.49	-91.03	-85.01
prob.	0.0219	0.9781	0.0000	0.0000

Table 8.2: Marginal Log-likelihood Values and Posterior Model Odds

Our model comparison analysis contains important results which are in line with the standard NK-DSGE literature. We find that indexation does not improve the model fit. The poor performance of indexation is in a sense encouraging as this feature of the NK is ad hoc and vulnerable to the Lucas critique. The existence of habit by contrast is a plausible formulation of utility that addresses issues examined in the recent literature.⁹

A limitation of the likelihood race methodology is that the assessment of model fit is only relative to its other rivals with different restrictions. The outperforming model in the space of competing models may still be poor (potentially misspecified) in capturing the important dynamics in the data. To further evaluate the absolute performance of one particular model against data, in a later section we compare the model's implied characteristics with those of the actual data and with a benchmark DSGE-VAR model.

8.8 SECOND MOMENT COMPARISONS WITH DATA

The focus on various alternative specifications seeks to address some of the concerns with Bayesian model comparisons pointed out by Sims (2003). By estimating a large number of model variants, this method intends to complete the space of competing models and to compute posterior odds that take into consideration other (seemingly irrelevant) aspects of the specification. One obvious pitfall or limitation of this methodology is that the assessment of how fit a model is only relative to its other rivals with different restrictions. The outperforming model in the space of competing models may still be poor in capturing the important dynamics in the data.

Sims (2003) argues when weighing the evidence in favor of a particular characteristic of the model failing to account for other aspects of the specification

⁹ In particular the "Easterin paradox", Easterlin (2003). See also Layard (2006) and Choudhary *et al.* (2011) for the role of external habit in the explanation of the paradox.

can lead to disparate inference. In other words, the Bayesian model comparison (based on Bayes factors and posterior odds) is criticized on the basis of the argument that the models considered are too sparse. In such cases, posterior odds may lead to extreme outcomes and may also be highly dependent on the prior distribution. Sims (2003) proposes that the possible solution is to ‘fill the space of models’ to make the model comparison more robust. Recent work by Del Negro *et al.* (2007) also seeks to address the lack of continuity in the model space when comparing DSGE models relative to BVARs, by combining data with artificial observations generated by the model. To further evaluate the absolute performance of one particular model (or information assumption) against data, in a later section we compare the model’s implied characteristics with those of the actual data and with a benchmark DSGE-VAR model.

The summary statistics such as first and second moments have been standard for researchers to use to validate models in the literature on DSGE models, especially in the RBC tradition. As the Bayes factors (or posterior model odds) are used to assess the relative fit amongst a number of competing models, the question of comparing the moments is: can the models correctly predict population moments, such as the variables’ volatility or their correlation, i.e. to assess the absolute fit of a model to macroeconomic data.

Following Schorfheide (2000), let y^{rep} be a sample of observation that one could have observed in the past or that one might observe in the future. One can derive the sampling distribution of y^{rep} given the current state of knowledge using the Bayes theorem: $p(y^{rep}|y) = \int L(y^{rep}|\theta)p(\theta|y)d\theta$. Assume that $T(y)$ is a test quantity that reflects an aspect of the data (moment) that one wants to check, e.g. correlation between output and inflation or the output volatility. In order to assess whether the estimated model can replicate population moments, one sequentially generates draws from the posterior distribution, $p(\theta|y)$ and the predictive distribution $p(y^{rep}|y)$ so that the predictive $T(y^{rep})$ can be computed.

For the simulation and computation of moments, Dynare assumes that the shocks follow a normal distribution. In a stochastic set-up, shocks are only allowed to be temporary. A permanent shock cannot be accommodated because of the need to stationarize the model. Also the expectations of future shocks in a stochastic model must be zero. But in Dynare we can make the effect of the shock propagate slowly throughout the economy by specifying the shock’s process and introducing a “latent shock variable”, that affects the model’s true exogenous variable, which is itself an AR(1). In a stochastic framework, these exogenous variables take random values in each period. In Dynare, these random values follow a normal distribution with zero mean, but we can (and

have to) specify the variability of these shocks (within the *shock;...end;* block). So setting *period=1000* when simulating the model specifies that the model is simulated over 1000 periods, where Dynare computes the path of variables over a 1000 period horizon by solving all the equations for every period, and this can be used to compute the (empirical) moments of the simulated variables (i.e. simulated model solutions).

To obtain the model-generated moments based on the real world data (i.e. posterior distribution), simply use the *stoch_simul* keyword after the estimation command. Table 8.3 presents some selected second moments implied by the above estimations and compares with those in the actual data. In particular, we compute these model-implied statistics by solving the models at the posterior means obtained from estimation. The results of the model's second moments are compared with the second moments in the actual data to evaluate the models' empirical performance.

	Standard Deviation		
	Output	Inflation	Interest rate
Data	0.5432	0.2392	0.5952
Model All	0.5928	0.3448	0.4205
Model Habit	0.5996	0.3113	0.4073
Model Index	0.6788	0.7212	0.6814
Model None	0.6886	0.5612	0.5318
Cross-correlation with Output			
Data	1.000	-0.2013	0.0323
Model All	1.000	-0.0788	-0.2847
Model Habit	1.000	-0.0969	-0.2646
Model Index	1.000	0.0420	-0.0245
Model None	1.000	0.0673	-0.0492
Autocorrelations (Order=1)			
Data	0.1526	0.5364	0.9462
Model All	0.0957	0.5714	0.8938
Model Habit	0.1150	0.5778	0.9074
Model Index	0.1487	0.8804	0.9667
Model None	0.1479	0.8418	0.9460

We have so far considered autocorrelation only up to order 1. To further illustrate how the estimated models capture the data statistics and persistence in particular, we now plot the autocorrelations up to order 10 of the actual data and those of the endogenous variables generated by the model variants in Figure 3. Using the *stoch_simul* keyword and argument, e.g., *ar=10* we produce autocorrelations up to order 10. All simulation outputs from Dynare are

stored in *FILENAME_results.mat* in the working directory. So we reload it to extract useful fields (stored in the struc. array *oo_*) For instance, the simulated autocorrelation function can be found on the diagonal of the field *oo_.autocorr*. To compute and plot the sample ACFs from the data, we need subfunctions **acfcomp.m** and **autocov.m**. Finally, in the working directory, **acfs_plot.m** plots the sample ACFs and estimated ACFs - see Figure 8.5.

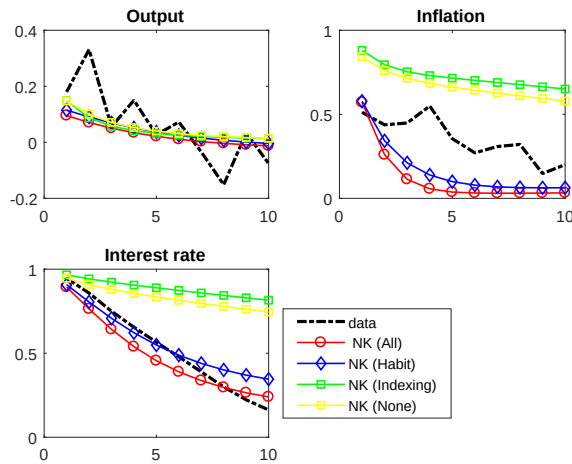


Figure 8.5: Autocorrelations of Observables in the Actual Data and in the Estimated Models

8.9 IMPROVING THE SECOND MOMENT FIT USING ENDOGENOUS PRIORS

Dynare implements a form of endogenous priors as in Christiano *et al.* (2011). The procedure is motivated by sequential Bayesian learning. Starting from independent initial priors on the parameters, specified in the *estimated_params* block, the standard deviations observed in a “pre-sample”, taken to be the actual sample, are used to update the initial priors. Thus, the product of the initial priors and the pre-sample likelihood of the standard deviations of the observables is used as the new prior (for more information, see the technical appendix of Christiano *et al.* (2011)). This procedure helps in cases where the regular posterior estimates, which minimize in-sample forecast errors, result

in a large over-prediction of model variable variances (a statistic that is not explicitly targeted, but often of particular interest to researchers). Del Negro and Schorfheide (2008) sets out a more general framework that extends to other second moments in addition to standard deviations.

Dynare implements a version of this procedure due to Christiano *et al.* (2011) except the “pre-sample” is the actual sample. Implementation is easy - simply add **endogenous_prior** to the estimation command. The product of the initial priors and the pre-sample likelihood of the standard deviations of the observables is used as the new prior. One can think of this procedure as a **hybrid** Simulated Method of Moments (SMM) - Bayesian Estimation Procedure. For more detail see the technical appendix of Christiano *et al.* (2011).

The following table indicates the endogenous prior feature produces *some* improvement in moment matching for the NKlinear model .

Standard Deviation			
	Output	Inflation	Interest rate
Data	0.5432	0.2392	0.5952
Model All (Previous)	0.5928	0.3448	0.4205
Model All (End. Priors)	0.5776	0.2738	0.3847
Cross-correlation with Output			
Data	1.000	-0.2013	0.0323
Model All (Previous)	1.000	-0.0788	-0.2847
Model All (End. Priors)	1.000	-0.0692	-0.2148
Autocorrelations (Order=1)			
Data	0.1526	0.5364	0.9462
Model All (Previous)	0.0957	0.5714	0.8938
Model All (End. Priors)	0.0603	0.5432	0.8890

Table 8.3: Selected Second Moments of the Model

8.10 IMPULSE RESPONSE FUNCTIONS

The following figure 8.6 depicts the mean responses corresponding to a positive one standard deviation technological shock. The endogenous variables of interest are the observables in the estimation and each response is for a 10 period (2.5 years) horizon. All impulse responses are computed simulating the vector of DSGE model parameters at the posterior mean values computed

from estimation. Figure 8.6 shows impulse response functions for a technology shock.

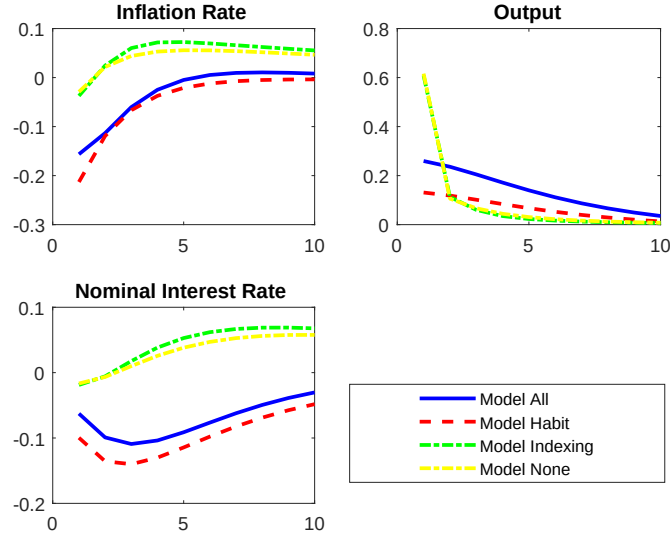


Figure 8.6: irfs- Technology Shock

8.11 VARIANCE AND HISTORICAL DECOMPOSITIONS

As explained above, impulse response functions allow us to analyse the effects of a shock to one of the endogenous variables on to the other variables. Variance decompositions, on the other hand, decomposes the variation in each endogenous variable into each shock to the system, thus providing information on the relative importance of each disturbance as a source of variation for each variable. Historical decompositions, in turn, can be used for counterfactual simulations. The data can be decomposed into the sum of a baseline forecast and the contribution of all shocks. This allow us to analyse how the data would have evolved if a shock or a combination of shocks are shut down (i.e., their contribution is zero).

The state-space representation of the model solution in linear form is given by:

$$\begin{aligned} X_{t+1} &= AX_t + B\varepsilon_{t+1} \\ Y_t &= CX_t \end{aligned}$$

where X_t is the potentially unobservable state vector and Y_t is the vector of the observables. The historical decomposition stems from the Moving Average (MA) representation of the model state space

$$Y_h = \sum_{j=0}^h d_j \varepsilon_{h-j} + CA^h X_0 \quad (8.18)$$

for $h = 1, \dots, T$. Note that each d_j is a matrix, with its i th column $d_{i,j}$ multiplying the i th shock; if we further define the effect of all the i th shocks on Y_h as $Y_{i,h} = \sum_{j=0}^h d_{i,j} \varepsilon_{i,h-j}$, then we can decompose Y_h as

$$Y_h = \sum_{i=0}^r Y_{i,h} + CA^h X_0 \quad (8.19)$$

where r is the number of shocks.

During the course of the estimation, Dynare automatically produces the Kalman filter estimates of all the $\{X_t\}$, and for the final set of parameters (either the mode, or the average over the MCMC iterations) it also calculates the smoothed estimates of the $\{X_t\}$. From these it also calculates the smoothed estimates of all the shocks. These latter, together with the smoothed estimates of X_0 are then used to calculate each of the individual terms of (8.19), with the last term, $CA^h X_0$, shown on the historical decomposition graphs as the effect of ‘initial values’.

In Dynare, variance decomposition for the specific periods (specified in [] after the option) can be carried out by using the option *conditional_variance_decomposition*. The decomposition for a given sample according to the estimated model can be computed using the command *shock_decomposition* which must be followed by the *estimation* statement (unless one specifies to use the calibrated parameters with the option *parameter_set*). The following figure provides, for inflation, historical decomposition of our estimation samples to each of the estimated shocks (based on the estimated posterior modes obtained using the NK linear model above). Figure 8.7 decomposes the historical inflation time series with the NK-DSGE model using the estimated parameters. In particular it presents the contribution of the respective shocks to the deviation of the smoothed inflation from its steady state with the coloured bars. Note that the figure below also plots the initial conditions which represent the distance between the rational expectation solution from its steady state before the shock arrives. In other words, these initial conditions refer to the part of the deviations from its steady state explained by the unknown initial value of the state variables. The influence of initial values usually dies out relatively quickly but its persistence depends on how much persistence the model has.

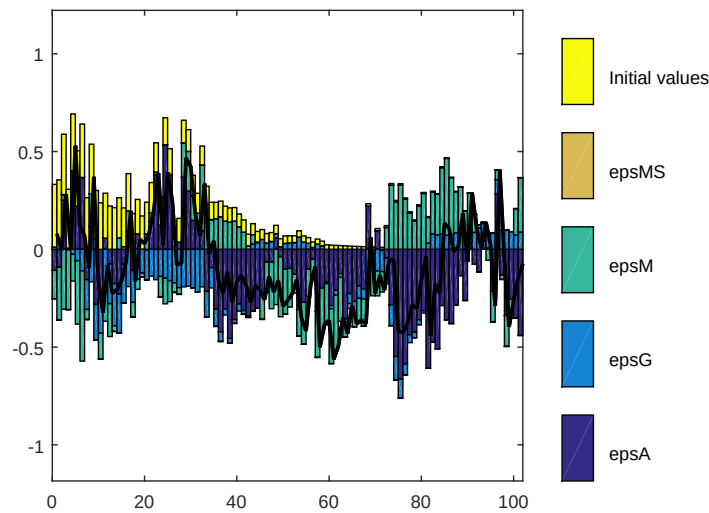


Figure 8.7: Historical Decomposition of Inflation

8.12 DYNARE ESTIMATION, COMPARISON AND VALIDATION EXERCISES

1. Estimation

- a) Using `us_data.mat`, `NKlinear_Est.mod` and the mode file saved as `NKlinear_Est_mode_saved` with the three observed variables chosen in the notes, first perform the identification and then estimate the NK model with a conventional Taylor rule. Use only 10,000 mcmc draws for now and re-run the results later with at least 100,000. Use the graph plotter from Days 1 and 2 provided to compare the impulse responses to the four shocks and `acfs_plot.m` to compare the ACFs. What do you notice about your results?
- b) Recall the problem arising in the identification from having four shocks and three observables. Address this by adding consumption growth dc as an observable. Now perform the identification and re-run the estimation (with an implementable rule). Compare irfs and ACFs with the original set-up. What do you notice about your results?

2. Comparison

- a) Repeat the model comparison exercise across for model specifications (all persistence mechanisms, only habit, only indexing and no persistence mechanisms) using an **implementable rule** rather than the conventional Taylor rule.
- b) For today's exercise find the posterior distribution based on 10,000 of sets MCMC-MH simulation. But after the course repeat the process with 100,000 draws.
- c) Use mode files from the conventional Taylor rule as initial modes in the estimation command. In most cases `mode_compute=4` should give a negative definite Hessian. If not you must use `mode_compute=6` which is designed to compute a negative definite Hessian, but it takes a lot of time!
- d) When you get this far you have LLs for eight models. Now conduct a likelihood race across these eight models.

3. Model Validation

- a) Now repeat the moment comparison exercise using an **implementable rule** rather than the conventional Taylor rule.
- b) From the stochastic simulation of the estimated models with an implementable monetary rule obtain the model-implied moments based on the estimated posterior modes and produce the IRFs.
- c) Use these results validate all eight model variants against the real word data by extending the Table on Slide 10
- d) Use **acfs_plot.m** to compare autocorrelations of the observables in the actual data and in the eight estimated models. (These use MATLAB files **acfcomp.m**, **autocov.m** in the folder)
- e) Use **irfs_plot.m** to compare the impulse response functions of the eight estimated models.

9

OPTIMAL POLICY

This section first considers the general optimal policy problem where the policymaker has a number of instruments and sets out to maximize a general discounted welfare criterion subject to the constraints of a DSGE model. If the policymaker is able to commit, the setting of instruments can be conducted in terms of the ex ante optimal policy. If the expected discounted household utility is chosen as the welfare criterion this becomes the well-known Ramsey problem. A problem with such a solution is that it involves a complex rule even for quite simple NK models. Much of the optimal policy literature therefore focuses on simple Taylor-type commitment rules that are optimized so as to come close to mimicking the Ramsey solution and this is the approach of this course. In the absence of an ability to commit the policymaker must set policy to be time-consistent. Before proceeding with our treatment of simple rules, we first briefly review the Ramsey and time-consistent solutions.

9.1 THE RAMSEY PROBLEM

We consider a model as a special case of the following general setup recognized by Dynare in non-linear form

$$Z_t = J(Z_{t-1}, X_t, w_t, \epsilon_t) \quad (9.1)$$

$$\mathbb{E}_t X_{t+1} = K(Z_t, X_t, w_t) \quad (9.2)$$

where Z_t, X_t are $(n - m) \times 1$ and $m \times 1$ vectors of backward and forward-looking variables, respectively, ϵ_t is a $\ell \times 1$ i.i.d shock variable and w_t is an $r \times 1$ vector of instruments. Under perfect information all variables dated t or earlier are observed at time t including shocks.

Now define

$$y_t \equiv \begin{bmatrix} Z_t \\ X_t \end{bmatrix}$$

Then, as in Dynare User Guide, chapter 7, (9.1) and (9.2) can be written

$$\mathbb{E}_t [f(y_t, y_{t+1}, y_{t-1}, w_t, \epsilon_{t+1})] = 0 \quad (9.3)$$

$$\begin{aligned}\mathbb{E}_t[\epsilon_{t+1}] &= 0 \\ \mathbb{E}_t[\epsilon_{t+1}\epsilon'_{t+1}] &= \Sigma_\epsilon\end{aligned}$$

This is quite general in that y_t can be enlarged to include lagged and forward-looking variables.

The general problem is to maximize at time 0, $\Omega_0 = E_0 [\sum_{t=0}^{\infty} \beta^t u(y_t, y_{t-1}, w_t)]$ subject to (9.3) given initial values Z_0 . To carry out this problem write the Lagrangian

$$L = E_0 [\sum \beta^t [u(y_t, y_{t-1}, w_t) + \lambda'_t f(y_t, y_{t+1}, y_{t-1}, w_t, \epsilon_{t+1})]] \quad (9.4)$$

where λ_t is a column vector of multipliers associated with the n constraints defining the model. First-order conditions are given by

$$\begin{aligned}E_0 \left[\frac{\partial L}{\partial w_t} \right] &= E_0 [u_3(y_t, y_{t-1}, w_t) + \lambda'_t f_4(y_t, y_{t+1}, y_{t-1}, w_t, \epsilon_{t+1})] = 0 \quad (9.5) \\ E_0 \left[\frac{\partial L}{\partial y_t} \right] &= E_0 [u_1(y_t, y_{t-1}, w_t) + \beta u_2(y_{t+1}, y_t, w_{t+1}) + \lambda'_t f_1(y_t, y_{t+1}, y_{t-1}, w_t, \epsilon_{t+1}) \\ &\quad + \frac{1}{\beta} \lambda'_{t-1} f_2(y_{t-1}, y_t, y_{t-2}, w_{t-1}, \epsilon_t) + \beta \lambda'_{t+1} f_3(y_{t+1}, y_{t+2}, y_t, w_{t+1}, \epsilon_{t+1})] \\ &= 0 \quad (9.6)\end{aligned}$$

where the subscripts in $\{u_i, f_i\}$ refer to the partial derivatives of the i th variable in u, f .

Now partition $\lambda_t = [\lambda_{1,t} \lambda_{2,t}]$ so that $\lambda_{1,t}$, the co-state vector associated with the backward-looking component of (9.5), namely (9.1), is of dimension $(n-m) \times 1$ and $\lambda_{2,t}$, the co-state vector associated with the forward-looking component of (9.5), namely (9.2), is of dimension $m \times 1$.

An important optimality condition is:

$$\lambda_{2,0} = 0; \text{ (ex ante optimal)} \quad (9.7)$$

$$\lambda_{2,0} = \lambda_2; \text{ ('timeless' solution)} \quad (9.8)$$

where λ_2 is the deterministic steady state of $\lambda_{2,t}$. To complete our solution we require $2n$ boundary conditions. Then together with (9.7) or (9.8), Z_0 given are n of these. The 'transversality condition' $\lim_{t \rightarrow \infty} \lambda_t = \lambda$ gives us the remaining n (see Currie and Levine (1993). chapter 4).

Thus in order to achieve optimality the policy-maker sets $\lambda_{2,0} = 0$ at time $t = 0$. As is well-known optimal policy sees an initial jump in the inflation rate even in the absence of any shocks. The timeless solution removes this feature

and imposes a time-invariance on the solution (see Adam (2011), for example). For the ex ante optimal policy, at time $t > 0$ there then exists a gain from reneging by resetting $\lambda_{2,t} = 0$. Thus there is an incentive to renege that exists at all points along the trajectory of the optimal policy by re-optimizing in this fashion. This essentially is the *time-inconsistency problem* facing stabilization policy in a model with structural dynamics.

9.2 TIME-CONSISTENT (DISCRETIONARY) POLICY

To evaluate the time-consistent (discretionary) policy we write the expected loss $\Omega_t(Z_t)$ at time t given observed Z_t , in Bellman form as¹

$$\Omega_t(Z_t) = \mathbb{E}_t \left[\sum_{\tau=t}^{\infty} \beta^{\tau-t} u(y_{\tau}, y_{\tau-1}, w_{\tau}) \right] = u(y_t, y_{t-1}, w_t) + \beta \mathbb{E}_t [\Omega_{t+1}(Z_{t+1})] \quad (9.9)$$

Define the value function

$$V(Z_t) = \max_{\{w_t\}} \{u(y_t, y_{t-1}, w_t) + \beta \mathbb{E}_t [V_{t+1}(Z_{t+1})]\} \quad (9.10)$$

Then Bellman's equation is

$$V_t(Z_t) = \max_{\{w_t\}} \mathbb{E}_t \{u(y_t, y_{t-1}, w_t) + \beta \mathbb{E}_t [V_{t+1}(Z_{t+1})]\} \quad (9.11)$$

In other words Ω_t is maximized at time t , subject to the model constraints, in the knowledge that a similar procedure will be used to minimize Ω_{t+1} at time $t + 1$. The dynamic programming solution then seeks a stationary *Markov Perfect* solution of the form $w_t = F(Z_t)$, and $X_t = G(Z_t)$.²

9.3 OPTIMIZED SIMPLE RULES

Optimal policy in the form of the Ramsey solution can be expressed as $w_t = f(Z_t, \lambda_{2,t})$. This poses problems for the implementability of policy in terms of complexity and the observability of elements of Z_t (such as the technology

¹ This applies only to the zero-growth steady state.

² See Currie and Levine (1993) and Söderlind (1999) for a LQ treatment of this problem under perfect information and Levine *et al.* (2012b) under imperfect information. But see Dennis and Kirsanova (2013) for the possibility of multiple equilibria.

process A_t , but more importantly $\lambda_{2,t}$). The macroeconomic policy literature therefore focuses on simple rules, using the Ramsey solution as a benchmark.

The general optimal policy problem seeks an optimized simple rule in which the vector of instruments w_t respond to an observed subset of macroeconomic variables in a prescribed (for example log-linear) fashion. All our rules take the log-linear form

$$\log w_t = D \log y_t \quad (9.12)$$

where we define $\log w_t \equiv [\log w_{1,t}, \log w_{2,t}, \dots, \log w_{r,t}]'$ over r instruments, and similarly for $\log y_t$, and the matrix D selects a subset of y_t from which to feed-back. Again this is quite general in that y_t can be enlarged to include lagged and forward-looking variables.

The optimized simple rules then defines the inter-temporal welfare loss at time t in Bellman form (9.9) (again ignoring long-term trend growth for now), sets steady-state values for instruments w_t , denoted by w , computes a second-order solution for a particular setting of w and solves the maximization problem at $t = 0$,

$$\max_{w,D} \Omega_0(Z_0, \Sigma, w, D) \quad (9.13)$$

given initial values Z_0 and the variance-covariance matrix of shocks Σ (a diagonal matrix in all the codes). In a purely stochastic problem we put $Z_0 = Z$, the steady state of Z_t , maximizing the conditional welfare at the steady state. In a purely deterministic problem there is no exogenous uncertainty and the optimization problem is driven by the need to return from Z_0 to its steady state, Z .

We now examine optimal monetary policy conducted in terms of the Ramsey solution, discretion and one of two simple Taylor interest rate rules used up to now, which are special cases of (9.12):

$$\log \left(\frac{R_{n,t}}{R_n} \right) = \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + \alpha_\pi \log \left(\frac{\Pi_t}{\Pi} \right) + \alpha_y \log \left(\frac{Y_t}{Y} \right) + \epsilon_t \quad (9.14)$$

$$\log \left(\frac{R_{n,t}}{R_n} \right) = \rho_r \log \left(\frac{R_{n,t-1}}{R_n} \right) + \alpha_\pi \log \left(\frac{\Pi_t}{\Pi} \right) + \alpha_y \log \left(\frac{Y_t}{Y_t^F} \right) + \epsilon_t \quad (9.15)$$

To recap from section 6, (9.14) is an ‘implementable’ rule that does not require knowledge of the output gap, (9.15) is a ‘conventional’ Taylor rule which does. In order to maximize efficiently with respect to feedback parameters we introduce the re-parametrization $\alpha_\pi \equiv (1 - \rho_r)\theta_\pi$ and $\alpha_y \equiv (1 - \rho_r)\theta_y$. Notice that

the special case of an integral rule ($\rho_r = 1$), $\alpha_y = 0$ with a zero-inflation steady state ($\Pi = 1$) is $\log R_{n,t} = \alpha_\pi \log P_t$ which is a *price level* rule.

9.4 LINEAR MODEL, QUADRATIC LOSS FUNCTION

Of the three policy rules discussed up to now, only the Ramsey solution is available for a non-linear model and a general objective function. Time consistent policy still requires a linear-quadratic approximation to the problem for medium-sized models. However *linear-quadratic problems* are manageable for all three forms of policy and are now examined.

Consider our first NK Model linearized around a zero-growth, zero-inflation steady state. Parameter values are those used in Section 5 and the standard deviation of shocks are set at 1%. A standard ad hoc quadratic loss function in deviation form is given by

$$\begin{aligned}\Omega_0 &= (1 - \beta)E_0 \left[\sum_{t=0}^{\infty} \beta^t (16\pi_t^2 + y_t^2 + r_{n,t}^2) \right] \\ &\simeq 16\text{var}(\pi_t) + \text{var}(y_t) + \text{var}(r_{n,t}) \text{ as } \beta \rightarrow 1\end{aligned}\quad (9.16)$$

where the variances above are unconditional.³ Dynare computes the Ramsey policy (optimal commitment), discretionary policy and optimized implementable simple rules in file **NKlinear_policy.mod** to be found in folder **Optimal**.

Rule	$\Omega_0 \times 10^4$	ρ	α_π	α_y	c_e	$\text{sd}(R_n)$
Ramsey	1.3095	n.a	n.a.	n.a.	0	0.3647
Discretion	3.5190	n.a	n.a.	n.a.	0.025	0.7729
Simple(Implementable)	1.3861	0.9933	1.701	0.031	0.0009	0.3779

Table 9.1: LQ Framework: Optimal Policy, Discretionary Policy and Optimized Rules

We report the expected inter-temporal loss as given by (9.16). To derive the household welfare in terms of a consumption equivalent *percentage* increase ($c_e \equiv \frac{\Delta C}{C} \times 10^2$), expanding $U(C_t, H_t)$ as a Taylor series

$$U(C_t, H_t) \approx U(C, H) + U_C(C_t - C) + U_H(H_t - H)$$

³ Since the model is quarterly, annual inflation $\pi^a \approx 4\pi$ so the period utility is $(\pi_t^a)^2 + y_t^2 + r_{n,t}^2$.

$$= U(C, H) + CU_C \frac{(C_t - C)}{C} + U_H(H_t - H)$$

Hence keeping H_t fixed at $H_t = H$ we have

$$\Delta U = U_C \Delta C = CU_C c_e \times 10^{-2} \quad (9.17)$$

In the mod file we have expressed the standard deviations in terms of proportions and in fact normalized the shocks at 0.01. Let X be the losses for discretion and the simple rule relative to the Ramsey case which by construction is the minimum achievable. Both sides of (9.17) are now comparable. Thus

$$c_e = \frac{X \times 10^2}{CU_C} \quad (9.18)$$

From the *non-linear* NK set-up we see that in the steady state of this model, $CU_C = 0.506878 \times 1.77213 = 0.8983$. It follows from (9.18) that a welfare loss difference of $X = 0.01$ gives a consumption equivalent percentage difference of $c_e = 1.1133\%$ which we use to arrive at the c_e column in Table 9.1. We also report the steady-state variances of the nominal interest rate as a measure of monetary activism, but also to highlight possible zero lower bound (ZLB) problems.

Figures 9.1-9.3 show the impulse responses to technology, government spending and a mark-up shock respectively comparing the Ramsey solution, discretionary policy and optimized simple rules). For a technology shock output immediately rises and, inflation falls. The optimal policy is to raise the interest rate a little initially to contain inflation, but then to commit to a sharp monetary relaxation before gradually returning to the steady state. Both consumption and leisure rise (the latter a familiar result in the NK literature) and hours fall. The productivity shocks results in a fall in the marginal cost, which is why inflation falls in the first place. The $\sqcup$ -shaped interest rate path is time-inconsistent. Only an increasing interest rate path after the initial fall will be time-consistent; the discretionary regime sees this happening with a larger drop in both the interest rate and inflation. Real variables - output, hours and consumption differ little between Ramsey and Discretion for all shocks which explains the small welfare differences for all shocks combined.

The mark-up shock is similar to the technology shock but with opposite effects; only the qualitative response of hours differ. The government spending shock however provides more interesting results. An increase in demand acts as a fiscal stimulus - in fact with $\frac{G}{Y} = 0.2$ in the steady state the impact multiplier is quite low, between $[0.5, 0.75]$.⁴ Commitment enables the policymaker

4 $\frac{\Delta Y_t}{\Delta G_t} = \frac{Y_t}{G_t} \times \text{irf}$.

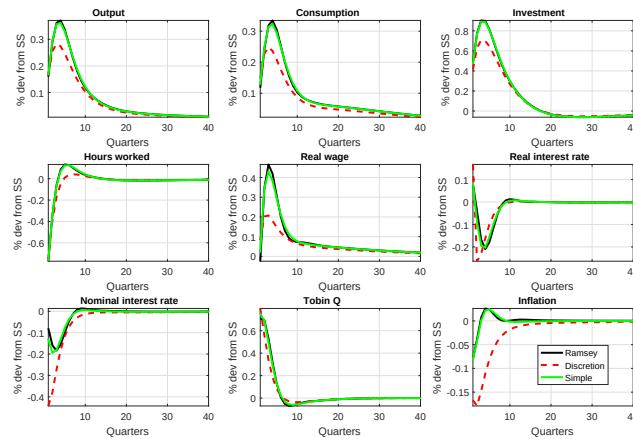


Figure 9.1: Impulse Responses to a Technology Shock

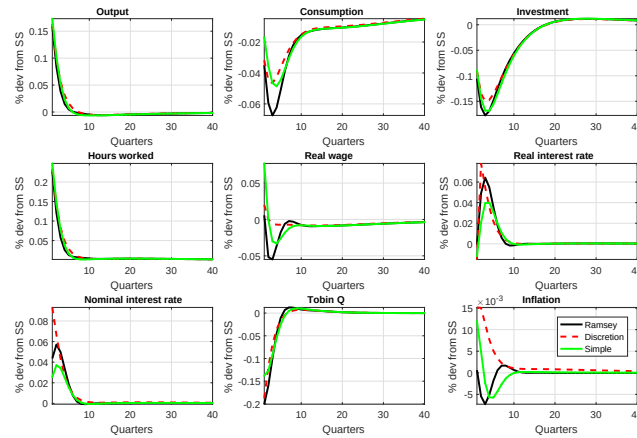


Figure 9.2: Impulse Responses to a Gov Spending Shock

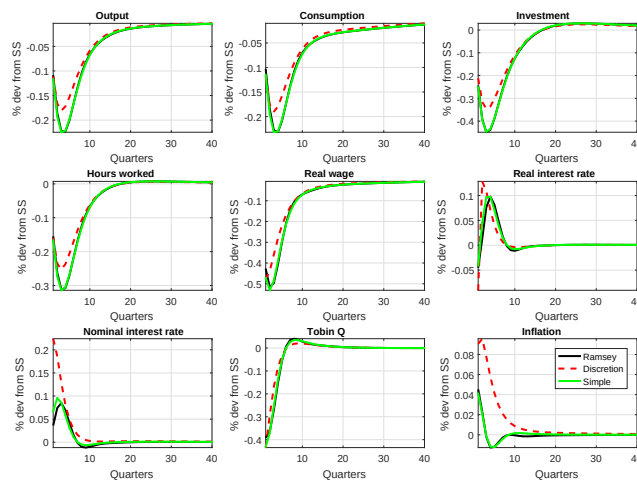


Figure 9.3: Impulse Responses to a Mark-up Shock

to contain inflation which under discretion rises substantially. This is a major source of the welfare differences noted previously.

9.5 DYNARE CODES

Dynare mod file **NKlinear_policy.mod** provides options for the Ramsey, Discretion and Implementable Simple Rules LQ exercises with a preprocessor command to select your choice. Matlab file **graphs_irfs_compare_NK_optimal_policy** compares irfs for these three policy regimes.

9.6 EXERCISES

1. Rework the LQ analysis in Table 9.1 for a ‘conservative banker’ with welfare loss given by

$$\begin{aligned}\Omega_0 &= (1 - \beta)\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t (32\pi_t^2 + y_t^2 + r_{n,t}^2) \right] \\ &\simeq 32\text{var}(\pi_t) + \text{var}(y_t) + \text{var}(r_{n,t}) \text{ as } \beta \rightarrow 1\end{aligned}\quad (9.19)$$

2. Rework Table 9.1 using the conventional Taylor rule. What do you notice about the results?

10

CRITIQUE OF DSGE MODELS AND THE ROAD AHEAD

We conclude the notes to take you home with some thoughts on where DSGE modelling is going.¹ There have been a number of recent assessments of the “state of macro” and the contribution of DSGE models - a list that is by no means exhaustive would include: Pesaran and Smith (2011), Blanchard (2009), Blanchard *et al.* (2010), Driffill (2011), Pesaran and Smith (2011), Blanchard *et al.* (2013), Blanchard (2016), Blanchard and Summers (2017), Vines and Wils (2018), Christiano *et al.* (2018).

This survey structured around a number of criticisms regarding DSGE models and ways in which the community of DSGE macro-modellers are responding. The rest of the survey is organized follows. Sections 10.1–10.6 describe six main areas of criticism and the response to these. Section 10.7 summarizes a sharp alternative to the DSGE approach to macroeconomic modelling that is motivated by these concerns, but provides very distinct solutions. Section 10.8 discusses how DSGE models can be used for macro-economic policy design and Section 10.9 concludes.

There are a number of key criticisms levelled at DSGE models. The first is fundamental and common to macroeconomics and microeconomics alike – namely, problems with rationality and Expected Utility Maximization (EUM). The second is that DSGE models examine fluctuations about an exogenous balanced growth path and there is no role for endogenous growth. The third consists of a number of concerns associated with estimation. The fourth is another fundamental problem with any micro-founded macro-model – that of heterogeneity and aggregation. The fifth and sixth concerns focus on the rudimentary nature of earlier models that lacked unemployment and a banking sector. We consider these in turn.

¹ This Chapter is taken from Levine (2020).

10.1 RATIONALITY

The assumption of rationality in general and that of rational expectations in macro-models in particular has naturally generated a lively debate in economics and the social sciences. We consider these two aspects in turn.

10.1.1 Rationality in General

The assumption of perfect rationality has come under scrutiny since the 1950s when Herbert A. Simon claimed that agents are not realistically so rational so as to aspire to pay-off maximization. Instead he proposed 'bounded rationality' as a more realistic alternative to the assumption of rationality, incorporating players' inductive reasoning processes. This is the route that the Agent-Based models take (see, Section 10.7). Certainly, experimental studies of decision-making show human behaviour to be regularly inconsistent and contradictory to the assumption of perfect rationality. However the question that arises is whether economic agents can *learn* to be rational, so rationality may well be a reasonable empirical postulate to describe behaviour near a long-run steady state. We return to this question in Section 10.1.2 below.

Models can only be beaten by alternative models. A model of irrationality has to pin down why one decision is preferred to another and here we observe that analytically tractable theories of the inconsistency and irrationality in human behaviour simply have not yet been fully developed. Hence our best analytical models are based on the rationality assumption as we unfortunately have nothing superior on offer. However we can be more positive than that at least when it comes to competitive behaviour. Darwinian selection helps rational (that is, profit-maximizing) firms (profit-maximizing) to succeed in competition.

Perhaps the most convincing argument for adopting the rationality assumption is provided by Myerson (1999). If we view the aim of social sciences is not only to predict human behaviour in the abstract, but also, crucially, to analyze social institutions and assess proposals for their reform, it is useful to evaluate these institutions under the assumption of perfect rationality. In this way, we can solve for flaws as either defects in the institutional structure (and thereby institutional reform is the required solution) or as flaws in the rationality of the agents (which begs for improved education and/or provision of information for individuals). Accordingly this has become a logical and useful assumption for economists in order to see with more clarity when social problems must be

solved by institutional reform. This argument can be refined to illustrate why this individual perfection assumption should be one of intelligent rational maximization, as in the models of non-cooperative game theory. Thus an argument for reform of social institutions (rather than for re-education of individuals) is most persuasive when it is based on a model which assumes that individuals intelligently understand their environment and rationally act to maximize their own welfare.

Even if we accept utility maximization, in an uncertain environment there still is an issue of whether it should be *expected* utility maximization (EUM). An alternative supported by experiments is *Prospect Theory* pioneered by Kahneman and Tversky (1979) and Kahneman and Tversky (1992). Prospect theory takes into account the empirical finding of experiments that people behave as if extremely improbable events are impossible and extremely probable events are certain (see Shiller (1999) and Barberis (2013)). Prospect theory can explain phenomena such as the equity premium puzzle. Woodford (2012) shows that reference-dependent choice of the kind captured by prospect theory may be understood as an efficient approach to rational choice with limited information processing capacity as in the ‘rational inattention’ literature discussed below in Section 10.1.2. However it is extremely difficult to incorporate into general equilibrium modelling; in the words of Shiller “EUM can be a workhorse for some sensible research”.

10.1.2 Rational Expectations in Macroeconomics

Staying broadly within the rational expectations paradigm a number of refinements are on offer that assume that agents are not able to perfectly observe states that define the economy. Levine *et al.* (2012a) propose a general framework for introducing information limitations at the point agents form expectations. The ‘rational inattention’ literature (Sims (2005), Luo and Young (2009), Luo (2008)) and the ‘sticky information’ approach of Reis (2009) also fits into this agenda. The basic idea is that agents can process information subject to a constraint that places an upper bound on the information flow. Borrowing from information theory (which in turn borrows from statistical physics) the idea is formalized by an upper bound on the decrease in entropy that ensues as agents proceed from a prior to a posterior of a signal.

A more radical deviation from rational expectations is provided by the statistical rational learning literature pioneered by Evans and Honkapohja (2001a). This introduces a specific form of bounded rationality in which utility-maximizing

agents make forecasts in each period based on standard econometric techniques such as least squares. In many cases this learning behaviour converges to a rational expectations equilibrium and much of this literature studies the conditions for this to happen.

In this genre of models there then exists a choice of learning model: *Euler* versus the *anticipated utility* approach (following Kreps (1998)) – henceforth EL and AU. In both approaches agents cannot form model-consistent expectations. Under EL agents forecast their own one-period ahead decisions whereas under AU agents form beliefs over the future infinite time horizon of aggregate states and prices which are exogenous to their decisions. The two approaches then differ with respect to what agents learn about – their own future one-period ahead decision for EL and variables exogenous to the agents for AU.

AU, also known the “infinite time-horizon” framework, is closely related to the “internal rationality” (IR) approach of Adam and Marcet (2011). Under both IR and AU agents maximize utility, given their constraints and a consistent set of probability beliefs about payoff-relevant variables that are *external*. Then with IR, beliefs take the form of a well-defined probability measure over a stochastic process (the ‘fully Bayesian’ plan). See Eusepi and Preston (2011) for an RBC BR model with AU, Preston (2005), Woodford (2013) and Hommes *et al.* (2017) who adopt a behavioural NK framework and Branch and McGough (2018) for a recent discussion of EL versus AU. Cogley and Sargent (2008) compares the IR vs AU and find that AU can closely approximate the fully Bayesian optimization. There are other agent-level learning alternatives such as shadow value and finite-horizon learning. See Branch *et al.* (2013), Woodford (2018) and Evans and McGough (2018) for recent reviews.

A considerable advance in the literature on bounded rationality came in Brock and Hommes (1997), which embeds a simple heterogeneous expectations mechanism into a cobweb model of partial equilibrium. As in the standard cobweb model, firms have to forecast the equilibrium price before they set their output level. To do so, they have the choice of using a simple adaptive expectations predictor at zero cost, or perfect foresight at positive cost. The authors argue that firms will choose predictors that result in higher net profits, where the probability of choosing a given predictor is determined by a logit model. This choice is justified by an appeal to the discrete choice model described in Manski and McFadden (1981), which is widely used in microeconomics and econometrics. A similar approach is used in the reinforcement learning literature described in Young (2004).

The insights of Brock and Hommes (1997) were slowly incorporated into the New Keynesian literature in the early 2000s. Branch and McGough (2004)

studied the impact of heterogeneous expectations on the existence of sunspot equilibria in rational expectations models, and Branch and Evans (2007) examined discrete choice dynamics of the form considered in Brock and Hommes (1997) in the context of a simple macroeconomic model. Heterogeneous expectations in a New Keynesian framework, albeit without discrete choice, were then examined in considerable detail in Branch and McGough (2009).

Heterogeneous expectations and discrete choice were fully incorporated into the New Keynesian three equation model around the time that the USA and Europe were recovering from the effects of the 2008 financial crisis, in the papers of Branch and McGough (2010) and De Grauwe (2011). The basic framework has become known as the behavioural New Keynesian model, largely the result of De Grauwe's book length treatment of the subject, (De Grauwe 2012). Via the expectational mechanism of Brock and Hommes (1997), the behavioural New Keynesian (BNK) approach incorporates bounded rationality and heterogeneity into the standard New Keynesian (NK) three equation model. This embeds an intuitive form of complexity into the standard approach, where strategy switching generates recurring bouts of instability. Specifically, households and firms have the choice between two (or more) predictors of output and inflation. Both predictors can be simple non-rational rules as in De Grauwe (2011), or one can be rational as in Hommes *et al.* (2017). Around the steady state of a BNK model, both predictors are equally accurate, but one predictor becomes increasingly accurate relative to the other as the economy moves away from the steady state. Any exogenous shock that moves the economy away from the steady state can then lead to agents rapidly switching from one predictor to the other. This creates an endogenous amplification effect which can explain the existence of excess kurtosis and stochastic volatility observed in macroeconomic data. A full review of the behavioural New Keynesian model literature is provided by Calvert Jump and Levine (2018).

10.2 INTEGRATING ENDOGENOUS GROWTH AND BUSINESS CYCLES

Turning to our second limitation – the lack of a role for endogenous growth. As Lucas (1987) pointed out the welfare gains from eliminating business cycle fluctuations in the standard RBC model are very small and are dwarfed by the gains from increased growth. It is true that adding New Keynesian frictions

significantly increases the gains from stabilization policy, but they still remain small compared with those from increased growth.

10.2.1 DSGE Models with Endogenous Long-term Growth

Recently a number of papers have introduced long-run growth into DSGE models. Wang and Wen (2011) and Annicchiarico *et al.* (2011) do so within the simple AK approach. This literature establishes the existence of a relationship between growth and volatility with the important policy implications that monetary rules designed to stabilize short-run fluctuations affects the long-run balanced-growth path of the economy.

10.2.2 DSGE Models with Endogenous Technological Change

Closer to existing DSGE models is a literature that models of R&D led endogenous productivity about a balanced-growth path that remains exogenous, but new intermediate goods arrive exogenously. Leading examples of this are Comin and Gertler (2006), Comin (2009), Holden (2011), Comin *et al.* (2014), and Comin *et al.* (2016). We focus on the most recent of these cited papers as it suggests a general method of incorporating endogenous technical change and medium-term cycles into a range of DSGE models.

Comin *et al.* (2016) then develops and estimates a macroeconomic model modelled to allow for endogenous technology via R&D and adoption. The endogenous productivity mechanism is based on Comin and Gertler (2006), which uses the approach to connect business cycles to growth. The Comin/Gertler work, in turn, is a variant of the Romer (1986) expanding variety model of technological change, modelled to include an endogenous pace of technology adoption.

At the heart of the model is an aggregate production function for the final good which is the product of a productivity term that reflects endogenous variation and one reflecting exogenous variation. In modelling the former the paper focuses on two types of productivity enhancing investments: (i) the creation of new technologies through research and development (R&D) and (ii) the diffusion of new technologies via adoption expenditures. Thus the model allows for both endogenous and endogenous movements in total factor productivity and the empirical strategy is to let the data determine the importance of each. R&D led changes in productivity can encompass the basic NK model

with only the exogenous form and a likelihood race can assess the empirical relevance of each form.

10.3 EMPIRICAL CONCERNS

Our third limitation centres on the empirical dimension. Although Bayesian Maximum-Likelihood estimation is a giant step forward from the calibration methods of earlier RBC models there are a number of concerns. Many of the empirical issues are discussed in the review, Fernandez-Villaverde (2009), and monograph Herbst and Schorfheide (2015). Concerns are associated with identification, choice of priors, pre-filtering of data, the relationship between VARS and the solution of DSGE models and non-linearities including those associated with occasionally binding constraints. We consider these in turn.

10.3.1 Identification

Any likelihood-based system estimation faces a potential identification problem originating from the complexity of mapping from many parameters to data and a resulting flat likelihood function. Techniques for testing for weak and strong non-identification issues is active area of research; see Canova and Sala (2009b), Iskrev (2008b), Ratto (2008b) and Koop *et al.* (2013) for example, research that is feeding into toolboxes available in Dynare.

Bayesian rather than classical maximum likelihood estimation does address this problem, if we had confidence in the priors. This brings us to the choice of priors.

10.3.2 Priors

In many cases, the justification for the choice of priors reflects more the prior that some previous eminent researcher has got her priors right, often for a different model specification anyway. Del Negro and Schorfheide (2008) proposes an easily implementable method to obtain prior distributions for DSGE model parameters from data for moments of observable variables.

They divide the parameters into three groups, which reflect the information used to construct the prior. The first group contains the parameters that determine the steady states. The second group includes the utility, technology, and policy parameters governing the DSGE model's endogenous propagation

mechanism. For many of these parameters prior information comes from unrelated data sets, e.g. the prior for the labor supply elasticity parameter comes from micro-level studies on labor supply, the one for the price stickiness parameters from studies on price changes, etc. The third group consists of parameters describing the propagation mechanism of exogenous shocks. They propose a method of “endogenous priors” that translates priors about reasonable magnitudes for second moments of observables into a joint prior distribution for these parameters. Such priors may come from pre-sample evidence, for instance, and are assumed to be invariant across different DSGE model specifications.

10.3.3 Pre-filtering of Data

Another critique of Bayesian estimation is the method of pre-filtering the data. DSGE models are usually estimated with a two-step approach: data are first filtered and then structural parameters are estimated. This means that the choice of the statistical filter might be arbitrary and can affect the structural estimation. An alternative is to implement a hybrid one-step framework which links the observables to the model counterparts via a flexible specification which does not require that the cyclical component is solely located at business cycle frequencies and allows the non-cyclical component to take various time series patterns, see Filippo (2011), Canova (2013) and Cantore *et al.* (2015)

10.3.4 The Relationship between VARS and DSGE Models

Fernandez-Villaverde *et al.* (2007) show that for case where the number of shocks equals the number of observations then under quite weak ‘invertibility’ conditions the solution to the DSGE model can be approximated by a finite VAR. Levine *et al.* (2019) shows that these conditions become stronger in DSGE models where under rational expectations agents have imperfect information of the model’s state variables. Given that many DSGE models have more shocks than observations (including measurement errors ensures this) this brings into question a common practice of validating a DSGE model by comparing its impulse response functions with those of a data VAR, still more the estimation of such models carried out by matching these two sets of outcomes. Meenagh *et al.* (2018) provides a survey of this ‘indirect inference’ approach to estimating DSGE models.

Del Negro and Schorfheide (2004) and Del Negro *et al.* (2007) propose a method of either validating the model performance or improving the fit using a combination of an unrestricted VAR and the VAR implied by the estimated DSGE model. The DSGE-VAR approach then uses the DSGE model itself to construct a prior distribution for the VAR coefficients so that DSGE-VAR estimates are tilted toward DSGE model restriction, thus identifying the shocks for the IRFs.

This method constructs the DSGE prior by generating dummy observations from the DSGE model, and adding them to the actual data and leads to an estimation of the VAR based on a mixed sample of artificial and actual observations. The ratio of dummy over actual observations (called the hyperparameter λ) controls the variance and therefore the weight of the DSGE prior relative to the sample. For extreme values of this parameter (0 or ∞) either an unrestricted VAR or the DSGE is estimated. If λ is small the prior is diffuse. When $\lambda = \infty$, we obtain a VAR approximation of the log-linearized DSGE model. As λ becomes small the cross-equation restrictions implied by the DSGE model are gradually relaxed. Alternatively, one can simply find the ‘optimal’ value of λ by estimating this parameter as one of the deep parameters (see, for more details,). The optimal λ then represents how much the DSGE model is able to explain the real data. Details on the algorithm used to implement this DSGE-VAR are to be found in Del Negro and Schorfheide (2004), Del Negro *et al.* (2007) and Adjemian *et al.* (2008).

10.3.5 Non-Linearities

The aftermath of the Great Recession has seen the proliferation of DSGE models endowed with various forms of non-linearities. DSGE models are now used to explore numerous topics such as the effects of the nominal interest rate zero lower bound, occasionally binding constraints, uncertainty shocks, higher volatility, big or asymmetric shocks, risk-premia and behavioral models with highly non-linear learning mechanisms. In a linear world, DSGE model solutions can be represented by linear state-space systems of equations with Gaussian random variables. Since, in general, linear combinations of Gaussian variables are still Gaussian, it is possible to use the Kalman filter compute the likelihood function, and, thus, to make inference by means of Bayesian estimation methods. By contrast, higher-order approximations result in non-linear state-space representations. Therefore, it is no longer possible to map the path

of state variables over time, and consequently standard estimation techniques cannot be used for estimating the deep parameters of the model.

The literature offers two different approaches to make inference on the parameters of a non-linear DSGE model depending on whether or not filtering techniques are used to approximate the possible evolution of the state variables. Among the non-filtering-based estimators, the GMM method was adapted in various ways to match the moments of non-linear models. However with this method the econometrician needs to make an arbitrary choice about which moments should be assigned higher priority. By contrast filtering techniques are needed to approximate the likelihood of being in a certain state given the observables, and ultimately use Bayesian inference in a non-linear context. There are two approaches to approximate the posterior density of the model, global and local approaches.

Global approaches, such as the particle filter (PF), can produce unbiased estimates of the likelihood. Fernandez-Villaverde and Rubio-Ramirez (2004), Fernandez-Villaverde and Rubio-Ramirez (2015) have adapted the Metropolis-Hasting algorithm to estimate non-linear models using the PF. However the computational burden is very high and grows exponentially with high dimensional state-space models - the curse of dimensionality problem.

Although relying on restrictive assumptions on the distribution of latent states, local methods enable linear filtering techniques to be applied on non-linear models; this reduces the computational efforts and guarantees higher precision relative to the PF with few particles. The cubature Kalman filter - Arasaratnam and Haykin (2009)- assumes the posterior distribution of the states to take a priori a Gaussian form, which results in a gain in efficiency because the non-linear functions needs to be evaluated in a limited number of efficiently selected nodes.

In Kollmann (2013) an estimation method is adopted that avoids the need for a filter altogether by restricting the model to have the number of observed variables (used for estimation) to be equal to the number of exogenous shocks in the DSGE model. Exogenous innovations are extracted recursively by inverting the observation equation, which allows easy computation of the sample likelihood.

Kollmann (2017) then considers the general case where there are more shocks than observables (inevitable if observations are made with measurement errors). He uses a second-order perturbation approximation and the 'pruning' scheme of Kim *et al.* (2008) under which second-order terms are replaced by products of the linearized solution. Unless the pruning algorithm is used, second-order approximated models often generate exploding simulated time

paths. Pruning is therefore crucial for applied work based on second-order (or higher) approximated models. Kollmann assumes that the pruned second-order approximated model is the true data generating process (DGP). The method exploits the fact that the pruned system is linear in a state vector that consists of variables solved to second- and first-order accuracy, and of products of first-order accurate variables. Having set up the model in a linear state-space form standard, Bayesian estimation can then be used.

Meyer-Gohde (2014) in what he calls ‘risky linear approximations’ reconciles the linear framework with risk by constructing approximations of the policy functions of DSGE models that are linear in states but that account for risk in the points and slopes used to construct the linear approximation. Then two different approximations are constructed, one around the stochastic steady state and one around the ergodic mean. The method is sufficiently efficient to be used in estimation. Due to the linearity in states and under the assumption of normally distributed shocks, the Kalman filter is operational for the risky linear approximation. The paper finds that the risky linear approximation using the Kalman filter is as equally successful as standard perturbation PF estimation, both with the state space and nonlinear moving average policy function representations, in identifying parameters outside the reach of standard linear approximations.

Arioli (2018) assesses the performance these local approximation filtering techniques in estimating non-linear DSGE models. The main focus consists in evaluating the accuracy and efficiency of these filtering techniques through a Monte-Carlo study on a small New-Keynesian model with a trend in inflation.

10.4 HETEROGENEOUS AGENTS AND AGGREGATION

Finally we turn to what is perhaps the most important challenge for DSGE macroeconomics –that of heterogeneous agents and aggregation. The first generation of DSGE models, the RBC models stayed within the representative agent paradigm. The next wave of New Keynesian models made only a small deviation from this framework by assuming consumers have access to complete markets. Although they may differ in their initial tastes and are subject to staggered wage contracts and idiosyncratic shocks, they still face a common budget constraint. Then the economy in aggregate does not depend on the distribution of individual qualities.

10.4.1 HA and HANK Models

Contemporary dynamic general equilibrium modelling has gone a long way in answering critiques of representative agent assumptions, particularly with the work on incomplete market models that started in the real business cycle literature (e.g. Aiyagari (1994), Krusell and Smith (1998a); see Heathcote *et al.* (2009) for a survey). Overlapping generations models also remain important - although more in growth theory than short run macroeconomics - and there also exist DSGE models that incorporate asset market segmentation and multiple countries (see e.g. Alvarez *et al.* (2002) for a model which incorporates both of these elements).

A growing literature has emerged in recent years that aims at re-examining some important macro questions through models that assume the presence of idiosyncratic shocks to individuals' income, together with the existence of incomplete markets and borrowing constraints. Those features are often combined with the kind of nominal rigidities and monetary non-neutralities that are the hallmark of New Keynesian models. Following Kaplan *et al.* (2016) we refer to those models as HANK models (for "Heterogenous Agent New Keynesian" models).

The seminal contribution to the heterogeneous agents (HA) literature is by Krusell and Smith (1998b). This paper investigates how movements in the distribution of income and wealth affect the macroeconomy. They construct a calibrated version of the stochastic growth model with partially uninsurable idiosyncratic risk and movements in aggregate productivity. They make two contributions.

First, they develop a solution method referred to as *aggregate approximation* that has become very popular in the literature. The main computational difficulty of dynamic heterogeneous-agent models is the dependence of aggregate variables on the income and wealth distributions of agents. In order to predict prices, for example, consumers need to keep track of these distributions. Krusell and Smith show that in equilibrium, despite the fact that the state of the economy at any point in time is an infinite-dimensional object, all aggregate variables can be almost perfectly described as a function of two simple statistics: the mean of the wealth distribution and the aggregate productivity shock.

Second, they find a significant departure from permanent income behavior which stands in contrast to standard representative-agent models. Adding only uninsurable idiosyncratic risk to the representative-agent model implies an unrealistic wealth distribution in their model: both the mass of poor agents and

the concentration of wealth among the very richest is below what we can be observed in the data. When they introduce preference (discount factor) heterogeneity to the model, then it succeeds quite well in replicating the observed wealth distribution. Although, aggregate wealth is mainly in the hands of the rich in the model, poor agents have a large influence on aggregate consumption. Since these agents are also impatient on average, they can be characterized as “hand-to-mouth” consumers which leads to the observed departure from permanent income behavior.

Several lessons have been drawn from the HA and HANK literature. Taking into account agents’ heterogeneity has been shown to be important in order to understand the transmission of monetary policy. Several authors have emphasized how the transmission of monetary policy and its aggregate effects may vary significantly depending on the prevailing fiscal policy, as the latter determines how the implementation of monetary policy affects the distribution of individual income and wealth among agents with different marginal propensities to consume.

But solving for the equilibrium of HANK economies requires the use of computational techniques that keep track of the wealth distribution and tackle the difficulties arising from the presence of occasionally binding borrowing constraints. This limits their usefulness for large and even medium-size NK models.

10.4.2 TANK Models

Debortoli and Galí (2018) provides a simpler two-agent NK (TANK) framework that is computationally easy to implement and even allows for some analytical results. Both these features help to understand and quantify the implications of heterogeneity for aggregate fluctuations. It distinguishes between two types of households at each point in time, which are labelled as “unconstrained” or “constrained”, depending on whether their consumption satisfies or not a consumption Euler equation. Having made that distinction, the paper identifies three dimensions of heterogeneity that explain differences in aggregate fluctuations between a HANK economy and its representative agent counterpart (RANK, for short): (i) changes in the average consumption gap between constrained and unconstrained households, (ii) variations in consumption dispersion within unconstrained households, and (iii) changes in the share of constrained households. They show that the previous three factors are captured through additive “wedges” showing up in a log-linearized Euler equation for

aggregate consumption, and which determine the differential behavior of a HANK economy relative to its RANK counterpart. Furthermore, by tracing their responses to aggregate shocks, they are able to assess the quantitative significance of each of those heterogeneity factors in shaping aggregate output fluctuations.

A further contribution of this paper is to assess the ability of Two Agent New Keynesian (TANK) models to approximate the role of heterogeneity in richer HANK models. The two types of consumers – constrained and unconstrained – have constant shares in the population, while allowing only for aggregate shocks (i.e. disregarding idiosyncratic shocks). A unconstrained subset of households are assumed to have full access to financial markets (including markets for stocks and bonds), while constrained households are assumed to behave in a “hand-to-mouth” fashion, consuming their current income at all times. This will be the case if they do not have access to financial markets, find themselves continuously against a binding borrowing constraint, or display a pure myopic behavior.

Debertoli and Gali show that TANK model, which only captures one dimension of heterogeneity (the one which we refer to as the gap component), approximates reasonably well the predictions of a baseline HANK model regarding the effects of aggregate shocks on aggregate variables, as well as its predictions regarding the consequences of changes in the environment, once the fraction of constrained (and the transfer rule) are calibrated accordingly. Nonetheless, it should be clear that a simple TANK model will never be able to address many other questions involving heterogeneity (such as the effects of monetary policy on income and wealth distribution and, possibly, welfare) for which a richer HANK model is needed.

10.5 UNEMPLOYMENT AND DISEQUILIBRIUM IN DSGE MODELS

Until recently as with their RBC antecedents the New Keynesian forms still omitted involuntary unemployment. We are now seeing labour markets models with unemployment in both RBC and DSGE NK models (for the latter, see for example Blanchard and Galí (2010), Thomas (2008) and Cantore *et al.* (2014)).

We now turn to the remaining area highlighted earlier, concerning the criticism that DSGE models fail to deal with disequilibrium. As Howitt (2012) puts it,

“...the macroeconomic learning literature of Sargent (1999), Evans and Honkapohja (2001b) and others goes a long way towards understanding disequilibrium dynamics. But understanding how the system works goes well beyond this. For in order to achieve the kind of coordinated state that general equilibrium analysis presumes, someone has to find the right prices for the myriad of goods and services in the economy, and somehow buyers and sellers have to be matched in all these markets. ”

10.5.1 Search and Match in DSGE Models

Given the above, the behavioural NK models surveyed here can incorporate this type of disequilibrium, but do not usually do so. In addition, this basic critique of temporary equilibrium is often conflated with criticisms of the more restrictive Walrasian equilibria used in early DSGE models and real business cycle models, which tends to be replaced with search and matching, buffer stocks, and imperfect competition in contemporary models. The incorporation of search and matching mechanisms started with the contributions of Mortensen and Pissarides (1994) and Pissarides (2000), and there has been significant progress in embedding Mortensen-Pissarides search-matching (MPSM) frictions into otherwise standard DSGE models. Examples include Campolmi *et al.* (2010), Faia *et al.* (2010) and Monacelli *et al.* (2010). Many models featuring MPSM frictions focus only on the extensive margin, but there also are examples of models (for instance Thomas (2008), Krause *et al.* (2008) and Cantore *et al.* (2014)) that model the intensive margin in addition to the unemployment rate.

10.5.2 Disequilibrium

Regarding output market frictions and buffer stocks of goods and services, only recently have these been incorporated into RBC or DSGE models. Examples are Khan and Thomas (2007), which provides a micro-founded theory of inventories that succeeds in reproducing stylized facts regarding inventory investment in the USA, and Den Haan (2014), which combines an inventory

model with a MPSM model of labour markets. These are equilibrium models in both the Nash sense and the sense described above, but disequilibrium models in the Walrasian sense. They also assume rational expectations; combining the goods and labour market frictions in these models with bounded rationality as above is a possible route for behavioural NK models to take.

10.6 A FINANCIAL SECTOR AND FINANCIAL FRICTION

Another major lacuna in the earlier generation of DSGE models was been the absence of a banking sector. The monetary transmission mechanism existed simply through one nominal interest on a riskless bond, set by the central bank. The seminal work on financial frictions by Bernanke *et al.* (1999) introduced a risk premium paid by firms with an implicit intermediary financial institution. But it is only very recently that a comprehensive banking sector has appeared – see Gertler and Kiyotaki (2010a) as a representative example of this development.

Since the recent financial crisis, the number of papers studying the importance of financial frictions on macroeconomic outcomes and policy implications has grown considerably, commonly building on the mechanisms proposed in the Kiyotaki and Moore (2007) (KM) collateral constraints model, or the Bernanke *et al.* (1999) (BGG) costly state verification model. In KM the propagation and amplification comes from the fluctuations in asset prices, while in BGG from fluctuation of agents net worth. The KM approach was extended to study the effects of financial constraints on the banking sector in Gertler and Kiyotaki (2010b) (GK) where the limited commitment problem of KM introduces an agency problem between depositors and banks; when the value of bank capital declines, the borrowing constraint tightens and limits the amount of deposits the bank can raise and subsequently, the level of investment. Another extension proposed in Gertler and Karadi (2011) uses this approach to analyse the role of unconventional monetary policy. It is assumed the central bank can perform financial intermediation at a cost, but when the borrowing constraint tightens sufficiently, this cost is less than the inefficiency introduced by the agency problem. Macro-prudential regulation is studied in Gertler *et al.* (2012) where the government offers banks a subsidy per unit of outside equity issued and finances the subsidy with a tax on total assets. In equilibrium the

tax is set to make the subsidy revenue neutral and is chosen so as to internalize the external benefits of outside equity issuance.

The KM and BGG approaches have both been applied to the housing market. Impatient households post housing as collateral as in KM to secure mortgage loans in Iacoviello and Neri (2010) where the mechanism of Iacoviello (2005) is focused on the demand-side of the economy, and shown to have important effects on the business cycle. The constraints arise in Forlati and Lambertini (2011) due to a BGG costly state verification mechanism which is applied to household credit by assuming households observe a private housing-value shock that can lead to default when households are insolvent. The authors emphasise increased housing investment risk in highly leveraged economies.

The large influence of the BGG and KM approaches to the financial frictions literature might be partly due to the simplicity of applying the frictions to a representative agent, rational expectations model, solved using linear approximation techniques. It has been argued in Holden *et al.* (2016) that this rules out *ex ante* the possibility of explaining a number of key stylized facts, such as the large positive skew in the interest spreads, and negative skews in investment. There have been a number of papers that do study the non-linear and asymmetric effects of financial frictions which are much better suited to explain such phenomena, usually using global solution methods, or the perturbation based methods of Holden (2016) and Guerrieri and Iacoviello (2015).

10.7 AGENT-BASED MACRO-MODELS

These fundamental concerns discussed up to now are now driving research into an Agent-Based (AB) alternative. This approach represents economic agents as well as various social and environmental phenomena as autonomous virtual entities that interact during simulation experiments following pre-defined rules. In standard macroeconomic models agents' decisions consist of behavioural or, in the case of DSGE models, micro-founded first-order conditions satisfying a dynamic optimization problem, that are continuous functions of the current and past state of the economy. The AB approach provides a potentially more flexible way of modelling the cognitive capabilities of decision makers and their responses to both the macro- and individual micro-environment. In AB economies can represent out-of-equilibrium behaviour and be regarded as "evolving systems of autonomous interacting agents" (Tessfatsion (2003)). Hence, while DSGE assumes that agents have very sophisti-

cated computational capabilities and they live in very simple environments, AB models assume that people use simple behavioural rules to cope with complex and dynamic environments (Howitt (2011)).

ACE models (again see LeBaron and Tesfatsion (2008)) certainly tackle aggregation head-on and dispense with the rationality problem by ditching rational expectations altogether. But should central banks go down this path for their models? To quote LeBaron and Tesfatsion they (ACE models) “raise some practical complications for the applied econometrician... computational methods such a method of moments might be too computationally costly to undertake ... Researchers at central banks might never decide to fit giant ACE macro models to data.”

This comment may prove to be too pessimistic. Although macroeconomic AB models still remain few in number, interest in this area has been growing recently. Perhaps the most developed AB macroeconomic model is Dawid *et al.* (2011) (see also Ashraf *et al.* (2011b)) which has the standard economic agents found in current DSGE models (consumption and capital goods producers, households, banks and government). There is also an increasing number of AB models of banks and interbank payment systems both in the academic literature (Arciero *et al.* (2009), Markose *et al.* (2011), Ashraf *et al.* (2011a)) and the working paper series of central banks (Galbiati and Soramaki (2008)). A good reference to a rapidly growing literature is Leigh Tesfatsion at Iowa State who maintains a large software database, bibliography and link repository at <http://www2.econ.iastate.edu/tesfatsi/ace.htm>. Dilaver *et al.* (2018) provides a review and synthesis of this literature and recent attempts to incorporate insights from ACE into DSGE models thus bridging the gap between the approaches.

10.8 MACROECONOMIC POLICY

This section first considers the general optimal policy problem where the policymaker has a number of instruments and sets out to maximize a general discounted welfare criterion subject to the constraints of a DSGE model. It then moves onto the optimal design of robust policy.

10.8.1 Macroeconomic Policy Design using DSGE Models

If the policymaker is able to commit, the setting of instruments can be conducted in terms of the ex ante optimal policy. If the expected discounted household utility is chosen as the welfare criterion this becomes the well-known Ramsey problem. Note that the Ramsey solution is not the same thing as the *social planner's problem* in any model with some market failure. A problem with such a solution is that it involves a complex rule even for quite simple NK models. Much of the optimal policy literature therefore focuses on simple Taylor-type commitment rules that are optimized so as to come close to mimicking the Ramsey solution and this is the approach of this course.

In the absence of an ability to commit the policymaker must set policy to be time-consistent. Then the expected discounted household utility is maximized at time t , subject to the model constraints, in the knowledge that the same maximization procedure will be used to minimize at time $t + 1$. A dynamic programming solution then seeks a stationary *Markov Perfect* solution of the equilibrium. See Currie and Levine (1993) and Söderlind (1999) for a LQ treatment of this problem under perfect information and Levine *et al.* (2012b) under imperfect information. But see Dennis and Kirsanova (2013) for the possibility of multiple equilibria.

10.8.2 Robust Policy Across Contrasting Models

All models are wrong, but some are useful. **Box (1979).**

DSGE models estimated by Bayesian-Maximum-Likelihood methods can be considered as probability models in the sense described by Sims (2007) and be used for risk-assessment and policy design. We now consider the general policy question: how when faced with the existence of multiple competing and contrasting models, all of which are believed to be misspecified, should policymakers set macroeconomic policy? Building on Levine and Pearlman (2010), Cogley *et al.* (2011) and Levine *et al.* (2012c), Deak *et al.* (2019) proposes general framework that uses a pool of contrasting models for the policy design problem. The methodology uses Bayesian estimation to weight alternative models to design optimized Taylor-type rules that are robust in a sense described below. A crucial requirement is to provide a k -period ahead predictive density, given macro-economic data. The predictive density characterizes out-of-sample observations that have not been used up to that point in time to estimate the posterior density of the parameter vector. As such this provides

predictions about future observations that fully incorporate the information regarding within-model uncertainty (defined below) in the data.

Two further requirements to apply the methodology are that the models in the pool to share the same policy instrument under investigation and to have a welfare criterion to rank alternative policies. The models in the pool do not need to share the estimated parameter vector, nor even the observables; they can be nested as well as non-nested. Thus the methodology can be applied to a wide range of macroeconomic models from mainstream DSGE, behavioural to agent-based, and indeed to other non-macroeconomic settings as long as these three requirements are met. The challenge for new models is to meet these basic conditions to be useful for guiding policy.

This methodology of then studies robust policy design where the policy-maker faces three forms of uncertainty. The first derives from uncertain future shocks. This is a standard problem in the optimal policy literature. The second is parameter uncertainty within each competing model. The third source of uncertainty, “across-model uncertainty”, we have alluded to: the existence of multiple competing models at her disposal. The paper then investigates the welfare consequences of a standard Taylor-type monetary policy rules using three medium-scale New Keynesian DSGE models: the Smets-Wouters model in Smets and Wouters (2007), the workhorse model widely used in policy-making institutions for forecasting and policy analysis, and the other two models that add variants of financial frictions. Hence, the model pool can be motivated by considering a policy maker who is uncertain how to incorporate financial frictions into a DSGE model or if they should be incorporated at all.

10.9 CONCLUDING COMMENTS

The opening comment in Blanchard (2016) expresses the view: “I see the current DSGE models as seriously flawed, but they are eminently improvable and central to the future of macroeconomics”. This survey has highlighted a number of perceived flaws and described research that is addressing these concerns. I conclude by highlighting three challenges for the construction, estimation and policy analysis of DSGE models.

First, much of the DSGE literature has focused on the closed economy, but a more recent growing body of research now extends the models to open economies; see Galí (2015) and Schmitt-Grohe and Uribe (2017) for excellent

textbook treatments. However the long tradition of comparing model properties to stylized facts has been thus far less common for emerging economies, which usually face a data-related ‘triple-whammy’: short spans of data, quarterly data of low quality for some variables and several crises, structural breaks and changes in policy regimes.

Nevertheless, an RBC-DSGE literature focusing on emerging economies is gradually developing and a few stylized facts, alongside some empirical debates, have emerged. These include: higher output volatility than developed economies; deeper and steeper recessions, but faster recoveries; Consumption and real wage volatility and real wage exceeding output volatility; real interest rates: counter-cyclical and leading the cycle; net exports strongly counter-cyclical and more volatile; dramatic "sudden stops" in capital inflows; a weaker role of trade openness as a transmission mechanism; the importance of informality. Modelling features discussed up to now that are particularly relevant include the credit-constrained and Ricardian household distinction and therefore TANK models; Cross country financial frictions; and the need to integrate growth and business cycles. Some first steps in addressing these issues are Batini *et al.* (2007), Gabriel *et al.* (2012), Veigh (2013) and Anand *et al.* (2015).

Secondly, there appears to be a consensus that Bayesian systems estimation is the way to proceed, and the case is strengthened for emerging economies with less reliable macro-data. However the development of Bayesian estimation techniques that can incorporate occasionally binding constraints (OBC) remains a major challenge. OBC arise in a number of ways in DSGE models: financial constraints discussed in Section 10.6 need not be continuously binding; policy constraints include the zero-lower bound on the nominal interest rate, a lower bound on the capital requirement of macro-prudential regulation and an upper bound on government debt. Although the solution of DSGE models with OBC is computationally possible, techniques that are sufficiently efficient for Bayesian estimation are as yet unavailable.

Thirdly, the use of the robustness methodology set out above has potential for arriving at a consensus on robust policy prescriptions for the full range of monetary, fiscal and macro-prudential policy issues outlined in Blanchard *et al.* (2010), Blanchard *et al.* (2013) and Blanchard and Summers (2017).

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APPENDICES

1 THE HOUSEHOLD PROBLEM

Households choose between work and leisure and therefore how much labour they supply. Let L_t be the total time available for work (say 16 hours per day) that consists of leisure time and H_t the proportion of this time spent at work. Then clearly $L_t + H_t = 1$ The single-period utility is

$$U = U(C_t, L_t) \quad (\text{A.1})$$

and we assume that¹

$$U_C > 0, U_L > 0, U_{CC} \leq 0, U_{LL} \leq 0 \quad (\text{A.2})$$

In a stochastic environment, the value function of the representative household at time t is given by

$$V_t = V_t(B_{t-1}) = \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta^s U(C_{t+s}, L_{t+s}) \right] \quad (\text{A.3})$$

The household's problem at time t is to choose paths for consumption $\{C_t\}$, leisure, $\{L_t\}$, labour supply $\{H_t = 1 - L_t\}$, capital stock $\{K_t\}$, investment $\{I_t\}$ and bond holdings to maximize V_t given by (A.3) given its budget constraint in period t

$$B_t = R_{t-1}B_{t-1} + W_tH_t - C_t - I_t - T_t \quad (\text{A.4})$$

where B_t is the given net stock of financial assets at the end of period t , r_t^K is the rental rate, w_t is the wage rate and R_t is the gross interest rate paid on assets held at the beginning of period t , I_t is investment and T_t are lump-sum taxes; and given that capital stock accumulates according to

$$K_t = (1 - \delta)K_{t-1} + (1 - S(X_t))I_t; \quad (\text{A.5})$$

$$X_t \equiv \frac{I_t}{I_{t-1}}; \quad S', S'' \geq 0; \quad S(1) = S'(1) = 0 \quad (\text{A.6})$$

¹ Our notation is $U_C \equiv \frac{\partial U}{\partial C}$, $U_{CC} \equiv \frac{\partial^2 U}{\partial C^2}$ etc.

In (A.6), $S(X_t)$ are investment adjustment costs, I_t units of output converts to $(1 - S(X_t))I_t$ of new capital sold at a real price Q_t (Tobin's Q). All variables are expressed in real terms relative to the price of output.

To solve the household problem we form a Lagrangian

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta^s \left(U(C_{t+s}, L_{t+s}) \right. \right. \\ & + \lambda_{t+s} \left[R_{t+s-1} B_{t+s-1} + W_{t+s} (1 - L_{t+s}) + r_{t+s}^K K_{t+s-1} - C_{t+s} - I_{t+s} - T_{t+s} - B_{t+s} \right] \\ & \left. \left. + Q_{t+s} [(1 - \delta) K_{t+s-1} + (1 - S(X_{t+s})) I_{t+s} - K_{t+s}] \right] \right] \end{aligned}$$

Then the first-order conditions with respect to $\{C_{t+s}\}$, $\{B_{t+s-1}\}$, $\{K_{t+s-1}\}$, $\{I_{t+s}\}$ and $\{L_{t+s}\}$ are respectively

$$\begin{aligned} \{C_{t+s}\} & : \mathbb{E}_t [U_{C,t+s} - \lambda_{t+s}] = 0; s \geq 0 & (A.22) \\ \{B_{t+s-1}\} & : \mathbb{E}_t [\beta^s \lambda_{t+s} R_{t+s-1} - \beta^{s-1} \lambda_{t+s-1}] = 0; s > 0 \text{ (} B_{t-1} \text{ given)} & (A.23) \\ \{K_{t+s-1}\} & : \mathbb{E}_t [\beta^s \lambda_{t+s} r_{t+s}^K + \beta^s \lambda_{t+s} Q_{t+s} (1 - \delta) - \beta^{s-1} \lambda_{t+s-1} Q_{t+s-1}] = 0; s > 0 \text{ (} K_{t-1} \text{ given)} & (A.24) \end{aligned}$$

$$\begin{aligned} \{I_{t+s}\} & : \mathbb{E}_t \left[\lambda_{t,t+s} Q_{t+s} (1 - S(I_{t+s}/I_{t+s-1})) - 1 - Q_{t+s} S'(I_{t+s}/I_{t+s-1}) \frac{I_{t+s}}{I_{t+s-1}} \right. \\ & \quad \left. - \beta \lambda_{t,t+s+1} Q_{t+s+1} S'(I_{t+s}/I_{t+s-1}) \times \left(-\frac{I_{t+s+1}}{I_{t+s}^2} I_{t+s+1} \right) \right] = 0; s \geq 0 & (A.25) \end{aligned}$$

$$\{L_{t+s}\} : \mathbb{E}_t [U_{L,t+s} - \lambda_{t+s} W_{t+s}] = 0; s \geq 0 \quad (A.26)$$

Putting $s = 0$ in (A.22), (A.25) and (A.26) and $s = 1$ in (A.23) and (A.24) and defining the stochastic discount factor as $\Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t}$ we now have:

$$\text{Euler Consumption} : 1 = R_t \mathbb{E}_t [\Lambda_{t,t+1}] \quad (A.12)$$

$$\text{Stochastic Discount Factor} : \Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t} \quad (A.13)$$

$$\text{where} : \lambda_t = U_{C,t} \quad (A.14)$$

$$\text{Labour Supply} : \frac{U_{H,t}}{\lambda_t} = -\frac{U_{L,t}}{\lambda_t} = -W_t \quad (A.15)$$

$$\text{Leisure and Hours} : L_t \equiv 1 - H_t \quad (A.16)$$

$$\begin{aligned} \text{Investment FOC} : & Q_t (1 - S(X_t) - X_t S'(X_t)) \\ & + \mathbb{E}_t [\Lambda_{t,t+1} Q_{t+1} S'(\Xi_{t+1}) \Xi_{t+1}^2] = 1 \end{aligned} \quad (A.17)$$

$$\text{Capital Supply} : \mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}^K] = 1 \quad (A.18)$$

where R_t^K is the gross return on capital given by

$$R_t^K = \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}} \quad (\text{A.19})$$

Note that the investment decision can be taken by separate capital producing firms who convert raw output into new capital at a cost and sold at a real price Q_t . They maximize with respect to $\{I_t\}$ expected discounted profits

$$\mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} [Q_{t+s}(1 - S(I_{t+s}/I_{t+s-1}))I_{t+s} - I_{t+s}] \quad (\text{A.20})$$

where $\Lambda_{t,t+s} = \beta^s \left(\frac{\lambda_{t+s}}{\lambda_t} \right)$ is the real stochastic discount rate over the interval $[t, t+s]$. This leads to the same first-order decision as (A.17).

1.1 Solution of the Household Problem with JR Preferences

Now form a Lagrangian

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_t \left[\sum_{s=0}^{\infty} \beta^s \left(U(C_{t+s}, L_{t+s}, X_{t+s}) \right. \right. \\ & + \lambda_{t+s} [R_{t+s-1}B_{t+s-1} + W_{t+s}(1 - L_{t+s}) + r_{t+s}^K K_{t+s-1} - C_{t+s} - I_{t+s} - T_{t+s} - B_{t+s}] \\ & \left. \left. + \lambda_{t+s} Q_{t+s} [(1 - \delta)K_{t+s-1} + (1 - S(X_{t+s}))I_{t+s} - I_{t+s}] + \mu_{t+s} [X_{t+s} - C_{t+s}^\gamma X_{t+s-1}^{1-\gamma}] \right) \right] \end{aligned}$$

Then the first-order conditions with respect to $\{X_{t+s}\}$, $\{C_{t+s}\}$, $\{B_{t+s-1}\}$, $\{K_{t+s-1}\}$, $\{I_{t+s}\}$ and $\{L_{t+s}\}$ are respectively

$$\{X_{t+s}\} : \mathbb{E}_t [U_{X,t+s} + \mu_{t+s} - \beta(1 - \gamma)\mu_{t+s+1}C_{t+s+1}^\gamma X_{t+s}^{-\gamma}] = 0; s \geq 0 \quad (\text{A.21})$$

$$\{C_{t+s}\} : \mathbb{E}_t [U_{C,t+s} - \lambda_{t+s} - \gamma\mu_{t+s}C_{t+s-1}^{\gamma-1}X_{t+s-1}^{1-\gamma}] = 0; s \geq 0 \quad (\text{A.22})$$

$$\{B_{t+s-1}\} : \mathbb{E}_t [\beta^s \lambda_{t+s} R_{t+s-1} - \beta^{s-1} \lambda_{t+s-1}] = 0; s > 0 \text{ (} B_{t-1} \text{ given)} \quad (\text{A.23})$$

$$\{K_{t+s-1}\} : \mathbb{E}_t [\beta^s \lambda_{t+s} r_{t+s}^K + \beta^s \lambda_{t+s} Q_{t+s}(1 - \delta) - \beta^{s-1} \lambda_{t+s-1} Q_{t+s-1}] = 0; s > 0 \text{ (} K_{t-1} \text{ given)} \quad (\text{A.24})$$

$$\{I_{t+s}\} : \mathbb{E}_t [\lambda_{t,t+s} Q_{t+s}(1 - S(I_{t+s}/I_{t+s-1})) - 1 - Q_{t+s} S'(I_{t+s}/I_{t+s-1}) \frac{I_{t+s}}{I_{t+s-1}}]$$

$$- \beta \lambda_{t,t+s+1} Q_{t+s+1} S' (I_{t+s} / I_{t+s-1}) \times \left(-\frac{I_{t+s+1}}{I_{t+s}^2} I_{t+s+1} \right)] = 0; s \geq 0 \quad (\text{A.2})$$

$$\{L_{t+s}\} : \mathbb{E}_t[U_{L,t+s} - \lambda_{t+s} W_{t+s}] = 0; s \geq 0 \quad (\text{A.2})$$

Putting $s = 0$ in (A.21), (A.22), (A.25) and (A.26) and $s = 1$ in (A.23) and (A.24) and defining the stochastic discount factor as $\Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t}$ we now have:

$$\text{Euler Consumption} : 1 = R_t \mathbb{E}_t [\Lambda_{t,t+1}] \quad (\text{A.27})$$

$$\text{Stochastic Discount Factor} : \Lambda_{t,t+1} \equiv \beta \frac{\lambda_{t+1}}{\lambda_t} \quad (\text{A.28})$$

$$\text{where} : \lambda_t = U_{C,t} - \gamma \mu_t \frac{X_t}{C_t} \quad (\text{A.29})$$

$$\text{and} : \mu_t = -U_{X,t} + \beta(1 - \gamma) \mathbb{E}_t \frac{\mu_{t+1} X_{t+1}}{X_t} \quad (\text{A.30})$$

$$\text{Labour Supply} : \frac{U_{H,t}}{\lambda_t} = -\frac{U_{L,t}}{\lambda_t} = -W_t \quad (\text{A.31})$$

$$\text{Leisure and Hours} : L_t \equiv 1 - H_t \quad (\text{A.32})$$

$$\begin{aligned} \text{Investment FOC} : Q_t(1 - S(\Xi_t) - \Xi_t S'(\Xi_t)) \\ + \mathbb{E}_t [\Lambda_{t,t+1} Q_{t+1} S'(\Xi_{t+1}) \Xi_{t+1}^2] = 1 \end{aligned} \quad (\text{A.33})$$

$$\text{Capital Supply} : \mathbb{E}_t [\Lambda_{t,t+1} R_{t+1}^K] = 1 \quad (\text{A.34})$$

where R_t^K is the gross return on capital given by

$$R_t^K = \frac{[r_t^K + (1 - \delta)Q_t]}{Q_{t-1}} \quad (\text{A.35})$$

2 DIXIT-STIGLITZ AGGREGATION

This Appendix sets out a work-horse model of monopolistic and oligopolistic price-setting model, originating from Dixit and Stiglitz (1977), used in very many branches of theoretical economics. The model is the basis for the NK model, but is also encountered in the endogenous growth literature with an expanding number of varieties, and in the two-country general equilibrium model of international monetary policy transmission of Obstfeld and Rogoff.

There are many other areas where this model raises its head: for instance, the ‘new trade’ theory of Krugman, (not so new now) and in spatial economics (see Fujita *et al.* (1999)). Sometimes the number of firms will be treated as a continuous variable (OK if the number of firms is large). For the more general model where the number of firms can be small (the ‘oligopoly model’) this is not appropriate and the discrete-number of firms case must be used. This is the case we consider. As the number of firms go to infinity we approach the monopolistic model in the NK model.²

The following sections first sets out the sequence of events and then proceeds to solve for a subgame-perfect equilibrium by backward induction.

2.1 Sequence of Events

1. Free-Entry Equilibrium: Firms enter the market until profits are forced down to zero.
2. Bertrand Equilibrium: Given the firm number firms set prices in a Bertrand equilibrium
3. Households Decisions: Given prices and income households choose quantities.

2.2 Solution of Stage 3

Households derive utility $U(C_0, C)$ from a quantity C_0 of a numeraire good and from an index of consumption of differentiated goods given by

$$C = \left[\sum_{i=1}^n c_i^\epsilon \right]^{\frac{1}{\epsilon}} ; \epsilon \in (0, 1) \quad (\text{B.1})$$

where $c_i, i = 1, 2, \dots, n$ is the quantity consumed of the i th differentiated good. (B.1) is a CES utility function with elasticity of substitution $\theta = \frac{1}{1-\epsilon}$. As $\epsilon \rightarrow 1$

² Sbordone (2008) studies competition effects by generalizing the NK open-economy model to allow for both a finite number of firms and Kimball aggregation.

the goods become perfect substitutes.³The households optimization problem is to maximize $U(C, C_0)$ subject to a budget constraint

$$C_0 + \sum_{i=1}^n p_i c_i = Y \quad (\text{B.2})$$

where total household income Y is exogenous.

We solve this problem in two stages. First, given expenditure $Y - C_0$ on the differentiated good we maximize C subject to a constraint $p_i c_i = Y - C_0$. First define a Lagrangian

$$C - \lambda \left(\sum_{i=1}^n p_i c_i - (Y - C_0) \right)$$

where $\lambda \geq 0$ is a Lagrange multiplier. Then the first-order conditions are:

$$\frac{1}{\epsilon} \left[\sum_{i=1}^n c_i^\epsilon \right]^{\frac{1}{\epsilon}-1} \epsilon c_i^{\epsilon-1} = \lambda p_i \quad (\text{B.3})$$

Returning to (B.3), dividing the j th equation by the k th equation we have

$$\left(\frac{c_j}{c_k} \right)^{\epsilon-1} = \frac{p_j}{p_k}$$

Substituting back into $\sum_{k=1}^n p_k c_k = Y - C_0$ we get

$$\sum_{k=1}^n p_k c_j \left(\frac{p_k}{p_j} \right)^{-\frac{1}{1-\epsilon}} = p_j^\theta c_j \sum_{k=1}^n p_k^{1-\theta} = Y - C_0$$

where recall that $\theta = \frac{1}{1-\epsilon} > 1$. This results in the demand for good $j = 1, 2, \dots, n$ given by

$$c_j = \frac{Y - C_0}{p_j^\theta \sum_{k=1}^n p_k^{1-\theta}} \quad (\text{B.4})$$

To interpret and manipulate (B.4) it is convenient to define a price index

$$P = \left[\sum_{k=1}^n p_k^{1-\theta} \right]^{\frac{1}{1-\theta}} \quad (\text{B.5})$$

³ Note that as $\epsilon \rightarrow 0$, $\left[\frac{1}{n} \sum_{i=1}^n x_i^\epsilon \right]^{\frac{1}{\epsilon}} \rightarrow [x_1 x_2 \cdots x_n]^{\frac{1}{n}}$, so the form of the utility function (B.1) ‘almost’ tends to Cobb-Douglas as $\theta \rightarrow 1$.

Now (B.4) and (B.5) can be written

$$c_j = \frac{Y - C_0}{p_j^\theta P^{1-\theta}} \quad (\text{B.6})$$

The importance of (B.6) is that given P , the elasticity of demand for variety j on the world market with respect to price is constant with elasticity $-\theta$.

Next substitute (B.6) into (B.1) to obtain

$$C = \frac{Y - C_0}{P^{1-\theta}} \left[\sum_{i=1}^n p_i^{-\epsilon\theta} \right]^{\frac{1}{\epsilon}} \quad (\text{B.7})$$

$$= \frac{Y - C_0}{P^{1-\theta}} \left[\sum_{i=1}^n p_i^{1-\theta} \right]^{\frac{1}{\epsilon}} \quad (\text{B.8})$$

$$= (Y - C_0) P^{\theta-1+\frac{1-\theta}{\epsilon}} \quad (\text{B.9})$$

$$= (Y - C_0) P^{-1} \quad (\text{B.10})$$

The importance of this result is that the budget constraint of the household can now be written in terms of the composite consumption index C and the composite price index P as

$$C_0 + PC = Y \quad (\text{B.11})$$

Also note that (B.6) now may be written

$$c_i = \left[\frac{p_j}{P} \right]^{-\theta} C \quad (\text{B.12})$$

a result we call upon later in both growth theory and open-economy macroeconomics.

This completes the first stage of the households' decision. The second stage is to maximize $U(C_0, C)$ given (B.11). To do this again define a Lagrangian

$$U(C_0, C) - \lambda(C_0 + PC - Y) \quad (\text{B.13})$$

where λ is (different) Lagrange multiplier. The foc are

$$\begin{aligned} U_{C_0} &= \lambda \\ U_C &= \lambda P \end{aligned}$$

Hence

$$\frac{U_{C_0}}{U_C} = \frac{1}{P} \quad (\text{B.14})$$

which together with the budget constraint (B.11) completely characterizes the allocation of consumption between the numeraire good and the differentiated goods. This completes the household's consumption choice given prices p_i , $i = 1, 2, \dots, n$.

In general a change in the price of a single differentiated good will shift demand between these goods and between differentiated goods as a whole and the numeraire good. For the special case where $U(C_0, C)$ is Cobb-Douglas, i.e., $U(C_0, C) = C^a C_0^{1-a}$ we have a very convenient result that $C_0 = aY$ and $PC = (1 - a)Y$; that is, *the proportion of income spent on the differentiated goods is constant*. In what follows we confine ourselves to this case.

2.3 Solution of Stage 2

Each firm produces a single differentiated good so there are n firms. We assume the cost structure is identical across these firms. The marginal costs, m , are fixed and firms face a set-up cost of F . Firm $i = 1, 2, \dots, n$ sets its price p_i and produces a quantity x_i resulting in profits π_i given by

$$\pi_i = (p_i - m)x_i - F \quad (\text{B.15})$$

where from stage 3, in a market equilibrium

$$x_i = c_i = \frac{(1 - a)Y}{p_i^\theta P^{1-\theta}} \quad (\text{B.16})$$

In a Bertrand equilibrium firm i maximize profits π_i given by (B.15), with respect to p_i , subject to (B.16) given the prices p_j , $j \neq i$ set by the other firms. The foc for this problem is

$$x_i + (p_i - m) \frac{\partial x_i}{\partial p_i} = 0 \quad (\text{B.17})$$

where from (B.16) we have

$$\frac{\partial x_i}{\partial p_i} = -\frac{\theta x_i}{p_i} + \underbrace{\frac{(1 - a)Y}{p_i^\theta} \frac{\partial (P^{\theta-1})}{\partial p_i}}_{\text{strategic interaction term}} \quad (\text{B.18})$$

In working out the effect of a change in the price of its good firm i considers two effects: the first term takes the total price index of goods in the market,

P , as given. The second *strategic* term considers the effect of its own price on the overall price index. In the standard model of monopolistic competition there are so many firms that we can ignore this strategic effect. However as the number becomes small this is no longer the case. Then using

$$\frac{\partial(P^{\theta-1})}{\partial p_i} = \frac{(\theta-1)p_i^{-\theta}}{(\sum_{i=1}^n p_i^{1-\theta})^2} \quad (\text{B.19})$$

we can proceed to derive the *symmetric* Bertrand equilibrium (since firms are identical) where $p_1 = p_2 = \dots = p$, $x_1 = x_2 = \dots = x$. Then a little algebra gives

$$\frac{\partial x_i}{\partial p_i} = -\frac{\theta x}{p} + \frac{x(\theta-1)}{np} \quad (\text{B.20})$$

so the foc (B.17) becomes

$$x \left[1 + \frac{p-m}{p} \left(-\theta + \frac{(\theta-1)}{n} \right) \right] = 0 \quad (\text{B.21})$$

Hence for any non-zero demand we arrive at the *Lerner index* giving the mark-up of price over marginal costs

$$L = \frac{p-m}{p} = \frac{1}{\theta - \frac{(\theta-1)}{n}} = \frac{n}{\theta(n-1) + 1} \quad (\text{B.22})$$

Thus as n becomes large we have the familiar monopolistic competition result that $L = \frac{1}{\theta}$; i.e., the mark-up depends only on the elasticity θ and tends to zero as goods become homogenous ($\theta \rightarrow \infty$). Noting that $\theta >$, for the oligopoly (small n) case the *mark-up decreases as the number of firms increases*.

2.4 Solution of Stage 1

In a free-entry subgame-perfect equilibrium of the entire game firms enter until their participation constraint(PC) is forced below zero. Then the symmetric equilibrium number of firms is the largest integer n^* such that

$$\pi = (p-m)x - F \geq 0 \quad (\text{B.23})$$

To find n^* first solve the following set of equations obtained from stages 1, 2 and 3 respectively, for n , x and p where n is not an integer:

$$\text{Lerner Index} : \frac{p-m}{p} = \frac{n}{\theta(n-1) + 1} \quad (\text{B.24})$$

$$\text{Firm's PC} : (p - m)x = F \quad (\text{B.25})$$

$$\text{Households' Demand} : pnx = (1 - a)Y \quad (\text{B.26})$$

Then dividing PC by BC and substituting into the Lerner relation we arrive at the non-integer solution $n = \hat{n}$:

$$\hat{n} = 1 + \frac{1}{\theta} \left[\frac{(1 - a)Y}{F} - 1 \right] \quad (\text{B.27})$$

Then the integer number of firms is the largest integer $n^* \leq \hat{n}$. From (B.27) we can see that the equilibrium number of firms increases with demand (Y) and the elasticity of substitution θ falls. The intuition here is that a fall in θ increases monopoly power because households are less inclined to switched into an alternative good when the price of its competing good rises. The response of non-incumbents is then to enter the market. This completes the model of oligopoly.

3 LINEARIZATION: THE EULER EQUATION

This section demonstrates the use of the Taylor series to linearize the Euler consumption equation about its steady state. The non-linear form is given by

$$U_{C,t} = \beta R_t \mathbb{E}_t [U_{C,t+1}] \quad (\text{C.1})$$

$$U_{C,t} = (1 - \varrho) C_t^{(1-\varrho)(1-\sigma_c)-1} ((1 - H_t)^{\varrho(1-\sigma_c)}) \quad (\text{C.2})$$

about a no-growth steady state $C_t = C$, $R_t = R$, $H_t = H$.

Let's now apply the Taylor-Series approximation to (C.1). Applying

$$f(R_t, U_{C,t+1}) \approx f(R, U_C) + \frac{\partial f}{\partial R_t} (R_t - R) + \frac{\partial f}{\partial U_{C,t+1}} (U_{C,t+1} - U_C)$$

with the partial derivatives evaluated at the steady state, to the right-hand-side of (C.1), we have

$$\begin{aligned} U_{C,t} &\approx U_C + \beta U_C (R_t - R) + \beta R \mathbb{E}_t [U_{C,t+1} - U_C] \\ &\approx U_C + U_C \frac{(R_t - R)}{R} + \mathbb{E}_t [U_{C,t+1} - U_C] \end{aligned} \quad (\text{C.3})$$

using the steady-state relationship $\beta R = 1$. Hence

$$u_{C,t} \equiv \frac{U_{C,t} - U_C}{U_C} \approx r_t + \mathbb{E}_t [u_{C,t+1}] \quad (\text{C.4})$$

as stated in section 5.3. Note that $r_t \equiv \frac{R_t - R}{R} \approx \log \frac{R_t}{R}$.

To linearize (C.2) it is quickest to first take logs to obtain

$$\log U_{C,t} = ((1 - \varrho)(1 - \sigma_c) - 1) \log C_t + \varrho(1 - \sigma_c) \log(1 - H_t) + \text{constant} \quad (\text{C.5})$$

Then subtracting the steady state from both sides of the equation we have

$$\log U_{C,t} - \log U_C = ((1 - \varrho)(1 - \sigma_c) - 1)(\log C_t - \log C) + \varrho(1 - \sigma_c)(\log(1 - H_t) - \log(1 - H)) \quad (\text{C.6})$$

Now again use the Taylor series expansion to approximate

$$\log(1 - H_t) - \log(1 - H) \approx -\frac{1}{1 - H} (H_t - H) = -\frac{H}{1 - H} \frac{H_t - H}{H} \equiv -\frac{H}{1 - H} h_t \quad (\text{C.7})$$

Hence we arrive at the linearization of section 5.3.

$$u_{C,t} \approx -((\sigma_c - 1)(1 - \varrho) + 1)c_t + (\sigma_c - 1)\varrho \frac{H}{1 - H} h_t \quad (\text{C.8})$$

Useful analysis can be done in the case of a logarithmic utility function ($\sigma_c = 1$). In this case $u_{C,t} \approx -c_t$ so the linear NK Euler equation:

$$\mathbb{E}_t[u_{C,t+1}] = u_{C,t} - r_{t+1} \Rightarrow c_t = \mathbb{E}_t[c_{t+1}] - r_{t+1} \quad (\text{C.9})$$

where $r_t = r_{n,t-1} - \pi_t$. Solving this forward in time gives

$$c_t = -\mathbb{E}_t[r_{t+1}] - \mathbb{E}_t[r_{t+2}] - \mathbb{E}_t[r_{t+3}] - \cdots = -\sum_{i=1}^{\infty} \mathbb{E}_t[r_{t+i}] \quad (\text{C.10})$$

Thus the proportional deviation of consumption from its steady state is minus the accumulated sum of expected future deviations of the nominal interest rate about its steady state. If interest rates are expected to be above the steady state for a long period in the future, then consumption falls and savings rise.

4 LINEARIZATION: THE NK PHILLIPS CURVE

The non-linear price dynamics are given by

$$JJ_t - \xi\beta\mathbb{E}_t[\Pi_{t+1}^{\zeta-1}JJ_{t+1}] = Y_t U_{C,t} \quad (\text{D.11})$$

$$J_t - \xi\beta\mathbb{E}_t[\Pi_{t+1}^{\zeta}J_{t+1}] = \left(\frac{1}{1 - \frac{1}{\zeta}}\right) Y_t U_{C,t} MC_t \quad (\text{D.12})$$

$$1 = \xi \Pi_t^{\zeta-1} + (1 - \xi) \left(\frac{J_t}{JJ_t} \right)^{1-\zeta} \quad (\text{D.13})$$

Expanding (D.11) as a Taylor series:

$$\begin{aligned} JJ + JJ_t - JJ - \xi \beta \mathbb{E}_t[\Pi^{\zeta-1} JJ] &+ (\zeta - 1) \Pi^{\zeta-2} JJ (\Pi_{t+1} - \Pi) + \Pi^{\zeta-1} (JJ_{t+1} - JJ) \\ &= YU_C + U_C(Y_t - Y) \end{aligned} \quad (\text{D.14})$$

Hence cancelling out the constants on both sides, putting $\Pi = 1$ and dividing by JJ we have that

$$jj_t \equiv \frac{JJ_t - JJ}{JJ} = \xi \beta \mathbb{E}_t[(\zeta - 1)\pi_{t+1} + jj_{t+1}] + \frac{YU_C}{JJ}(y_t + u_{C,t}) \quad (\text{D.15})$$

Similarly linearizing (D.12) we arrive at

$$j_t \equiv \frac{J_t - J}{J} = \xi \beta \mathbb{E}_t[\zeta \pi_{t+1} + j_{t+1}] + \frac{YU_C MC}{J \left(1 - \frac{1}{\zeta}\right)}(y_t + u_{C,t} + mc_t) \quad (\text{D.16})$$

Next we linearize (D.13) and put $\Pi = 1$ to obtain

$$\xi(\zeta - 1)\pi_t + (1 - \xi)(1 - \xi) \left(\frac{J}{JJ} \right)^{-\zeta} (j_t - jj_t) = 0 \quad (\text{D.17})$$

We then use the steady state relationships (6.27) – (6.91), (D.16) and (D.17) to obtain

$$jj_t = \xi \beta \mathbb{E}_{t+1}[(\zeta - 1)\pi_{t+1} + h_{t+1}] + y_t + u_{C,t} \quad (\text{D.18})$$

$$j_t = \xi \beta \mathbb{E}_{t+1}[\zeta \pi_{t+1} + j_{t+1}] + y_t + u_{C,t} + mc_t \quad (\text{D.19})$$

$$\xi \pi_t = (1 - \xi)(j_t - jj_t) \quad (\text{D.20})$$

Then subtracting (D.19) from (D.18) and using (D.20) we arrive at the *linear NK Phillips Curve*

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \frac{(1 - \xi)(1 - \xi \beta)}{\xi} mc_t \quad (\text{D.21})$$

As for the previous linearized Euler equation this can be solved forward in time to give

$$\pi_t = \frac{(1 - \xi)(1 - \xi \beta)}{\xi} \sum_{i=0}^{\infty} \beta^i mc_{t+i} \quad (\text{D.22})$$

telling us that in the NK model in proportional deviation terms about the steady state, inflation is proportional to the discounted sum of expected future deviations of marginal costs.

5 INFLATION AND PRICE DISPERSION DYNAMICS

This Appendix shows how first order conditions expressed as summations, as in Calvo contracts, can be expressed as difference equations suitable for coding in Dynare. Then the dynamic form of price dispersion, Δ_t is derived.

5.1 A Useful Lemma

In the first order conditions for Calvo contracts and expressions for value functions we are confronted with expected discounted sums of the general form

$$\Omega_t = \mathbb{E}_t \left[\sum_{k=0}^{\infty} \beta^k X_{t,t+k} Y_{t+k} \right] \quad (\text{E.1})$$

where $X_{t,t+k}$ has the property $X_{t,t+k} = X_{t,t+1} X_{t+1,t+k}$ and $X_{t,t} = 1$ (for example an inflation, interest or discount rate over the interval $[t, t+k]$).

Lemma

Ω_t can be expressed as

$$\Omega_t = Y_t + \beta \mathbb{E}_t [X_{t,t+1} \Omega_{t+1}] \quad (\text{E.2})$$

Proof

$$\begin{aligned} \Omega_t &= X_{t,t} Y_t + \mathbb{E}_t \left[\sum_{k=1}^{\infty} \beta^k X_{t,t+k} Y_{t+k} \right] \\ &= Y_t + \mathbb{E}_t \left[\sum_{k'=0}^{\infty} \beta^{k'+1} X_{t,t+k'+1} Y_{t+k'+1} \right] \\ &= Y_t + \beta \mathbb{E}_t \left[\sum_{k'=0}^{\infty} \beta^{k'} X_{t,t+1} X_{t+1,t+k'+1} Y_{t+k'+1} \right] \\ &= Y_t + \beta \mathbb{E}_t [X_{t,t+1} \Omega_{t+1}] \quad \square \end{aligned}$$

5.2 Price Dynamics

Then (6.10) can be written

$$\frac{P_t^O}{P_t} = \frac{J_t}{J_t} \quad (\text{E.3})$$

and summations JJ_t and J_t are of the form considered in the Lemma above. Applying the Lemma we then have the recursive form:

$$\begin{aligned}
JJ_t - \xi \mathbb{E}_t[\Lambda_{t,t+1} \Pi_{t+1}^{\zeta-1} JJ_{t+1}] &= Y_t \\
J_t - \xi \mathbb{E}_t[\Lambda_{t,t+1} \Pi_{t+1}^{\zeta} J_{t+1}] &= \frac{1}{1 - \frac{1}{\zeta}} Y_t MC_t MS_t \\
1 &= \xi \Pi_t^{\zeta-1} + (1 - \xi) \left(\frac{J_t}{JJ_t} \right)^{1-\zeta} \\
MC_t &= \frac{P_t^W}{P_t} = \frac{W_t}{P_t F_{H,t}} \\
\Delta_t &\equiv \frac{1}{n} \sum_{j=1}^n (P_t(j)/P_t)^{-\zeta} = \xi \Pi_t^{\zeta} \Delta_{t-1} + (1 - \xi) \left(\frac{J_t}{JJ_t} \right)^{-\zeta}
\end{aligned} \tag{E.4}$$

as in the main text.

With *indexing* by an amount $\gamma \in [0, 1]$, the optimal price-setting first-order condition for a firm j setting a new optimized price $P_t^O(j)$ is now given by

$$P_t^O(j) = \frac{\frac{1}{(1-\gamma/\zeta)} \mathbb{E}_t \left[\sum_{k=0}^{\infty} \xi^k \Lambda_{t,t+k} P_{t+k} MC_{t+k} MS_{t+k} Y_{t+k}(j) \right]}{\mathbb{E}_t \left[\sum_{k=0}^{\infty} \xi^k D_{t,t+k} Y_{t+k}(j) \left(\frac{P_{t+k-1}}{P_{t-1}} \right)^{\gamma} \right]}$$

Price dynamics are now given by

$$\begin{aligned}
\frac{P_t^O}{P_t} &= \frac{J_t}{JJ_t} \\
JJ_t - \xi \beta \mathbb{E}_t[\tilde{\Pi}_{t+1}^{\zeta-1} JJ_{t+1}] &= Y_t U_{C,t} \\
J_t - \xi \beta \mathbb{E}_t[\tilde{\Pi}_{t+1}^{\zeta} J_{t+1}] &= \frac{1}{1 - \frac{1}{\zeta}} MC_t MS_t Y_t U_{C,t} \\
\tilde{\Pi}_t &\equiv \frac{\Pi_t}{\Pi_{t-1}^{\gamma}}
\end{aligned}$$

5.3 Dynamics of Price Dispersion

Price dispersion lowers aggregate output as follows. As with consumption goods, the demand equations for each differentiated good m with price $P_t(m)$ forming aggregate investment and public services takes the form

$$I_t(m) = \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} I_t; \quad G_t(m) = \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} G_t \tag{E.5}$$

Hence equilibrium for good m gives

$$Y_t(m) = A_t H_t(m) \left(\frac{K_t(m)}{Y_t(m)} \right)^{\frac{1-\alpha}{\alpha}} = (C_t + I_t + G_t) \left(\frac{P_t(m)}{P_t} \right)^{-\zeta} \quad (\text{E.6})$$

where $Y_t(m)$, $H_t(m)$ and $K_t(m)$ are the quantities of output, hours and capital needed in the wholesale sector to produce good m in the retail sector. Since the capital-labour ratio is constant integrating over m , and using $H_t = \int_0^1 H_t(m) dm$ we obtain

$$Y_t = \frac{F(A_t, H_t, K_t)}{\Delta_t} \quad (\text{E.7})$$

as in the main text.

Price dispersion is linked to inflation as follows. Assuming as before that the number of firms is large we obtain the following dynamic relationship:

$$\Delta_t = \xi \Pi_t^\zeta \Delta_{t-1} + (1 - \xi) \left(\frac{J_t}{J J_t} \right)^{-\zeta} \quad (\text{E.8})$$

Proof

In the next period, ξ of these firms will keep their old prices, and $(1 - \xi)$ will change their prices to P_{t+1}^O . By the law of large numbers, we assume that the distribution of prices among those firms that do not change their prices is the same as the overall distribution in period t . It follows that we may write

$$\begin{aligned} \Delta_{t+1} &= \xi \sum_{j \text{ no change}} \left(\frac{P_t(j)}{P_{t+1}} \right)^{-\zeta} + (1 - \xi) \left(\frac{J_{t+1}}{J J_{t+1}} \right)^{-\zeta} \\ &= \xi \left(\frac{P_t}{P_{t+1}} \right)^{-\zeta} \sum_{j \text{ no change}} \left(\frac{P_t(j)}{P_t} \right)^{-\zeta} + (1 - \xi) \left(\frac{J_{t+1}}{J J_{t+1}} \right)^{-\zeta} \\ &= \xi \left(\frac{P_t}{P_{t+1}} \right)^{-\zeta} \sum_j \left(\frac{P_t(j)}{P_t} \right)^{-\zeta} + (1 - \xi) \left(\frac{J_{t+1}}{J J_{t+1}} \right)^{-\zeta} \\ &= \xi \Pi_{t+1}^\zeta \Delta_t + (1 - \xi) \left(\frac{J_{t+1}}{J J_{t+1}} \right)^{-\zeta} \quad \square \end{aligned}$$

6 DATA SOURCES AND CONSTRUCTION

The original data series are summarized below:

- NGDP: Gross domestic product; BEA table 1.1.5, line 1
- RGDP: Real Gross domestic product; BEA table 1.1.6, line 1
- PCE: Personal Consumption expenditure; BEA Table 1.1.5, line 2
- PFI: Private Fixed Investment; BEA Table 5.3.5; line 1
- CNP16OV: Civilian non institutional population, FRED database, BLS
- $CNP16OV_index$: $CNP16OV(2005:2)=1$
- FFR: Federal Fund Rate, Fred Database, FED St. Louis
- $GDP_Deflator$: $NGDP/RGDP*100$
- BAA: Moody's seasoned Baa corporate bond yields, FED Board.

The three/four observables used in the estimation are then constructed as follows:

- $$\log Y_t^{obs} = LN \left(\frac{RGDP}{CNP16OV_index} \right) * 100$$
- $$\Pi_t^{obs} = LN \left(\frac{GDP_Deflator_t}{GDP_Deflator_{t-1}} \right) * 100$$
- $$\log C_t^{obs} = LN(PCE/index/GDP_Deflator) * 100$$
- $$\log I_t^{obs} = LN(PFI/index/GDP_Deflator) * 100$$
- $$R_{n,t}^{obs} = FFR/4$$
- $$R_{kn}^{obs} = BAA/4$$